

Mathematical Preliminaries

Vector and tensor algebra, fields, integral theorems,
and full solutions to the chapter problems

A from-scratch, definition–theorem–proof companion to
Steigmann and Shirani, *Principles of Continuum Mechanics*,
Chapter 1

(§ 1.1.1 through § 1.3, in the book’s order and numbering)

Each subsection opens with plain expository ground, in the spirit of the text, before any formal statement. Nothing in the chapter is skipped; every algebraic step, including those the book leaves to the reader, is carried out. The chapter problems are all solved in §1.3.

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Chapter 1

Mathematical Preliminaries

A working knowledge of vector and tensor algebra and analysis is a prerequisite for continuum mechanics. This chapter assembles the results we will need. The style is deliberately slow: we begin each topic with the naive question that forces the next idea, then make it precise and prove it. Throughout, $\mathcal{V}_n$ is an n dimensional real vector space, $\mathbb{E}^n$ the same carrying an inner product, and $\mathbb{E}^3$ the three dimensional case; $\{\underline{e}_i\}$ is a fixed orthonormal basis, indices run over $\{1, 2, 3\}$ unless stated otherwise, and a repeated index in a term is summed. The symbol δ_{ij} is Kronecker's, ε_{ijk} the permutation symbol, and Lin the space of second order tensors. Appendix [A](#) maps every numbered book equation to the place it is produced here.

1.1 Vector and tensor theory

1.1.1 Vector spaces

Displacements in space, velocities, and forces share exactly two features: any two of them may be added tip to tail, and any one may be scaled by a real number. Nothing else about “arrows” is used in the algebra that follows, so instead of drawing pictures we simply write down those two operations and the rules they obey. The reward is leverage: a theorem

proved from the rules holds at once for displacements, forces, and later for tensors and fields. The list of rules is the definition of a vector space.

DEFINITION 1.1 (Vector space). A real vector space is a set $\mathcal{V}$ with maps $+$: $\mathcal{V} \times \mathcal{V} \rightarrow \mathcal{V}$ and $\cdot$: $\mathbb{R} \times \mathcal{V} \rightarrow \mathcal{V}$ such that, for all $\underline{a}, \underline{b}, \underline{c} \in \mathcal{V}$ and $\alpha, \beta \in \mathbb{R}$,

- (a) $\alpha \underline{a} + \beta \underline{b} \in \mathcal{V}$,
- (b) $(\underline{a} + \underline{b}) + \underline{c} = \underline{a} + (\underline{b} + \underline{c})$,
- (c) $\underline{a} + \underline{b} = \underline{b} + \underline{a}$,
- (d) $\exists \underline{0} : \underline{a} + \underline{0} = \underline{a}$,
- (e) $\exists (-\underline{a}) : \underline{a} + (-\underline{a}) = \underline{0}$,
- (f) $(\alpha\beta)\underline{a} = \alpha(\beta\underline{a})$,
- (g) $(\alpha + \beta)\underline{a} = \alpha\underline{a} + \beta\underline{a}$,
- (h) $\alpha(\underline{a} + \underline{b}) = \alpha\underline{a} + \alpha\underline{b}$,
- (i) $1\underline{a} = \underline{a}$,
- (j) $0\underline{a} = \underline{0}$.

Vectors are set in bold; real numbers (scalars) in light face.

We want to pin any vector down by a finite list of numbers. For that we need a set of reference vectors that neither wastes effort duplicating each other nor leaves gaps. “No duplication” is linear independence; “no gaps” together with independence is a basis.

DEFINITION 1.2 (Linear dependence). $\{\underline{v}_1, \dots, \underline{v}_p\} \subset \mathcal{V}$ is linearly independent iff $\alpha_i \underline{v}_i = \underline{0} \Rightarrow \alpha_1 = \dots = \alpha_p = 0$, and linearly dependent otherwise.

DEFINITION 1.3 (Dimension, basis). $\mathcal{V}$ is finite dimensional of dimension n , written $\mathcal{V}_n$, iff every linearly independent set has at most n elements and some such set has exactly n . Any linearly independent set of n elements is a basis.

THEOREM 1.4 (Coordinates exist and are unique). *Let $\{\underline{u}_1, \dots, \underline{u}_n\}$ be a basis of $\mathcal{V}_n$. Then every $\underline{v} \in \mathcal{V}_n$ has a representation $\underline{v} = \alpha_i \underline{u}_i$ with unique coefficients α_i .*

Proof. Existence. $\{\underline{u}_1, \dots, \underline{u}_n, \underline{v}\}$ has $n + 1 > n$ elements, hence is dependent: $\lambda \underline{v} + \lambda_i \underline{u}_i = \underline{0}$ with the coefficients not all zero. If $\lambda = 0$ then $\lambda_i \underline{u}_i = \underline{0}$ forces every $\lambda_i = 0$, a contradiction; so $\lambda \neq 0$ and $\underline{v} = (-\lambda_i/\lambda) \underline{u}_i$. *Uniqueness.* $\alpha_i \underline{u}_i = \beta_i \underline{u}_i \Rightarrow (\alpha_i - \beta_i) \underline{u}_i = \underline{0} \Rightarrow \alpha_i = \beta_i$ by independence. ■

The axioms can add and scale, but they cannot yet say how *long* a vector is or what *angle* two of them make: those words never appear in (a)–(j). To measure we add one operation that eats two vectors and returns a number, is symmetric, is linear in each slot, and is strictly positive on nonzero vectors. From it flow length, angle, orthogonality, and all of Euclidean geometry.

DEFINITION 1.5 (Inner product, norm). $\mathbb{E}^n$ is $\mathcal{V}_n$ with a map $\cdot : \mathbb{E}^n \times \mathbb{E}^n \rightarrow \mathbb{R}$ such that, for all $\underline{u}, \underline{v}, \underline{w}$ and $\alpha \in \mathbb{R}$,

$$\begin{aligned} \text{(a)} \quad \underline{u} \cdot \underline{v} &= \underline{v} \cdot \underline{u}, & \text{(b)} \quad \underline{u} \cdot (\underline{v} + \underline{w}) &= \underline{u} \cdot \underline{v} + \underline{u} \cdot \underline{w}, & \text{(c)} \quad (\alpha \underline{u}) \cdot \underline{v} &= \alpha(\underline{u} \cdot \underline{v}), \\ \text{(d)} \quad \underline{u} \cdot \underline{u} &\geq 0, & \underline{u} \cdot \underline{u} = 0 &\iff \underline{u} = \underline{0}. \end{aligned}$$

The norm is $|\underline{u}| = (\underline{u} \cdot \underline{u})^{1/2}$; $\underline{u}$ is a unit vector iff $|\underline{u}| = 1$; $\underline{u} \perp \underline{v}$ iff $\underline{u} \cdot \underline{v} = 0$.

Before we may define the angle by $\cos \theta = \underline{u} \cdot \underline{v} / (|\underline{u}| |\underline{v}|)$, the right side must lie in $[-1, 1]$, or the arccosine is meaningless. That bound is not a new assumption: positivity already forces it, because a particular squared length can never be negative.

THEOREM 1.6 (Cauchy–Schwarz). $|\underline{u} \cdot \underline{v}| \leq |\underline{u}| |\underline{v}|$ for all $\underline{u}, \underline{v} \in \mathbb{E}^n$, with equality iff $\underline{u}, \underline{v}$ are parallel.

Proof. If $\underline{v} = \underline{0}$ both sides vanish. Otherwise positivity (d) applied to $\underline{u} - \lambda \underline{v}$ gives, for every $\lambda \in \mathbb{R}$,

$$0 \leq |\underline{u} - \lambda \underline{v}|^2 = |\underline{u}|^2 - 2\lambda \underline{u} \cdot \underline{v} + \lambda^2 |\underline{v}|^2,$$

using bilinearity and symmetry. The right side is least at $\lambda = \underline{u} \cdot \underline{v} / |\underline{v}|^2$; substituting, $0 \leq |\underline{u}|^2 - (\underline{u} \cdot \underline{v})^2 / |\underline{v}|^2$, i.e. $(\underline{u} \cdot \underline{v})^2 \leq |\underline{u}|^2 |\underline{v}|^2$. Equality forces $\underline{u} - \lambda \underline{v} = \underline{0}$ by (d), i.e. $\underline{u} \parallel \underline{v}$. ■

COROLLARY 1.7 (Triangle inequality). $|\underline{u} + \underline{v}| \leq |\underline{u}| + |\underline{v}|$.

Proof. $|\underline{u} + \underline{v}|^2 = |\underline{u}|^2 + 2\underline{u} \cdot \underline{v} + |\underline{v}|^2 \leq |\underline{u}|^2 + 2|\underline{u}| |\underline{v}| + |\underline{v}|^2 = (|\underline{u}| + |\underline{v}|)^2$ by Theorem 1.6; take square roots. ■

DEFINITION 1.8 (Angle). For $\underline{u}, \underline{v} \neq \underline{0}$, $\theta = \arccos(\underline{u} \cdot \underline{v} / (|\underline{u}||\underline{v}|)) \in [0, \pi]$, well defined by Theorem 1.6.

A general basis reproduces vectors but scrambles the metric: components are not projections, and $\underline{u} \cdot \underline{v}$ is not $u_i v_i$. A basis of mutually orthogonal unit vectors removes every cross term, so a component becomes a projection and the inner product becomes the plain sum of products. This is why all computation below uses an orthonormal basis.

DEFINITION 1.9 (Orthonormal basis). $\{\underline{e}_1, \dots, \underline{e}_n\}$ is orthonormal iff $\underline{e}_i \cdot \underline{e}_j = \delta_{ij}$.

PROPOSITION 1.10. *Every orthonormal set is linearly independent.*

Proof. $\alpha_i \underline{e}_i = \underline{0} \Rightarrow 0 = (\alpha_i \underline{e}_i) \cdot \underline{e}_j = \alpha_i \delta_{ij} = \alpha_j$. ■

PROPOSITION 1.11 (Components and the dot product). *Relative to an orthonormal basis $\{\underline{e}_i\}$, every $\underline{v}$ satisfies $\underline{v} = v_i \underline{e}_i$ with $v_i = \underline{v} \cdot \underline{e}_i$, and for $\underline{u} = u_i \underline{e}_i$, $\underline{v} = v_i \underline{e}_i$, $\underline{u} \cdot \underline{v} = u_i v_i$.*

Proof. By Theorem 1.4, $\underline{v} = v_i \underline{e}_i$ uniquely; then $\underline{v} \cdot \underline{e}_j = v_i \delta_{ij} = v_j$, and $\underline{u} \cdot \underline{v} = u_i v_j \delta_{ij} = u_i v_i$. ■

REMARK 1.12 (the summation convention, stated once). In $v_i \underline{e}_i$ the index i occurs twice and is summed; it is a dummy and may be relabelled, $v_i \underline{e}_i = v_j \underline{e}_j$. An index occurring once per term is free and must match across every term, as j in $\underline{v} \cdot \underline{e}_j = v_j$. A free index must never share a label with a dummy in the same term, and a symbol appearing three or more times in a term is undefined. Intuition: the dummy index is a hidden $\sum$; the free index is the “address” of the component the equation is really about.

REMARK 1.13 (projection, and why $\underline{0}$ is orthogonal to all). $\underline{v} = (v \cdot \underline{e}_k) \underline{e}_k$ exhibits $v_i = \underline{v} \cdot \underline{e}_i$ as the orthogonal projection of $\underline{v}$ on $\underline{e}_i$: geometrically, drop a perpendicular from the tip of $\underline{v}$ onto the $\underline{e}_i$ axis. Axiom (d) with $\underline{u} = \underline{0}$ gives $\underline{0} \cdot \underline{v} = 0$ for all $\underline{v}$: the zero vector, having no direction, is perpendicular to everything.

PROPOSITION 1.14 (Gram–Schmidt). *Every basis $\{\underline{u}_1, \dots, \underline{u}_n\}$ yields an orthonormal basis $\{\underline{e}_i\}$ with*

$$\text{span}\{\underline{e}_1, \dots, \underline{e}_k\} = \text{span}\{\underline{u}_1, \dots, \underline{u}_k\}$$

for each k ; hence there are infinitely many orthonormal bases when $n \geq 1$.

Proof. Recursively $\underline{w}_1 = \underline{u}_1$ and $\underline{w}_k = \underline{u}_k - \sum_{i < k} (\underline{u}_k \cdot \underline{e}_i) \underline{e}_i$, $\underline{e}_k = \underline{w}_k / |\underline{w}_k|$. Independence of the $\underline{u}_i$ gives $\underline{w}_k \neq \underline{0}$, and for $j < k$, $\underline{w}_k \cdot \underline{e}_j = \underline{u}_k \cdot \underline{e}_j - \underline{u}_k \cdot \underline{e}_j = 0$. Infinitude: for any orthogonal Q (Definition 1.47), $\{Q\underline{e}_i\}$ is again orthonormal, and such Q form a continuum. ■

Many physical quantities are linear scalar functions of a vector: the work of a fixed force along a variable displacement, a single component, a directional derivative. Any such function is fixed by its values on the basis, and reassembling those values yields one vector whose dot product reproduces the function. Isolating this fact now pays off later, when the increment of a field becomes its gradient.

THEOREM 1.15 (Riesz representation). *$f : \mathbb{E}^n \rightarrow \mathbb{R}$ is linear iff there is a unique $\underline{a} \in \mathbb{E}^n$ with $f(\underline{v}) = \underline{a} \cdot \underline{v}$ for all $\underline{v}$; explicitly $\underline{a} = f(\underline{e}_i) \underline{e}_i$.*

Proof. ($\Leftarrow$) $\underline{v} \mapsto \underline{a} \cdot \underline{v}$ is linear by (b),(c). ($\Rightarrow$) By linearity and Proposition 1.11, $f(\underline{v}) = f(v_i \underline{e}_i) = v_i f(\underline{e}_i) = a_i v_i = \underline{a} \cdot \underline{v}$ with $a_i := f(\underline{e}_i)$. Uniqueness: $\underline{a} \cdot \underline{v} = \underline{a}' \cdot \underline{v} \forall \underline{v} \Rightarrow (\underline{a} - \underline{a}') \cdot \underline{v} = 0$; take $\underline{v} = \underline{a} - \underline{a}'$ and use (d). ■

EXAMPLE 1.16 (the Riesz construction in action). Suppose $f : \mathbb{E}^3 \rightarrow \mathbb{R}$ is linear and we know its values on the standard basis: $f(\underline{e}_1) = 3$, $f(\underline{e}_2) = -1$, $f(\underline{e}_3) = 2$. Set $a_i := f(\underline{e}_i)$, so $\underline{a} = 3\underline{e}_1 - \underline{e}_2 + 2\underline{e}_3$. Then for any $\underline{v} = v_1 \underline{e}_1 + v_2 \underline{e}_2 + v_3 \underline{e}_3$,

$$f(\underline{v}) = f(v_i \underline{e}_i) = v_i f(\underline{e}_i) = 3v_1 - v_2 + 2v_3 = \underline{a} \cdot \underline{v}.$$

The abstract function f is now the concrete operation “dot with $(3, -1, 2)$ ”. Every linear scalar function on $\mathbb{E}^3$ hides a single vector; finding that vector is the entire content of the Riesz theorem.

REMARK 1.17 (why Riesz is the engine of continuum calculus). The pattern “linear functional $\Rightarrow$ dot product with a fixed vector” will reappear every time we define a derivative. The gradient of a scalar field (§1.2.2) is the Riesz representer of the linear part of the increment. For a vector field the increment is linear and vector valued, so its representer is a *tensor* (§1.2.5). Recognising this template once makes every later gradient obvious.

1.1.2 Change-of-basis formulas

A vector is a geometric object, but its components $v_i = \underline{v} \cdot \underline{e}_i$ are read off a chosen basis. Turn the basis and the same $\underline{v}$ shows different numbers. Mechanics must track that change exactly, so that physical statements can be made basis independent. The bookkeeping is a single matrix of dot products between the old and new axes, and it turns out to be orthogonal.

DEFINITION 1.18 (Direction cosines). For two orthonormal bases $\{\underline{e}_i\}$, $\{\underline{e}'_i\}$, set $a_{ij} = \underline{e}_i \cdot \underline{e}'_j$.

THEOREM 1.19 (Transformation law and orthogonality). *With $v_i = \underline{v} \cdot \underline{e}_i$, $v'_i = \underline{v} \cdot \underline{e}'_i$,*

$$v'_j = a_{ij}v_i, \quad v_i = a_{ij}v'_j, \quad a_{ik}a_{jk} = \delta_{ij} = a_{ki}a_{kj}.$$

Proof. Expanding each old axis in the new basis, $\underline{e}_i = \underline{e}_i \cdot \underline{e}'_j \underline{e}'_j = a_{ij} \underline{e}'_j$, so $\underline{v}'_j \underline{e}'_j = \underline{v} = v_i \underline{e}_i = v_i a_{ij} \underline{e}'_j$, whence $\underline{v}'_j = a_{ij} v_i$ by independence of $\{\underline{e}'_j\}$; swapping the two bases gives the inverse. Orthogonality: $a_{ik}a_{jk} = \underline{e}_i \cdot \underline{e}'_k \underline{e}_j \cdot \underline{e}'_k = \underline{e}_i \cdot (\underline{e}_j \cdot \underline{e}'_k \underline{e}'_k) = \underline{e}_i \cdot \underline{e}_j = \delta_{ij}$, and $a_{ki}a_{kj} = \delta_{ij}$ likewise. ■

REMARK 1.20 (what “orthogonal matrix” means here). $a_{ik}a_{jk} = \delta_{ij}$ says the rows of (a_{ij}) are orthonormal; it is the algebraic echo of the geometric fact that both frames are orthonormal. Nothing is rotated in space, only the labels; the vector itself is untouched. This is the passive view of a change of basis, to be contrasted later with the active view, where an orthogonal *tensor* genuinely moves vectors (§1.1.5).

1.1.3 Vector products in $\mathbb{E}^3$

Two vectors span a parallelogram. Its *area*, and the *normal* direction to its plane, are geometric data the inner product cannot supply. In exactly three dimensions a plane has a unique normal line, so “area times normal” can be packed into one vector, the cross product; feeding a third vector then measures the signed *volume* of the spanned parallelepiped, the box product. These are the tools behind area, volume, orientation, determinants, and the axial vector of a rotation.

DEFINITION 1.21 (Cross and box products). $\times : \mathbb{E}^3 \times \mathbb{E}^3 \rightarrow \mathbb{E}^3$ satisfies, for all $\underline{u}, \underline{v}, \underline{w}$ and α, β ,

$$(a) \ \underline{u} \times \underline{v} = -\underline{v} \times \underline{u}, \quad (b) \ |\underline{u} \times \underline{v}| = |\underline{u}||\underline{v}| \sin \theta,$$

$$(c) \ (\alpha \underline{u} + \beta \underline{v}) \times \underline{w} = \alpha \underline{u} \times \underline{w} + \beta \underline{v} \times \underline{w},$$

$$(d) \ [\underline{u}, \underline{v}, \underline{w}] := \underline{u} \cdot (\underline{v} \times \underline{w}) = \underline{v} \cdot (\underline{w} \times \underline{u}) = \underline{w} \cdot (\underline{u} \times \underline{v}).$$

PROPOSITION 1.22. $\underline{u} \times \underline{u} = \underline{0}$; $\underline{u} \times \underline{v} \perp \underline{u}, \underline{v}$; and $[\cdot, \cdot, \cdot]$ is *trilinear and alternating* (it changes sign under any transposition of its arguments).

Proof. (a) gives $\underline{u} \times \underline{u} = -\underline{u} \times \underline{u} = \underline{0}$; then (d) gives $\underline{u} \cdot (\underline{u} \times \underline{v}) = [\underline{u}, \underline{u}, \underline{v}] = 0$ and $\underline{v} \cdot (\underline{u} \times \underline{v}) = [\underline{v}, \underline{u}, \underline{v}] = 0$. Trilinearity in slot one is (c) via (d); the other slots follow from cyclicity in (d). Alternation: (a) gives $[\underline{u}, \underline{v}, \underline{w}] = -\underline{u} \cdot \underline{w} \times \underline{v} = -[\underline{u}, \underline{w}, \underline{v}]$, and cyclicity extends this to every transposition. ■

DEFINITION 1.23 (Right-handed basis; permutation symbol). An orthonormal basis of $\mathbb{E}^3$ is right handed iff $\underline{e}_1 \times \underline{e}_2 = \underline{e}_3$ and cyclically. Then

$$\varepsilon_{ijk} := [\underline{e}_i, \underline{e}_j, \underline{e}_k] = \begin{cases} +1, & (i, j, k) \text{ even perm. of } (1, 2, 3), \\ -1, & (i, j, k) \text{ odd perm.}, \\ 0, & \text{repeated index.} \end{cases}$$

PROPOSITION 1.24 (Components of the products). *In a right-handed orthonormal basis, $\underline{e}_i \times \underline{e}_j = \varepsilon_{ijk} \underline{e}_k$, $(\underline{a} \times \underline{b})_k = \varepsilon_{ijk} a_i b_j$, and $[\underline{a}, \underline{b}, \underline{c}] = \varepsilon_{ijk} a_i b_j c_k$.*

Proof. $\underline{e}_i \times \underline{e}_j = (\underline{e}_i \times \underline{e}_j) \cdot \underline{e}_k \underline{e}_k = \varepsilon_{ijk} \underline{e}_k$; then $\underline{a} \times \underline{b} = a_i b_j \varepsilon_{ijk} \underline{e}_k$ and $[\underline{a}, \underline{b}, \underline{c}] = (\underline{a} \times \underline{b}) \cdot \underline{c} = \varepsilon_{ijk} a_i b_j c_k$. ■

LEMMA 1.25 (Alternating trilinear forms are one dimensional). *If $\omega : \mathbb{E}^3 \times \mathbb{E}^3 \times \mathbb{E}^3 \rightarrow \mathbb{R}$ is trilinear and alternating, then $\omega(\underline{a}, \underline{b}, \underline{c}) = \omega(\underline{e}_1, \underline{e}_2, \underline{e}_3) [\underline{a}, \underline{b}, \underline{c}]$.*

Proof.

$$\begin{aligned} \omega(\underline{a}, \underline{b}, \underline{c}) &= a_i b_j c_k \omega(\underline{e}_i, \underline{e}_j, \underline{e}_k) \\ &= a_i b_j c_k \varepsilon_{ijk} \omega(\underline{e}_1, \underline{e}_2, \underline{e}_3) \\ &= \omega(\underline{e}_1, \underline{e}_2, \underline{e}_3) [\underline{a}, \underline{b}, \underline{c}], \end{aligned}$$

using $\omega(\underline{e}_i, \underline{e}_j, \underline{e}_k) = \varepsilon_{ijk} \omega(\underline{e}_1, \underline{e}_2, \underline{e}_3)$ from alternation. ■

REMARK 1.26 (the workhorse). Lemma 1.25 is the one fact that makes determinants and invariants painless: *any* quantity linear in three vectors and sign reversing under a swap is a fixed multiple of the box product. Each time we invoke it, a page of index juggling collapses to a line. Intuitively, in three dimensions there is essentially one way to measure oriented volume, and everything alternating and trilinear must be proportional to it.

DEFINITION 1.27 (Determinant). For $B = (b_{ij}) \in \mathbb{R}^{3 \times 3}$, $\det B := \varepsilon_{ijk} b_{i1} b_{j2} b_{k3}$.

PROPOSITION 1.28. $\varepsilon_{mnp} \det B = \varepsilon_{ijk} b_{im} b_{jn} b_{kp}$, hence $\varepsilon_{ijk} \varepsilon_{mnp} = \det(\delta_{\bullet\bullet})$, the 3×3 determinant of the Kronecker array with rows i, j, k and columns m, n, p .

Proof. $(\underline{x}, \underline{y}, \underline{z}) \mapsto \varepsilon_{ijk} x_i y_j z_k$ is trilinear and alternating in its three columns, so by Lemma 1.25 equals $[\underline{x}, \underline{y}, \underline{z}]$ times its value 1 at $(\underline{e}_1, \underline{e}_2, \underline{e}_3)$. Applied to columns $(b_{\cdot m}, b_{\cdot n}, b_{\cdot p})$ this gives $\varepsilon_{ijk} b_{im} b_{jn} b_{kp} = \varepsilon_{mnp} \det B$. Putting $b_{ij} = \delta_{ij}$ (so $\det B = 1$) yields the second identity. ■

THEOREM 1.29 (ε - δ identity). $\varepsilon_{ijk}\varepsilon_{mnk} = \delta_{im}\delta_{jn} - \delta_{in}\delta_{jm}$, $\varepsilon_{ijk}\varepsilon_{mjk} = 2\delta_{im}$, $\varepsilon_{ijk}\varepsilon_{ijk} = 6$.

Proof. Expand the determinant of Proposition 1.28, set $p = k$, sum ($\delta_{kk} = 3$): $\varepsilon_{ijk}\varepsilon_{mnk} = \delta_{im}(3\delta_{jn} - \delta_{jn}) - \delta_{in}(3\delta_{jm} - \delta_{jm}) + (\delta_{in}\delta_{jm} - \delta_{im}\delta_{jn}) = \delta_{im}\delta_{jn} - \delta_{in}\delta_{jm}$. Set $n = j$, sum: $\varepsilon_{ijk}\varepsilon_{mjk} = 3\delta_{im} - \delta_{im} = 2\delta_{im}$. Set $m = i$, sum: $\varepsilon_{ijk}\varepsilon_{ijk} = 2\delta_{ii} = 6$. ■

COROLLARY 1.30 (Determinant, contracted).

$$\det B = \frac{1}{6} \varepsilon_{ijk}\varepsilon_{mnp} b_{im}b_{jn}b_{kp}.$$

Proof. Contract $\varepsilon_{ijk}b_{im}b_{jn}b_{kp} = \varepsilon_{mnp} \det B$ with ε_{mnp} and use $\varepsilon_{mnp}\varepsilon_{mnp} = 6$. ■

1.1.4 Tensors

Riesz handled linear rules returning a *number*. Physics abounds in linear rules returning a *vector*: stress sends a surface normal to a traction, the inertia tensor sends angular velocity to angular momentum, the deformation gradient sends a material fibre to its deformed image. The object encoding such a rule is a second order tensor. Its simplest instance, $\underline{a} \otimes \underline{b}$, is the rule “project onto $\underline{b}$, then point along $\underline{a}$ ”.

DEFINITION 1.31 (Tensor). A (second order) tensor is a linear map $\underline{A} : \mathbb{E}^3 \rightarrow \mathbb{E}^3$. Equality, scalar multiple and sum are defined by $\underline{A} = \underline{B} \iff \underline{A}\underline{v} = \underline{B}\underline{v} \forall \underline{v}$, $(\alpha \underline{A})\underline{v} = \alpha(\underline{A}\underline{v})$, $(\underline{A} + \underline{B})\underline{v} = \underline{A}\underline{v} + \underline{B}\underline{v}$. The set Lin is a vector space; $\underline{Q}\underline{v} = \underline{0}$ and $\underline{I}\underline{v} = \underline{v}$ for all $\underline{v}$.

DEFINITION 1.32 (Tensor product). For $\underline{a}, \underline{b} \in \mathbb{E}^3$, $(\underline{a} \otimes \underline{b})\underline{v} := (\underline{b} \cdot \underline{v}) \underline{a}$.

THEOREM 1.33 (Component representation). $\{\underline{e}_i \otimes \underline{e}_j\}$ is a basis of Lin , and $\underline{A} = A_{ij}\underline{e}_i \otimes \underline{e}_j$ with $A_{ij} = \underline{e}_i \cdot \underline{A}\underline{e}_j$; hence $\dim \text{Lin} = 9$.

Proof. Spanning. For $\underline{v} = v_i \underline{e}_i$, $\underline{A}\underline{v} = v_i \underline{A}\underline{e}_i = v_i \underline{e}_j \cdot \underline{A}\underline{e}_i \underline{e}_j = A_{ji}(\underline{e}_i \cdot \underline{v}) \underline{e}_j = (A_{ji} \underline{e}_j \otimes \underline{e}_i)\underline{v}$, so $\underline{A} = A_{ij}\underline{e}_i \otimes \underline{e}_j$ with $A_{ij} = \underline{e}_i \cdot \underline{A}\underline{e}_j$. *Independence.* $\alpha_{ij}\underline{e}_i \otimes \underline{e}_j = \underline{Q}$ applied to $\underline{e}_k$ gives $\alpha_{ik}\underline{e}_i = \underline{0}$, so $\alpha_{ik} = 0$. ■

EXAMPLE 1.34 (reading off components of tensor products). (i) *The tensor product $\underline{a} \otimes \underline{b}$.* By Definition 1.32, $(\underline{a} \otimes \underline{b})\underline{e}_j = (\underline{b} \cdot \underline{e}_j)\underline{a} = b_j\underline{a}$. Dotting with $\underline{e}_i$,

$$(\underline{a} \otimes \underline{b})_{ij} = \underline{e}_i \cdot (\underline{a} \otimes \underline{b})\underline{e}_j = \underline{e}_i \cdot b_j\underline{a} = b_j\underline{e}_i \cdot \underline{a} = a_ib_j.$$

Thus the component matrix is the outer product of the column (a_i) with the row (b_j) :

$$((\underline{a} \otimes \underline{b})_{ij}) = \begin{pmatrix} a_1b_1 & a_1b_2 & a_1b_3 \\ a_2b_1 & a_2b_2 & a_2b_3 \\ a_3b_1 & a_3b_2 & a_3b_3 \end{pmatrix}.$$

Such a matrix has rank ≤ 1 : every row is a multiple of (b_1, b_2, b_3) . Consequently $\det(\underline{a} \otimes \underline{b}) = 0$ unless $\underline{a}$ or $\underline{b}$ is $\underline{0}$.

(ii) *The identity tensor.* $\underline{I}\underline{e}_j = \underline{e}_j$, so $I_{ij} = \underline{e}_i \cdot \underline{e}_j = \delta_{ij}$. In the basis expansion, $\underline{I} = \delta_{ij}\underline{e}_i \otimes \underline{e}_j = \underline{e}_1 \otimes \underline{e}_1 + \underline{e}_2 \otimes \underline{e}_2 + \underline{e}_3 \otimes \underline{e}_3 = \underline{e}_i \otimes \underline{e}_i$, with matrix $(I_{ij}) = \text{diag}(1, 1, 1)$.

(iii) *The zero tensor.* $\underline{O}\underline{v} = \underline{0}$ for all $\underline{v}$, so $O_{ij} = \underline{e}_i \cdot \underline{0} = 0$; every entry of the matrix is zero.

REMARK 1.35 (outer versus inner product). Two operations eat a pair of vectors and return very different objects. The *inner product* $\underline{a} \cdot \underline{b} = a_ib_i$ sums all matching pairs and returns one number: it measures overlap and forgets direction. The *outer product* (tensor product) stores every pair, returning nine numbers arranged in the matrix (a_ib_j) : it remembers both directions. In $\mathbb{E}^3$ one may think of $\underline{a} \otimes \underline{b}$ as the instruction “measure how much $\underline{v}$ aligns with $\underline{b}$ (via $\underline{b} \cdot \underline{v}$), then lay that amount along $\underline{a}$ ”. The trace connects the two: $\text{tr}(\underline{a} \otimes \underline{b}) = a_ib_j\delta_{ij} = a_ib_i = \underline{a} \cdot \underline{b}$, collapsing the nine entries back to one.

EXAMPLE 1.36 (a diagonal tensor: the reflection). Suppose a linear map $\underline{A}$ is specified by its action on the basis: $\underline{A}\underline{e}_1 = -\underline{e}_1$, $\underline{A}\underline{e}_2 = \underline{e}_2$, $\underline{A}\underline{e}_3 = \underline{e}_3$. We compute every component $A_{ij} = \underline{e}_i \cdot \underline{A}\underline{e}_j$ in turn.

Column $j = 1$: $\underline{A}\underline{e}_1 = -\underline{e}_1$, so $A_{11} = \underline{e}_1 \cdot -\underline{e}_1 = -1$, $A_{21} = \underline{e}_2 \cdot -\underline{e}_1 = 0$, $A_{31} = \underline{e}_3 \cdot -\underline{e}_1 = 0$.

Column $j = 2$: $\underline{A}\underline{e}_2 = \underline{e}_2$, so $A_{12} = 0$, $A_{22} = 1$, $A_{32} = 0$.

Column $j = 3$: $\underline{A}\underline{e}_3 = \underline{e}_3$, so $A_{13} = 0$, $A_{23} = 0$, $A_{33} = 1$.

Hence

$$(\underline{A}_{ij}) = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \text{diag}(-1, 1, 1),$$

and the tensor product form is $\underline{A} = -\underline{e}_1 \otimes \underline{e}_1 + \underline{e}_2 \otimes \underline{e}_2 + \underline{e}_3 \otimes \underline{e}_3$. For any $\underline{v} = v_i \underline{e}_i$, $\underline{A}\underline{v} = -v_1 \underline{e}_1 + v_2 \underline{e}_2 + v_3 \underline{e}_3$: the component along $\underline{e}_1$ is negated while those along $\underline{e}_2, \underline{e}_3$ are unchanged. This is precisely the *mirror reflection* in the plane perpendicular to $\underline{e}_1$.

REMARK 1.37 (why the diagonal form matters). A diagonal tensor $\underline{A} = \lambda_i \underline{e}_i \otimes \underline{e}_i$ (no sum) acts by scaling each basis direction independently: $\underline{A}\underline{v} = \lambda_1 v_1 \underline{e}_1 + \lambda_2 v_2 \underline{e}_2 + \lambda_3 v_3 \underline{e}_3$. The λ_i are the eigenvalues and the $\underline{e}_i$ the eigenvectors. In continuum mechanics, when $\underline{A}$ is the (symmetric) stretch tensor, the eigenvectors are the principal directions and $|\lambda_i|$ the principal stretches. The spectral theorem (Chapter 3) guarantees that every symmetric tensor can be written in this form for a suitable orthonormal basis.

LEMMA 1.38. $\underline{A}(\underline{a} \otimes \underline{b}) = (\underline{A}\underline{a}) \otimes \underline{b}$.

Proof. $[\underline{A}(\underline{a} \otimes \underline{b})]\underline{v} = \underline{A}[(\underline{b} \cdot \underline{v})\underline{a}] = (\underline{b} \cdot \underline{v})\underline{A}\underline{a} = [(\underline{A}\underline{a}) \otimes \underline{b}]\underline{v}$. ■

THEOREM 1.39 (Transformation law for tensor components). *With $a_{ij} = \underline{e}_i \cdot \underline{e}'_j$, $A'_{ij} = a_{ki} a_{lj} A_{kl}$.*

Proof. $\underline{A}\underline{e}'_n = (A_{kl} \underline{e}_k \otimes \underline{e}_l) \underline{e}'_n = A_{kl} a_{ln} \underline{e}_k$, so $A'_{jn} = \underline{e}'_j \cdot \underline{A}\underline{e}'_n = A_{kl} a_{ln} \underline{e}'_j \cdot \underline{e}_k = A_{kl} a_{kj} a_{ln}$. ■

REMARK 1.40 (order = number of cosine factors). Scalars, vectors and tensors transform with the direction cosine matrix zero, one and two times: $\varphi' = \varphi$, $v'_i = a_{ki} v_k$, $A'_{ij} = a_{ki} a_{lj} A_{kl}$. The count of a 's is the tensor's *order*, and is often taken as the definition of a Cartesian tensor. To illustrate: the trace $\text{tr } \underline{A} = A_{ii}$ carries zero free indices, hence zero cosines and transforms as a scalar ($\text{tr } \underline{A}' = A'_{ii} = a_{ki} a_{li} A_{kl} = \delta_{kl} A_{kl} = A_{kk} = \text{tr } \underline{A}$). The traction vector $t_i = A_{ij} n_j$ carries one free index,

hence one cosine factor ($t'_i = a_{ki}t_k$), and transforms as a vector. The tensor itself carries two. Invariants (§1.1.5) exist precisely because even combinations of cosines can pair up into Kronecker deltas, cancelling all basis dependence.

1.1.5 Algebraic operations with second-order tensors

Transpose and symmetry. In $\underline{u} \cdot \underline{A}\underline{v}$ one often needs to let $\underline{A}$ act on the other factor. The tensor legalising this, $\underline{u} \cdot \underline{A}\underline{v} = \underline{A}^t \underline{u} \cdot \underline{v}$, is the transpose. Whether a tensor equals or negates its transpose splits Lin into the two parts that rule mechanics: strain is symmetric, spin is skew.

DEFINITION 1.41 (Transpose). $\underline{A}^t$ is defined by $\underline{u} \cdot \underline{A}\underline{v} = \underline{v} \cdot \underline{A}^t \underline{u}$ for all $\underline{u}, \underline{v}$.

PROPOSITION 1.42. $(\underline{A}^t)_{ij} = A_{ji}$, $(\underline{A}^t)^t = \underline{A}$, $(\underline{A}\underline{B})^t = \underline{B}^t \underline{A}^t$.

Proof. $(\underline{A}^t)_{ij} = \underline{e}_i \cdot \underline{A}^t \underline{e}_j = \underline{e}_j \cdot \underline{A} \underline{e}_i = A_{ji}$. The rest follow from Definition 1.41 and uniqueness: $\underline{u} \cdot (\underline{A}\underline{B})\underline{v} = \underline{A}^t \underline{u} \cdot \underline{B}\underline{v} = \underline{B}^t \underline{A}^t \underline{u} \cdot \underline{v}$. ■

DEFINITION 1.43 (Symmetric, skew). $\underline{A}$ is symmetric iff $\underline{A}^t = \underline{A}$, skew iff $\underline{A}^t = -\underline{A}$.

PROPOSITION 1.44 (Axial vector of a skew tensor). *For skew $\underline{A}$ there is a unique $\underline{w}$ with $\underline{A}\underline{v} = \underline{w} \times \underline{v}$ for all $\underline{v}$; $A_{ij} = -\varepsilon_{ijk} w_k$, $w_k = -\frac{1}{2} \varepsilon_{kij} A_{ij}$.*

Proof. Skewness gives $A_{ii} = 0$ and three free entries; the general skew array is $A_{ij} = -\varepsilon_{ijk} w_k$. Then $(\underline{A}\underline{v})_i = -\varepsilon_{ijk} w_k v_j = \varepsilon_{ikj} w_k v_j = (\underline{w} \times \underline{v})_i$. Inversion by Theorem 1.29: $-\varepsilon_{ijm} A_{ij} = \varepsilon_{ijm} \varepsilon_{ijk} w_k = 2w_m$. ■

Product and inverse.

DEFINITION 1.45 (Product, inverse). $(\underline{A}\underline{B})\underline{v} := \underline{A}(\underline{B}\underline{v})$. $\underline{A}$ is invertible iff $\underline{A}\underline{u} = \underline{v}$ has a unique solution $\underline{u} = \underline{A}^{-1}\underline{v}$ for every $\underline{v}$.

PROPOSITION 1.46. $(\underline{A}\underline{B})_{ij} = A_{ik} B_{kj}$; in general $\underline{A}\underline{B} \neq \underline{B}\underline{A}$; and if $\underline{A}$ is invertible, $\underline{A}^{-1}\underline{A} = \underline{A}\underline{A}^{-1} = \underline{I}$.

Proof. $(\underline{A}\underline{B})\underline{e}_j = \underline{A}(B_{kj}\underline{e}_k) = B_{kj}A_{ik}\underline{e}_i$, so $(\underline{A}\underline{B})_{ij} = A_{ik}B_{kj}$; and $(\underline{A}^{-1}\underline{A})\underline{u} = \underline{A}^{-1}\underline{v} = \underline{u}$, $(\underline{A}\underline{A}^{-1})\underline{v} = \underline{A}\underline{u} = \underline{v}$. ■

Orthogonal tensors. Rigid motions neither stretch nor shear a body: they preserve every length and angle, hence every inner product. The tensors doing this are the orthogonal ones, and they are exactly those whose transpose is their inverse. Those with $\det = +1$ are rotations; $\det = -1$ adds a reflection.

DEFINITION 1.47 (Orthogonal). $\underline{A}$ is orthogonal iff $\underline{A}\underline{u} \cdot \underline{A}\underline{v} = \underline{u} \cdot \underline{v}$ for all $\underline{u}, \underline{v}$.

PROPOSITION 1.48. $\underline{A}$ orthogonal $\iff \underline{A}^t \underline{A} = \underline{I} \iff \underline{A} \underline{A}^t = \underline{I}$; then $\underline{A}^{-1} = \underline{A}^t$ and $\det \underline{A} = \pm 1$.

Proof. $\underline{A}\underline{u} \cdot \underline{A}\underline{v} = \underline{u} \cdot \underline{A}^t \underline{A}\underline{v}$, so orthogonality is $\underline{u} \cdot (\underline{A}^t \underline{A} - \underline{I})\underline{v} = 0 \forall \underline{u}, \underline{v}$; take $\underline{u} = (\underline{A}^t \underline{A} - \underline{I})\underline{v}$ to get $\underline{A}^t \underline{A} = \underline{I}$, then $\underline{A}^{-1} = \underline{A}^t$ and $\underline{A} \underline{A}^t = \underline{I}$. Finally $(\det \underline{A})^2 = \det \underline{A}^t \det \underline{A} = \det \underline{I} = 1$ (Theorem 1.52). ■

Principal invariants. The nine components of a tensor change with the basis; its physics does not. So we seek scalar combinations surviving every change of basis. Building them from the box product, itself geometric, makes invariance automatic. Three exist, and they are the coefficients of the characteristic polynomial (Problem 23), hence carry the eigenvalue data used throughout deformation analysis.

DEFINITION 1.49 (Invariants). For all $\underline{a}, \underline{b}, \underline{c}$,

$$\begin{aligned} I_1(\underline{A})[\underline{a}, \underline{b}, \underline{c}] &= [\underline{A}\underline{a}, \underline{b}, \underline{c}] + [\underline{a}, \underline{A}\underline{b}, \underline{c}] + [\underline{a}, \underline{b}, \underline{A}\underline{c}], \\ I_2(\underline{A})[\underline{a}, \underline{b}, \underline{c}] &= [\underline{A}\underline{a}, \underline{A}\underline{b}, \underline{c}] + [\underline{A}\underline{a}, \underline{b}, \underline{A}\underline{c}] + [\underline{a}, \underline{A}\underline{b}, \underline{A}\underline{c}], \\ I_3(\underline{A})[\underline{a}, \underline{b}, \underline{c}] &= [\underline{A}\underline{a}, \underline{A}\underline{b}, \underline{A}\underline{c}]. \end{aligned}$$

PROPOSITION 1.50 (Well-posedness). I_1, I_2, I_3 do not depend on $\underline{a}, \underline{b}, \underline{c}$.

Proof. Each right side is trilinear and alternating in $(\underline{a}, \underline{b}, \underline{c})$; by Lemma 1.25 it equals a fixed scalar times $[\underline{a}, \underline{b}, \underline{c}]$. ■

THEOREM 1.51 (Trace). I_1 is linear, $I_1(\underline{A}) = \text{tr } \underline{A} := A_{ii}$, $\text{tr}(\underline{e}_i \otimes \underline{e}_j) = \delta_{ij}$, $\text{tr}(\underline{u} \otimes \underline{v}) = \underline{u} \cdot \underline{v}$.

Proof. Definition 1.49₁ is linear in $\underline{A}$. With $[\underline{e}_1, \underline{e}_2, \underline{e}_3] = 1$,

$$\text{tr}(\underline{e}_i \otimes \underline{e}_j) = \delta_{j1}\varepsilon_{i23} + \delta_{j2}\varepsilon_{1i3} + \delta_{j3}\varepsilon_{12i} = \delta_{ij},$$

so $\text{tr } \underline{A} = A_{ij}\delta_{ij} = A_{ii}$ and $\text{tr}(\underline{u} \otimes \underline{v}) = u_i v_j \delta_{ij} = \underline{u} \cdot \underline{v}$. ■

THEOREM 1.52 (Determinant). $I_3(\underline{A}) = \det(A_{ij})$; $\underline{A}$ invertible $\iff \det \underline{A} \neq 0$; $\det(\underline{A}\underline{B}) = \det \underline{A} \det \underline{B}$; $\det \underline{A}^t = \det \underline{A}$.

Proof. $I_3(\underline{A}) = [\underline{A}\underline{e}_1, \underline{A}\underline{e}_2, \underline{A}\underline{e}_3] = \varepsilon_{ijk} A_{i1} A_{j2} A_{k3} = \det(A_{ij})$ by Proposition 1.24 and Definition 1.27. Multiplicativity and transpose invariance follow from Proposition 1.28 applied to $(A_{ik} B_{kj})$ and to (A_{ji}) . Invertibility: $\det \underline{A} \neq 0 \iff$ the columns $\underline{A}\underline{e}_i$ are independent $\iff \underline{A}$ is a bijection. ■

REMARK 1.53 (determinant = oriented volume ratio). When $[\underline{a}, \underline{b}, \underline{c}] > 0$ it is the volume of the parallelepiped on $\underline{a}, \underline{b}, \underline{c}$, so Definition 1.49₃ reads $\det \underline{A} = [\underline{A}\underline{a}, \underline{A}\underline{b}, \underline{A}\underline{c}]/[\underline{a}, \underline{b}, \underline{c}]$: $\det \underline{A}$ is the factor by which $\underline{A}$ scales oriented volume. Hence $\det > 0$ means orientation preserving, and later $\det$ of the deformation gradient is the local volume ratio, with $\det = 1$ signalling incompressibility.

DEFINITION 1.54 (Cofactor). $\underline{A}^*$ is defined by $\underline{A}^*(\underline{a} \times \underline{b}) = \underline{A}\underline{a} \times \underline{A}\underline{b}$.

PROPOSITION 1.55. $I_2(\underline{A}) = \text{tr } \underline{A}^*$; I_2 is quadratic and I_3 cubic in $\underline{A}$.

Proof. Applying the trace formula to $\underline{A}^*$ and $\underline{A}^*(\underline{e}_2 \times \underline{e}_3) = \underline{A}\underline{e}_2 \times \underline{A}\underline{e}_3$ etc. (see Problem 22a for the full computation) gives $\text{tr } \underline{A}^* = I_2(\underline{A})$. Degrees are read off Definition 1.49_{2,3}. ■

REMARK 1.56 (the cofactor maps area elements). If $\underline{a} \times \underline{b}$ is the area vector of a parallelogram, then $\underline{A}^*(\underline{a} \times \underline{b}) = \underline{A}\underline{a} \times \underline{A}\underline{b}$ is the area vector of the deformed parallelogram. Thus the determinant tells how $\underline{A}$ scales *volumes* (Remark 1.53), while the cofactor tells how $\underline{A}$ scales *oriented areas*. In the notation of continuum mechanics, the deformation gradient

$\underline{\underline{F}}$ satisfies $n \, da = \underline{\underline{F}}^*(N \, dA)$ (Nanson's formula): the cofactor converts a reference area element into its current image. Because area is one dimension lower than volume, the cofactor is quadratic in $\underline{\underline{A}}$ while the determinant is cubic; this is exactly what the degrees in Proposition 1.55 express. The identity $\underline{\underline{A}}^t \underline{\underline{A}}^* = (\det \underline{\underline{A}}) \underline{\underline{I}}$ (Problem 22c) ties the two together.

Inner product of tensors. Lin is nine dimensional; to do analysis on it, e.g. to call a stress increment small, it needs a length. The natural one pairs components entrywise and is generated intrinsically by trace and transpose.

DEFINITION 1.57 (Tensor inner product). $\underline{\underline{A}} \cdot \underline{\underline{B}} := \text{tr}(\underline{\underline{A}} \underline{\underline{B}}^t)$.

THEOREM 1.58. $\underline{\underline{A}} \cdot \underline{\underline{B}} = A_{ij} B_{ij} = \underline{\underline{B}} \cdot \underline{\underline{A}}$; $|\underline{\underline{A}}| = (A_{ij} A_{ij})^{1/2}$ makes Lin isometric to $\mathbb{E}^9$, so $|\underline{\underline{A}} \cdot \underline{\underline{B}}| \leq |\underline{\underline{A}}| |\underline{\underline{B}}|$ and $|\underline{\underline{A}} + \underline{\underline{B}}| \leq |\underline{\underline{A}}| + |\underline{\underline{B}}|$; and $\underline{\underline{A}} \cdot (\underline{\underline{u}} \otimes \underline{\underline{v}}) = \underline{\underline{u}} \cdot \underline{\underline{A}} \underline{\underline{v}}$.

Proof. $(\underline{\underline{A}} \underline{\underline{B}}^t)_{ik} = A_{ij} B_{kj}$, so $\text{tr}(\underline{\underline{A}} \underline{\underline{B}}^t) = A_{ij} B_{ij}$, symmetric. This is the Euclidean inner product of nine components, so the inequalities transfer from Theorem 1.6 and Corollary 1.7. Finally, by Lemma 1.38, $\underline{\underline{A}} \cdot (\underline{\underline{u}} \otimes \underline{\underline{v}}) = \text{tr}((\underline{\underline{A}} \underline{\underline{v}}) \otimes \underline{\underline{u}}) = \underline{\underline{A}} \underline{\underline{v}} \cdot \underline{\underline{u}} = \underline{\underline{u}} \cdot \underline{\underline{A}} \underline{\underline{v}}$. ■

DEFINITION 1.59 (Two orthogonal splittings). $\underline{\underline{A}} = \text{Sym } \underline{\underline{A}} + \text{Skw } \underline{\underline{A}}$ with $\text{Sym } \underline{\underline{A}} = \frac{1}{2}(\underline{\underline{A}} + \underline{\underline{A}}^t)$, $\text{Skw } \underline{\underline{A}} = \frac{1}{2}(\underline{\underline{A}} - \underline{\underline{A}}^t)$; and $\underline{\underline{A}} = \text{Sph } \underline{\underline{A}} + \text{Dev } \underline{\underline{A}}$ with $\text{Sph } \underline{\underline{A}} = \frac{1}{3}(\text{tr } \underline{\underline{A}}) \underline{\underline{I}}$, $\text{Dev } \underline{\underline{A}} = \underline{\underline{A}} - \frac{1}{3}(\text{tr } \underline{\underline{A}}) \underline{\underline{I}}$.

PROPOSITION 1.60. *Each summand is of the named type, the splittings are unique, $\text{tr}(\text{Dev } \underline{\underline{A}}) = 0$, and $\text{Sym } \underline{\underline{A}} \cdot \text{Skw } \underline{\underline{B}} = 0$, $\text{Sph } \underline{\underline{A}} \cdot \text{Dev } \underline{\underline{B}} = 0$ for all $\underline{\underline{A}}, \underline{\underline{B}}$. (Details in Problems 14–15.)*

Proof. $\frac{1}{2}(\underline{\underline{A}} + \underline{\underline{A}}^t)$ is symmetric, $\frac{1}{2}(\underline{\underline{A}} - \underline{\underline{A}}^t)$ skew, and their sum is $\underline{\underline{A}}$; $\text{tr}(\underline{\underline{A}} - \frac{1}{3}(\text{tr } \underline{\underline{A}}) \underline{\underline{I}}) = \text{tr } \underline{\underline{A}} - \text{tr } \underline{\underline{A}} = 0$. For symmetric $\underline{\underline{S}}$ and skew $\underline{\underline{W}}$, $\underline{\underline{S}} \cdot \underline{\underline{W}} = S_{ij} W_{ij} = S_{ji} (-W_{ji}) = -\underline{\underline{S}} \cdot \underline{\underline{W}} = 0$; and $\text{Sph } \underline{\underline{A}} \cdot \text{Dev } \underline{\underline{B}} = \frac{1}{3}(\text{tr } \underline{\underline{A}}) \text{tr}(\text{Dev } \underline{\underline{B}}) = 0$. Uniqueness is proved in Problems 14a, 15a. ■

1.2 Scalar, vector and tensor fields; gradient, divergence and curl; integral formulae

1.2.1 Euclidean point spaces

The algebra above treats *arrows*: they add, scale, and share a zero. Mechanics needs instead the *places* occupied by particles, and places behave differently. Two places cannot be added: “the doorway plus the window” names nothing. There is no zero place until someone plants a flag and calls it the origin. What *is* intrinsic is the displacement from one place to another, and a displacement does not move when the flag moves. So positions are not vectors, but their differences are. We therefore work with a set of points whose subtraction lands in a vector space.

DEFINITION 1.61 (Euclidean point space). A Euclidean point space is a set $\mathbb{E}^3$ of points $\mathbf{x}, \mathbf{y}, \dots$ with a subtraction $(\mathbf{x}, \mathbf{y}) \mapsto \mathbf{y} - \mathbf{x}$ into a three dimensional inner product space $\mathbb{E}^3$, the translation space, such that

(P1) (transport) for every $\mathbf{x}$ and $\underline{v} \in \mathbb{E}^3$ there is a unique $\mathbf{y}$ with $\mathbf{y} - \mathbf{x} = \underline{v}$, written $\mathbf{y} = \mathbf{x} + \underline{v}$;

(P2) (Chasles) $(\mathbf{y} - \mathbf{x}) + (\mathbf{z} - \mathbf{y}) = \mathbf{z} - \mathbf{x}$.

REMARK 1.62 (what each axiom buys). (P1) says the translation space acts freely and transitively on points: every displacement carries each point to a unique new one, and every pair of points is joined by a unique displacement. This is precisely what lets us later compare a field’s value at $\mathbf{x}$ with its values at neighbours $\mathbf{x} + s\underline{u}$, the comparison a derivative is built from. Setting $\mathbf{z} = \mathbf{x}$ in (P2), with $\mathbf{x} - \mathbf{x} = \underline{0}$ (the unique solution of $\mathbf{y} - \mathbf{x} = \underline{0}$ being $\mathbf{y} = \mathbf{x}$), gives $\mathbf{x} - \mathbf{y} = -(\mathbf{y} - \mathbf{x})$: reversing a displacement negates its arrow.

The definition is origin free, which is clean but useless for computation. Naming a reference place, the origin, is planting the flag; every

point then acquires one arrow with three components. The cost is dependence on the flag, measured precisely below.

PROPOSITION 1.63 (Position vector, Cartesian coordinates). *Fix an origin $\mathbf{O}$. Then (a) every point $\mathbf{x}$ determines a unique position vector $\underline{x} := \mathbf{x} - \mathbf{O}$, and $\mathbf{x} \mapsto \underline{x}$ is a bijection; (b) $\underline{x} = x_i \underline{e}_i$ with $x_i = \underline{x} \cdot \underline{e}_i$; (c) under $\mathbf{O} \rightarrow \mathbf{O}'$, $\underline{x}' = \underline{x} - \underline{c}$ with $\underline{c} = \mathbf{O}' - \mathbf{O}$, and $x'_i = x_i - c_i$.*

Proof. (a) $\underline{x} \in \mathbb{E}^3$ is well defined. Injective: $\mathbf{x}_1 - \mathbf{O} = \mathbf{x}_2 - \mathbf{O}$ gives, by (P2), $\underline{0} = (\mathbf{x}_1 - \mathbf{O}) + (\mathbf{O} - \mathbf{x}_2) = \mathbf{x}_1 - \mathbf{x}_2$. Surjective: $\mathbf{x} = \mathbf{O} + \underline{v}$ has $\underline{x} = \underline{v}$. (b) $\underline{x} \cdot \underline{e}_i = x_j \delta_{ji} = x_i$. (c) By (P2), $\mathbf{x} - \mathbf{O}' = (\mathbf{x} - \mathbf{O}) - \underline{c}$; dot with $\underline{e}_i$. ■

REMARK 1.64 (the deliberate abuse $\underline{x}(= \underline{x} - \underline{0})$). The book writes the position vector as a bare $\underline{x}$, so the same letter $\underline{x}$ names both the point and its position vector. Part (a) makes this legitimate *after* an origin is fixed, since the correspondence is then a bijection; part (c) is the warning label, that any formula with a bare $\underline{x}$ silently carries the chosen origin. Good continuum theory keeps its final physical statements independent of that choice.

DEFINITION 1.65 (Fields). A scalar field is $\varphi : \mathbb{E}^3 \rightarrow \mathbb{R}$, a vector field $\underline{v} : \mathbb{E}^3 \rightarrow \mathbb{E}^3$, a tensor field $\underline{A} : \mathbb{E}^3 \rightarrow \text{Lin}$; their argument is the position $\underline{x}$.

1.2.2 Gradients

In one variable a differentiable f satisfies $f(x_0 + h) = f(x_0) + f'(x_0)h +$ (small): near x_0 the graph is a straight line, and $f'(x_0)$ its slope. The single idea of differential calculus is that a smooth function is, up close, linear. We now transplant it to a field, where the increment depends on the *vector* $\underline{h} = \underline{x} - \underline{x}_0$, so the linear part is a linear scalar function of $\underline{h}$. Two things must be made precise: what “small” means for the remainder, and how to replace the abstract linear part by one computable vector.

DEFINITION 1.66 (Landau symbol). For r defined near 0, $r(\varepsilon) = o(\varepsilon)$ iff $r(\varepsilon)/\varepsilon \rightarrow 0$ as $\varepsilon \rightarrow 0$; equivalently, for every $\eta > 0$ there is $\delta > 0$ with $|r(\varepsilon)| \leq \eta\varepsilon$ for $0 < \varepsilon < \delta$.

REMARK 1.67 (why “faster than linear” removes ambiguity). If a field had two linear parts f_0, g_0 in (1.96), subtracting gives $f_0(\underline{h}) - g_0(\underline{h}) = -o(|\underline{h}|)$; putting $\underline{h} = s\underline{u}$ ($|\underline{u}| = 1$) and using linearity, $s[f_0(\underline{u}) - g_0(\underline{u})] = -o(s)$, so $f_0(\underline{u}) - g_0(\underline{u}) = -o(s)/s \rightarrow 0$. The left side is independent of s , hence 0 for every unit $\underline{u}$, so $f_0 = g_0$. Demanding the remainder die *faster* than $\underline{h}$ is exactly what pins the linear part down.

DEFINITION 1.68 (Differentiable scalar field). φ is differentiable at $\underline{x}_0$ iff there is a linear $f_0 : \mathbb{E}^3 \rightarrow \mathbb{R}$ with

$$\varphi(\underline{x}) = \varphi(\underline{x}_0) + f_0(\underline{x} - \underline{x}_0) + o(|\underline{x} - \underline{x}_0|) \quad (\underline{x} \rightarrow \underline{x}_0). \quad (1.96)$$

THEOREM 1.69 (Existence, uniqueness, naming of the gradient). *If φ is differentiable at $\underline{x}_0$, there is a unique $\nabla\varphi|_{\underline{x}_0} \in \mathbb{E}^3$ with $f_0(\underline{h}) = \nabla\varphi|_{\underline{x}_0} \cdot \underline{h}$, hence*

$$\varphi(\underline{x}) = \varphi(\underline{x}_0) + \nabla\varphi|_{\underline{x}_0} \cdot (\underline{x} - \underline{x}_0) + o(|\underline{x} - \underline{x}_0|). \quad (1.99)$$

Proof. f_0 is a linear functional on $\mathbb{E}^3$; by Riesz (Theorem 1.15) there is a unique $\underline{c}$ with $f_0(\underline{h}) = \underline{c} \cdot \underline{h}$. Explicitly, with $c_i := f_0(\underline{e}_i)$ and $\underline{h} = h_i \underline{e}_i$, $f_0(\underline{h}) = h_i f_0(\underline{e}_i) = h_i c_i = \underline{c} \cdot \underline{h}$. Name $\underline{c} =: \nabla\varphi|_{\underline{x}_0}$ and substitute into (1.96). ■

REMARK 1.70 (the gradient is a slope, and it points uphill). $\nabla\varphi|_{\underline{x}_0}$ is one vector attached to the place $\underline{x}_0$. By (1.99) the first order change under a step $\underline{h}$ is $\nabla\varphi \cdot \underline{h}$, largest when $\underline{h} \parallel \nabla\varphi$ (Cauchy–Schwarz). So the gradient points in the direction of steepest increase and its length is that steepest rate; its level sets are the directions of no first order change, orthogonal to $\nabla\varphi$.

REMARK 1.71 (the double construction reused throughout). The argument has a fixed three step shape: increment $\rightarrow$ linear part $\rightarrow$

Riesz representer. For a scalar field the representer is a vector, $\nabla\varphi$. Applied to a *vector* field the linear part is a linear map and Riesz upgrades to “a linear map is a tensor”, producing $\nabla\mathfrak{v} \in \text{Lin}$ (§1.2.5); applied to a flux it produces the divergence. Spotting this template once explains every later derivative at a glance.

1.2.3 Directional derivatives

The differentiability limit (1.96) lets $\mathfrak{x} \rightarrow \mathfrak{x}_0$ from all directions at once, a limit in three variables. The cheapest way to probe it is to fix a direction and move only along it: set $\mathfrak{v} = \mathfrak{x} - \mathfrak{x}_0$, let $s = |\mathfrak{v}|$ so that $\mathfrak{v} = s\mathfrak{u}$ with $|\mathfrak{u}| = 1$, and watch $f(s) = \varphi(\mathfrak{x}_0 + s\mathfrak{u})$ as an ordinary function of one real variable. The coefficient of s in its expansion is the rate of change of φ along $\mathfrak{u}$. Let us compute it and see that the gradient already contains the answer for every direction.

DEFINITION 1.72 (Directional derivative). For a unit vector $\mathfrak{u}$ and fixed $\mathfrak{x}_0$, with $f(s) = \varphi(\mathfrak{x}_0 + s\mathfrak{u})$, the derivative of φ at $\mathfrak{x}_0$ along $\mathfrak{u}$ is $f'(0)$.

THEOREM 1.73 (The gradient gives every directional derivative). *If φ is differentiable at $\mathfrak{x}_0$, then for every unit $\mathfrak{u}$*

$$\left(\frac{df}{ds}\right)\Big|_{s=0} = \mathfrak{u} \cdot \nabla\varphi\Big|_{\mathfrak{x}_0}. \quad (1.102)$$

Proof. With $\mathfrak{h} = s\mathfrak{u}$, $|\mathfrak{h}| = s$ (as $|\mathfrak{u}| = 1$, $s \geq 0$), (1.99) gives

$$f(s) = \varphi(\mathfrak{x}_0) + \nabla\varphi\Big|_{\mathfrak{x}_0} \cdot (s\mathfrak{u}) + o(s) = \varphi(\mathfrak{x}_0) + s(\mathfrak{u} \cdot \nabla\varphi\Big|_{\mathfrak{x}_0}) + o(s). \quad (1.100)$$

Subtract $f(0)$, divide by $s \neq 0$: $[f(s) - f(0)]/s = \mathfrak{u} \cdot \nabla\varphi\Big|_{\mathfrak{x}_0} + o(s)/s$ (1.101); let $s \rightarrow 0$. ■

Choosing the special directions $\mathfrak{e}_i$, which change only one coordinate, turns directional derivatives into ordinary partial derivatives and hands us the components of the gradient.

COROLLARY 1.74 (Partials as gradient components). *With $\hat{\varphi}(x_1, x_2, x_3) := \varphi(x_i\mathfrak{e}_i)$,*

$$\mathfrak{e}_i \cdot \nabla\varphi\Big|_{\mathfrak{x}_0} = \left(\frac{\partial\hat{\varphi}}{\partial x_i}\right)\Big|_{(x_1^0, x_2^0, x_3^0)} =: \hat{\varphi}_{,i}\Big|_{\mathfrak{x}_0}. \quad (1.103\text{--}1.106)$$

Proof. Take $\underline{u} = \underline{e}_i$ in Theorem 1.73; $\underline{x}_0 + s\underline{e}_i$ increments only the i th coordinate, so $\underline{e}_i \cdot \nabla \varphi|_{\underline{x}_0} = \lim_{s \rightarrow 0} \frac{1}{s} [\hat{\varphi}(\dots, x_i^0 + s, \dots) - \hat{\varphi}(\dots, x_i^0, \dots)] = \partial \hat{\varphi} / \partial x_i$. As $\underline{x}_0$ is arbitrary, $\underline{e}_i \cdot \nabla \varphi = \hat{\varphi}_{,i}$. ■

THEOREM 1.75 (Cartesian gradient). $\nabla \varphi = \hat{\varphi}_{,i} \underline{e}_i = \varphi_{,i} \underline{e}_i$. (1.107–1.108)

Proof. By orthonormal expansion (Proposition 1.11), $\underline{w} = (\underline{w} \cdot \underline{e}_i) \underline{e}_i$; take $\underline{w} = \nabla \varphi$ and use $\nabla \varphi \cdot \underline{e}_i = \hat{\varphi}_{,i}$. Dropping the caret gives (1.108). ■

REMARK 1.76 (caret and comma). $\varphi(\underline{x})$ eats a point; $\hat{\varphi}(x_1, x_2, x_3)$ eats three real numbers. Different domains make them different functions, yet they take equal values when $\underline{x} = x_i \underline{e}_i$. The comma abbreviates a Cartesian partial, $\varphi_{,i} = \partial \varphi / \partial x_i$, from here on; dropping the caret is the same controlled abuse as writing a bare $\underline{x}$ for the position (Remark 1.64).

1.2.4 Parameterized paths and the chain rule

Directional derivatives run along straight lines of fixed direction. A particle's trajectory, or a coordinate line in curvilinear coordinates, bends: its direction $\underline{x}'(u)$ varies along the way. To differentiate a field along such a curve we compose φ with the path and use the one variable derivative. The resulting rule, in its coordinate free form $d\varphi = \nabla \varphi \cdot d\underline{x}$, is powerful enough to generate $\nabla \varphi$ in any coordinate system.

THEOREM 1.77 (Chain rule). *For a smooth simple curve $\underline{x}(u)$ and $\psi(u) = \varphi(\underline{x}(u))$,*

$$\psi'(u_0) = \nabla \varphi|_{\underline{x}(u_0)} \cdot \underline{x}'(u_0). \quad (1.112)$$

Proof. With $\underline{x}_0 = \underline{x}(u_0)$, $\underline{x}_1 = \underline{x}(u_1)$, (1.99) gives

$$\psi(u_1) = \psi(u_0) + \nabla \varphi|_{\underline{x}_0} \cdot (\underline{x}_1 - \underline{x}_0) + o(|\underline{x}_1 - \underline{x}_0|). \quad (1.109)$$

Smoothness of the path gives

$$\underline{x}_1 - \underline{x}_0 = \underline{x}'(u_0)(u_1 - u_0) + o(u_1 - u_0), \quad (1.110)$$

so $|\underline{x}_1 - \underline{x}_0| = O(u_1 - u_0)$ and $o(|\underline{x}_1 - \underline{x}_0|) = o(u_1 - u_0)$. Insert (1.110) into (1.109), subtract $\psi(u_0)$, divide by $u_1 - u_0$:

$$\frac{\psi(u_1) - \psi(u_0)}{u_1 - u_0} = \nabla\varphi|_{\underline{x}_0} \cdot \underline{x}'(u_0) + \frac{o(u_1 - u_0)}{u_1 - u_0}, \quad (1.111)$$

and let $u_1 \rightarrow u_0$. ■

COROLLARY 1.78 (Coordinate free differential). *With $d\underline{x} = \underline{x}'(u) du$,*

$$\frac{d\varphi}{du} = \nabla\varphi \cdot \frac{d\underline{x}}{du}, \quad d\varphi = \nabla\varphi \cdot d\underline{x}. \quad (1.113\text{--}1.114)$$

This holds for every $d\underline{x}$ and uses no coordinates.

REMARK 1.79 (the power of (1.114)). Equation (1.114), $d\varphi = \nabla\varphi \cdot d\underline{x}$, is coordinate free: it never mentions a basis. Yet it can *generate* the gradient formula in *any* coordinate system. The recipe is always the same: write $d\varphi$ from the conventional chain rule, express the coordinate increments as dot products with $d\underline{x}$, read off $\nabla\varphi$ by matching with (1.114). The next two examples execute this recipe in Cartesian and cylindrical coordinates.

EXAMPLE 1.80 (recovering the Cartesian formula). *Step 1 (conventional chain rule).* With $\hat{\varphi}(x_1, x_2, x_3) = \varphi(x_i \underline{e}_i)$ and $\{\underline{e}_i\}$ fixed,

$$d\hat{\varphi} = \frac{\partial \hat{\varphi}}{\partial x_1} dx_1 + \frac{\partial \hat{\varphi}}{\partial x_2} dx_2 + \frac{\partial \hat{\varphi}}{\partial x_3} dx_3 = \hat{\varphi}_{,i} dx_i.$$

Step 2 (express dx_i via $d\underline{x}$). Since the basis is fixed, $\underline{x} = x_i \underline{e}_i$ gives $d\underline{x} = dx_i \underline{e}_i$. Dotting with $\underline{e}_i$ and using orthonormality, $dx_i = \underline{e}_i \cdot d\underline{x}$. Insert:

$$d\hat{\varphi} = \hat{\varphi}_{,i} (\underline{e}_i \cdot d\underline{x}) = (\hat{\varphi}_{,i} \underline{e}_i) \cdot d\underline{x}.$$

Step 3 (match with (1.114)). Setting this equal to $\nabla\varphi \cdot d\underline{x}$ gives $(\nabla\varphi - \hat{\varphi}_{,i} \underline{e}_i) \cdot d\underline{x} = 0$ for all $d\underline{x}$. Choose $d\underline{x} = \nabla\varphi - \hat{\varphi}_{,i} \underline{e}_i$ itself; then its squared norm $|\nabla\varphi - \hat{\varphi}_{,i} \underline{e}_i|^2 = 0$, so

$$\nabla\varphi = \hat{\varphi}_{,i} \underline{e}_i. \quad (1.115)$$

This is identical to (1.108), confirming that both routes give the same formula. The Cartesian gradient has no extra factors because $\{\underline{e}_i\}$ is fixed and orthonormal; when the basis varies with position (as in polar coordinates), extra terms appear.

EXAMPLE 1.81 (cylindrical polar coordinates). With $\underline{k} = \underline{e}_3$, define the rotating frame

$$\underline{e}_r(\theta) = \cos \theta \underline{e}_1 + \sin \theta \underline{e}_2, \quad \underline{e}_\theta(\theta) = \underline{k} \times \underline{e}_r = -\sin \theta \underline{e}_1 + \cos \theta \underline{e}_2, \quad (1.116)$$

orthonormal for each θ , with

$$\underline{e}'_r(\theta) = \underline{e}_\theta(\theta), \quad \underline{e}'_\theta(\theta) = -\underline{e}_r(\theta). \quad (\dagger)$$

A point has $\underline{x} = r \underline{e}_r(\theta) + z \underline{k}$ (1.117). Then, using $d\underline{e}_r = \underline{e}'_r(\theta)d\theta = \underline{e}_\theta d\theta$,

$$d\underline{x} = dr \underline{e}_r + r d\underline{e}_r + dz \underline{k} = dr \underline{e}_r + r d\theta \underline{e}_\theta + dz \underline{k}, \quad (1.119)$$

and dotting with the orthonormal $\{\underline{e}_r, \underline{e}_\theta, \underline{k}\}$,

$$dr = \underline{e}_r \cdot d\underline{x}, \quad r d\theta = \underline{e}_\theta \cdot d\underline{x}, \quad dz = \underline{k} \cdot d\underline{x}. \quad (1.120)$$

With $\varphi = \tilde{\varphi}(r, \theta, z)$, $d\tilde{\varphi} = \tilde{\varphi}_{,r}dr + \tilde{\varphi}_{,\theta}d\theta + \tilde{\varphi}_{,z}dz$; substituting (1.120),

$$d\tilde{\varphi} = (\tilde{\varphi}_{,r}\underline{e}_r + \frac{1}{r}\tilde{\varphi}_{,\theta}\underline{e}_\theta + \tilde{\varphi}_{,z}\underline{k}) \cdot d\underline{x}, \quad (1.121)$$

so by (1.114), dropping the tilde,

$$\nabla\varphi = \varphi_{,r} \underline{e}_r + \frac{1}{r} \varphi_{,\theta} \underline{e}_\theta + \varphi_{,z} \underline{k}. \quad (1.122)$$

The extra $1/r$, absent from the Cartesian (1.108), arises solely because $\{\underline{e}_r, \underline{e}_\theta, \underline{k}\}$ varies with position while $\{\underline{e}_i\}$ is fixed.

EXAMPLE 1.82 (gradient of the radius $r = |\underline{x}|$). *Index route.* Start from $r^2 = \underline{x} \cdot \underline{x} = x_j x_j$. Note that $x_{j,i} = \partial x_j / \partial x_i = \delta_{ji}$: the Cartesian coordinate x_j depends only on itself, so its derivative with respect to x_i is 1 when $j = i$ and 0 otherwise. Differentiate $r^2 = x_j x_j$ using the product rule:

$$2r r_{,i} = (x_j x_j)_{,i} = x_{j,i} x_j + x_j x_{j,i} = \delta_{ji} x_j + x_j \delta_{ji} = x_i + x_i = 2x_i, \quad (1.123)$$

where the Kronecker delta δ_{ji} collapsed the sum over j to its $j = i$ term: $\delta_{ji}x_j = x_i$. Divide both sides by $2r$: $r_{,i} = x_i/r$. Since $\nabla r = r_{,i}\underline{e}_i$ (by 1.108),

$$\nabla r = \frac{x_i}{r}\underline{e}_i = \frac{1}{r}\underline{x}. \quad (1.124)$$

Coordinate free route. The differential product rule for the inner product reads $d(\underline{a} \cdot \underline{b}) = d\underline{a} \cdot \underline{b} + \underline{a} \cdot d\underline{b}$ (proved by $d(a_i b_i) = da_i b_i + a_i db_i$). Apply it to $r^2 = \underline{x} \cdot \underline{x}$:

$$2r \, dr = d(\underline{x} \cdot \underline{x}) = d\underline{x} \cdot \underline{x} + \underline{x} \cdot d\underline{x} = 2 \underline{x} \cdot d\underline{x}. \quad (1.125)$$

Cancel 2 and divide by r : $dr = r^{-1}\underline{x} \cdot d\underline{x}$. By (1.114), $dr = \nabla r \cdot d\underline{x}$; matching the two gives $\nabla r = r^{-1}\underline{x}$. Both routes arrive at the same answer, but the coordinate free route never wrote a single index.

REMARK 1.83 (∇r points radially outward). Since $\underline{x}/r = \underline{x}/|\underline{x}|$ is the unit vector in the radial direction, ∇r is the unit radial vector: $|\nabla r| = 1$, pointing from the origin toward the point $\underline{x}$. The level sets of r are concentric spheres, and ∇r is indeed perpendicular to them, consistent with the general fact that the gradient is orthogonal to the level sets of the field (Remark 1.70).

1.2.5 Gradients of vector fields

Repeat the linearisation of §1.2.2 on a vector field. Now the increment $\underline{v}(\underline{x}) - \underline{v}(\underline{x}_0)$ is a *vector*, so its linear part is a linear map $\mathbb{E}^3 \rightarrow \mathbb{E}^3$, that is, a tensor. This is the double construction one rung up: the fixed object representing the linear part is $\nabla \underline{v} \in \text{Lin}$.

DEFINITION 1.84 (Differentiable vector field). $\underline{v}$ is differentiable at $\underline{x}_0$ iff there is a tensor $\underline{L}_0$ with

$$\underline{v}(\underline{x}) = \underline{v}(\underline{x}_0) + \underline{L}_0(\underline{x} - \underline{x}_0) + \underline{o}(\underline{x} - \underline{x}_0), \quad |\underline{o}(\underline{x} - \underline{x}_0)| = o(|\underline{x} - \underline{x}_0|), \quad (1.126\text{--}1.127)$$

and one writes $\underline{L}_0 = \nabla \underline{v}|_{\underline{x}_0}$.

THEOREM 1.85 (Cartesian representation). $\nabla \underline{v} = v_{i,j} \underline{e}_i \otimes \underline{e}_j$, i.e. $(\nabla \underline{v})_{ij} = v_{i,j}$. (1.132)

Proof. With $\underline{x} - \underline{x}_0 = s\underline{u}$, $|\underline{u}| = 1$, $\underline{L}_0 \underline{u} = \lim_{s \rightarrow 0} \frac{1}{s} [\underline{v}(\underline{x}_0 + s\underline{u}) - \underline{v}(\underline{x}_0)]$ (1.129). Take $\underline{u} = \underline{e}_j$ and $\underline{v} = \hat{v}_i \underline{e}_i$ with fixed $\underline{e}_i$: $\underline{L}_0 \underline{e}_j = \underline{e}_i \lim_{s \rightarrow 0} \frac{1}{s} [\hat{v}_i(\dots, x_j^0 + s, \dots) - \hat{v}_i(\dots)] = \hat{v}_{i,j} \underline{e}_i$ (1.130–1.131). Hence $L_{ij} = \underline{e}_i \cdot \underline{L}_0 \underline{e}_j = v_{i,j}$. ■

COROLLARY 1.86 (Chain rule for vector fields). $d\underline{v} = (\nabla \underline{v}) d\underline{x}$. (1.133)

Proof. $d\hat{v}_i = \hat{v}_{i,j} dx_j = \hat{v}_{i,j} \underline{e}_j \cdot d\underline{x}$, so $d\underline{v} = d\hat{v}_i \underline{e}_i = \hat{v}_{i,j} \underline{e}_i (\underline{e}_j \cdot d\underline{x}) = (\hat{v}_{i,j} \underline{e}_i \otimes \underline{e}_j) d\underline{x} = (\nabla \underline{v}) d\underline{x}$ (1.134). ■

EXAMPLE 1.87 ($\underline{u} = \mu \underline{x} + \nu(\underline{x} \otimes \underline{x}) \underline{a}$). The field $\underline{u}(\underline{x}) = \mu \underline{x} + \nu(\underline{x} \otimes \underline{x}) \underline{a}$ arises, for instance, in quadratic displacement fields. We compute its gradient by two routes.

Index route. Write $u_i = \mu x_i + \nu x_i x_j a_j$ (since $(\underline{x} \otimes \underline{x}) \underline{a} = \underline{x} \cdot \underline{a} \underline{x}$, its i th component is $x_i(x_j a_j)$). Differentiate with respect to x_k :

$$\begin{aligned} u_{i,k} &= \mu \frac{\partial x_i}{\partial x_k} + \nu \left(\frac{\partial x_i}{\partial x_k} x_j a_j + x_i \frac{\partial x_j}{\partial x_k} a_j \right) \\ &= \mu \delta_{ik} + \nu (\delta_{ik} x_j a_j + x_i \delta_{jk} a_j) \\ &= \mu \delta_{ik} + \nu (\underline{x} \cdot \underline{a}) \delta_{ik} + \nu x_i a_k \\ &= [\mu + \nu(\underline{x} \cdot \underline{a})] \delta_{ik} + \nu x_i a_k. \end{aligned} \quad (1.136)$$

Recognise the first term as $[\mu + \nu(\underline{x} \cdot \underline{a})] I_{ik}$ and the second as $(\underline{x} \otimes \underline{a})_{ik} = x_i a_k$, so

$$\nabla \underline{u} = [\mu + \nu(\underline{x} \cdot \underline{a})] \underline{I} + \nu \underline{x} \otimes \underline{a}. \quad (1.137)$$

Differential route. The product rule $d(\underline{b} \otimes \underline{c}) = d\underline{b} \otimes \underline{c} + \underline{b} \otimes d\underline{c}$ (proved by $d(b_i c_j) = db_i c_j + b_i dc_j$) gives $d(\underline{x} \otimes \underline{x}) = d\underline{x} \otimes \underline{x} + \underline{x} \otimes d\underline{x}$, so

$$\begin{aligned} d\underline{u} &= \mu d\underline{x} + \nu(d\underline{x} \otimes \underline{x} + \underline{x} \otimes d\underline{x}) \underline{a} \\ &= \mu d\underline{x} + \nu(\underline{x} \cdot \underline{a}) d\underline{x} + \nu \underline{x} (\underline{a} \cdot d\underline{x}) \\ &= \{[\mu + \nu(\underline{x} \cdot \underline{a})] \underline{I} + \nu \underline{x} \otimes \underline{a}\} d\underline{x}. \end{aligned} \quad (1.138)$$

In line two we used $(d\underline{x} \otimes \underline{x}) \underline{a} = (\underline{x} \cdot \underline{a}) d\underline{x}$ and $(\underline{x} \otimes d\underline{x}) \underline{a} = (d\underline{x} \cdot \underline{a}) \underline{x} = \underline{x} (\underline{a} \cdot d\underline{x})$. Comparing with $d\underline{u} = (\nabla \underline{u}) d\underline{x}$ confirms (1.137).

EXAMPLE 1.88 (affine field $\underline{u} = \underline{x} + (\underline{b} \cdot \underline{x})\underline{a}$). Here $\underline{a}, \underline{b}$ are fixed vectors. Write $\underline{u}(\underline{x}) = \underline{x} + \underline{a}(\underline{b} \cdot \underline{x})$; its differential is

$$d\underline{u} = d\underline{x} + \underline{a}(\underline{b} \cdot d\underline{x}) = d\underline{x} + \underline{a} \otimes \underline{b} d\underline{x} = (\underline{I} + \underline{a} \otimes \underline{b})d\underline{x},$$

where we recognised $\underline{a}(\underline{b} \cdot d\underline{x}) = (\underline{a} \otimes \underline{b})d\underline{x}$ by Definition 1.32. Thus

$$\nabla \underline{u} = \underline{I} + \underline{a} \otimes \underline{b}. \quad (1.141)$$

Notice that $\nabla \underline{u}$ is *uniform*: it does not depend on $\underline{x}$. This is no accident. Write $\underline{u}(\underline{x}) = (\underline{I} + \underline{a} \otimes \underline{b})\underline{x} =: \underline{A}\underline{x}$; then for any $\underline{x}_0$,

$$\underline{u}(\underline{x}) = \underline{A}\underline{x} = \underline{A}\underline{x}_0 + \underline{A}(\underline{x} - \underline{x}_0) = \underline{u}(\underline{x}_0) + \underline{A}(\underline{x} - \underline{x}_0),$$

which is already in the form (1.127) with $\underline{L}_0 = \underline{A}$ and *zero remainder*. Whenever the field is affine (linear in $\underline{x}$), the gradient is constant and the approximation (1.127) is exact everywhere.

EXAMPLE 1.89 (polar gradient of a vector field). For $\underline{u} = u_r \underline{e}_r + u_\theta \underline{e}_\theta + u_z \underline{k}$ with components functions of (r, θ) , using (†) and the product rule,

$$\begin{aligned} d\underline{u} &= [u_{r,r} dr + (u_{r,\theta} - u_\theta) d\theta] \underline{e}_r \\ &\quad + [u_{\theta,r} dr + (u_{\theta,\theta} + u_r) d\theta] \underline{e}_\theta \\ &\quad + [u_{z,r} dr + u_{z,\theta} d\theta] \underline{k}, \end{aligned}$$

where the $-u_\theta$ and $+u_r$ come from $d\underline{e}_r = \underline{e}_\theta d\theta$, $d\underline{e}_\theta = -\underline{e}_r d\theta$. Inserting $dr = \underline{e}_r \cdot d\underline{x}$, $d\theta = \frac{1}{r} \underline{e}_\theta \cdot d\underline{x}$ and $(\underline{e}_a \cdot d\underline{x}) \underline{e}_b = (\underline{e}_b \otimes \underline{e}_a) d\underline{x}$,

$$\begin{aligned} \nabla \underline{u} &= u_{r,r} \underline{e}_r \otimes \underline{e}_r + \frac{1}{r} (u_{r,\theta} - u_\theta) \underline{e}_r \otimes \underline{e}_\theta + u_{\theta,r} \underline{e}_\theta \otimes \underline{e}_r \\ &\quad + \frac{1}{r} (u_{\theta,\theta} + u_r) \underline{e}_\theta \otimes \underline{e}_\theta + u_{z,r} \underline{k} \otimes \underline{e}_r + \frac{1}{r} u_{z,\theta} \underline{k} \otimes \underline{e}_\theta. \end{aligned} \quad (1.145)$$

1.2.6 Divergence and curl

The tensor $\nabla \underline{v}$ carries nine numbers. Two coordinate free combinations carry the physics: its trace measures the local rate of volume change, the divergence, and its skew part measures local rotation, the curl. Defining each through the trace and box product machinery keeps them intrinsic, after which the component formulas drop out.

DEFINITION 1.90 (Divergence of a vector field). $\operatorname{div} \underline{v} := \operatorname{tr}(\nabla \underline{v})$.

PROPOSITION 1.91. $\operatorname{div} \underline{v} = v_{i,i}$.

Proof. $\operatorname{div} \underline{v} = \operatorname{tr}(v_{i,j} \underline{e}_i \otimes \underline{e}_j) = v_{i,j} \delta_{ij} = v_{i,i}$ (1.147), by $\operatorname{tr}(\underline{e}_i \otimes \underline{e}_j) = \delta_{ij}$. ■

DEFINITION 1.92 (Divergence of a tensor field). $\operatorname{div} \underline{A}$ is the vector field with $\underline{c} \cdot \operatorname{div} \underline{A} = \operatorname{div}(\underline{A}^t \underline{c})$ for every fixed $\underline{c}$.

PROPOSITION 1.93. $\operatorname{div} \underline{A} = A_{ij,j} \underline{e}_i$. (1.150)

Proof. Take $\underline{c} = \underline{e}_i$: $\underline{A}^t \underline{e}_i = A_{ij} \underline{e}_j$, so $\underline{e}_i \cdot \operatorname{div} \underline{A} = \operatorname{div}(A_{ij} \underline{e}_j) = A_{ij,j}$ (i fixed, j summed). ■

PROPOSITION 1.94 (Product rule). $\underline{u} \cdot \operatorname{div} \underline{A} = \operatorname{div}(\underline{A}^t \underline{u}) - \operatorname{tr}[\underline{A}^t(\nabla \underline{u})]$. (1.152)

Proof. The scalar product rule $u_i A_{ij,j} = (A_{ij} u_i)_{,j} - A_{ij} u_{i,j}$ (1.151) reads, term by term, $\underline{u} \cdot \operatorname{div} \underline{A} = \operatorname{div}(\underline{A}^t \underline{u}) - \operatorname{tr}[\underline{A}^t(\nabla \underline{u})]$, since $(\underline{A}^t \underline{u})_j = A_{ij} u_i$ and $\operatorname{tr}[\underline{A}^t \nabla \underline{u}] = A_{ij} u_{i,j}$. ■

DEFINITION 1.95 (Curl of a vector field). $\operatorname{curl} \underline{v}$ is the vector with $\underline{c} \cdot \operatorname{curl} \underline{v} = \operatorname{div}(\underline{v} \times \underline{c})$ for every fixed $\underline{c}$.

PROPOSITION 1.96. $\operatorname{curl} \underline{v} = \varepsilon_{kij} v_{i,k} \underline{e}_j$. (1.165)

Proof. Take $\underline{c} = \underline{e}_j$: $\underline{v} \times \underline{e}_j = v_i \varepsilon_{ijk} \underline{e}_k$, so $\underline{e}_j \cdot \operatorname{curl} \underline{v} = \operatorname{div}(v_i \varepsilon_{ijk} \underline{e}_k) = \varepsilon_{ijk} v_{i,k} = \varepsilon_{kij} v_{i,k}$ (1.164), using $\varepsilon_{ijk} = \varepsilon_{kij}$. ■

REMARK 1.97 (curl of a tensor field). $\operatorname{curl} \underline{A}$ is defined by $(\operatorname{curl} \underline{A}) \underline{c} = \operatorname{curl}(\underline{A}^t \underline{c})$. Taking $\underline{c} = \underline{e}_m$, $\underline{A}^t \underline{e}_m = A_{mi} \underline{e}_i$ has i th component A_{mi} , so $\operatorname{curl}(\underline{A}^t \underline{e}_m) = \varepsilon_{kij} A_{mi,k} \underline{e}_j$; matching $(\operatorname{curl} \underline{A}) \underline{e}_m = (\operatorname{curl} \underline{A})_{jm} \underline{e}_j$ gives $\operatorname{curl} \underline{A} = \varepsilon_{kij} A_{mi,k} \underline{e}_j \otimes \underline{e}_m$, equivalently $\varepsilon_{ilm} A_{jl,i} \underline{e}_m \otimes \underline{e}_j$ (Problem 28).

EXAMPLE 1.98 (divergence of a tensor field in polar coordinates). For $\underline{A} = A_{rr} \underline{e}_r \otimes \underline{e}_r + A_{r\theta} \underline{e}_r \otimes \underline{e}_\theta + \cdots$ with components functions of (r, θ) ,

decompose $\operatorname{div} \underline{A}$ on $\{\underline{e}_r, \underline{e}_\theta, \underline{k}\}$ and use (1.152). From (1.145) applied to the frame fields,

$$\nabla \underline{e}_r = \frac{1}{r} \underline{e}_\theta \otimes \underline{e}_\theta, \quad \nabla \underline{e}_\theta = -\frac{1}{r} \underline{e}_r \otimes \underline{e}_\theta, \quad (1.156)$$

and the scalar divergence of $\underline{w} = w_r \underline{e}_r + w_\theta \underline{e}_\theta + w_z \underline{k}$ is $\operatorname{div} \underline{w} = w_{r,r} + \frac{1}{r}(w_{\theta,\theta} + w_r)$ (1.158). Applying (1.152) with $\underline{A}^t \underline{e}_r = A_{rr} \underline{e}_r + A_{r\theta} \underline{e}_\theta + A_{rz} \underline{k}$ and the like, and $\operatorname{tr}[\underline{A}^t \nabla \underline{e}_r] = \frac{1}{r} A_{\theta\theta}$, $\operatorname{tr}[\underline{A}^t \nabla \underline{e}_\theta] = -\frac{1}{r} A_{r\theta}$,

$$\begin{aligned} \underline{e}_r \cdot \operatorname{div} \underline{A} &= A_{rr,r} + \frac{1}{r}(A_{rr} - A_{\theta\theta} + A_{r\theta,\theta}), \\ \underline{e}_\theta \cdot \operatorname{div} \underline{A} &= A_{\theta r,r} + \frac{1}{r}(A_{\theta r} + A_{r\theta} + A_{\theta\theta,\theta}), \\ \underline{k} \cdot \operatorname{div} \underline{A} &= A_{zr,r} + \frac{1}{r}(A_{zr} + A_{z\theta,\theta}). \end{aligned} \quad (1.162)$$

1.2.7 Integral formulae

Balance laws equate the rate of change of a quantity inside a region to fluxes through its boundary. Passing between the two needs a bridge from volume integrals of derivatives to surface integrals of values. That bridge is the divergence theorem; every conservation law later rests on it.

THEOREM 1.99 (Divergence theorem). *For a C^1 vector field $\underline{v}$ on a bounded region R with piecewise smooth orientable boundary ∂R and exterior unit normal $\underline{n}$,*

$$\int_R \operatorname{div} \underline{v} \, dv = \int_{\partial R} \underline{v} \cdot \underline{n} \, da. \quad (1.167)$$

The proof is classical and is taken as given. Three consequences, used constantly later, are immediate.

COROLLARY 1.100 (Component, tensor and scalar forms).

$$\int_R v_{i,i} \, dv = \int_{\partial R} v_i n_i \, da;$$

$$\int_R \operatorname{div} \underline{A} \, dv = \int_{\partial R} \underline{A} \underline{n} \, da; \quad \int_R \nabla \varphi \, dv = \int_{\partial R} \varphi \underline{n} \, da.$$

Proof. The first is (1.167) in components. For the tensor form, fix $\underline{c}$ and apply (1.167) to $\underline{A}^t \underline{c}$: $\underline{c} \cdot \int_R \operatorname{div} \underline{A} \, dv = \int_{\partial R} \underline{A}^t \underline{c} \cdot \underline{n} \, da = \underline{c} \cdot \int_{\partial R} \underline{A} \underline{n} \, da$, true for all $\underline{c}$. The scalar form is the tensor form with $\underline{A} = \varphi \underline{I}$, since $\operatorname{div}(\varphi \underline{I}) = \nabla \varphi$ and $\varphi \underline{I} \underline{n} = \varphi \underline{n}$. $\blacksquare$

THEOREM 1.101 (Stokes, for reference). *For a differentiable $\underline{v}$ on a piecewise smooth surface Ω bounded by a piecewise smooth closed curve $\partial\Omega$,*

$$\int_{\Omega} \operatorname{curl} \underline{v} \cdot \underline{n} \, da = \int_{\partial\Omega} \underline{v} \cdot d\underline{x}, \quad (1.168)$$

with $\underline{n}$ oriented so that $\underline{\nu} = \underline{t} \times \underline{n}$ is the outward normal to $\partial\Omega$ in the tangent plane. (Stated for completeness; not used in the sequel.)

1.3 Problems

All twenty eight problems of the chapter are solved below, in the book's order, with every algebraic step shown. Two identities recur:

$$\underline{a} \times (\underline{b} \times \underline{c}) = (\underline{a} \cdot \underline{c})\underline{b} - (\underline{a} \cdot \underline{b})\underline{c}, \quad \varepsilon_{ijk}\varepsilon_{imn} = \delta_{jm}\delta_{kn} - \delta_{jn}\delta_{km}. \quad (\star)$$

Problems 1–23 use only the algebra of §1.1; Problems 24–28 use the field calculus of §1.2.

PROBLEM 1 (Cauchy–Schwarz, triangle, Pythagoras). *Prove $|\underline{a} \cdot \underline{b}| \leq |\underline{a}||\underline{b}|$, the triangle inequality, and the Pythagorean theorem for orthogonal $\underline{a}, \underline{b}$.*

Solution. Cauchy–Schwarz. If $\underline{a} = \underline{0}$ then $\underline{a} \cdot \underline{b} = 0$ and the inequality reads $0 \leq 0$; assume $\underline{a} \neq \underline{0}$. Following the hint, form the vector

$$\underline{w} = (\underline{a} \cdot \underline{a})\underline{b} - (\underline{a} \cdot \underline{b})\underline{a}.$$

Positivity of the norm gives $0 \leq |\underline{w}|^2 = \underline{w} \cdot \underline{w}$. Expand the dot product by bilinearity, keeping every one of the four cross terms (write $\alpha = \underline{a} \cdot \underline{a}$,

$\beta = \underline{a} \cdot \underline{b}$ for brevity):

$$\begin{aligned}
|\underline{w}|^2 &= (\alpha \underline{b} - \beta \underline{a}) \cdot (\alpha \underline{b} - \beta \underline{a}) \\
&= \alpha^2 \underline{b} \cdot \underline{b} - \alpha\beta \underline{b} \cdot \underline{a} - \beta\alpha \underline{a} \cdot \underline{b} + \beta^2 \underline{a} \cdot \underline{a} \\
&= \alpha^2 \underline{b} \cdot \underline{b} - 2\alpha\beta^2 + \beta^2\alpha \quad (\text{using } \underline{b} \cdot \underline{a} = \underline{a} \cdot \underline{b} = \beta, \underline{a} \cdot \underline{a} = \alpha) \\
&= \alpha[\alpha \underline{b} \cdot \underline{b} - \beta^2] = (\underline{a} \cdot \underline{a})[(\underline{a} \cdot \underline{a})(\underline{b} \cdot \underline{b}) - (\underline{a} \cdot \underline{b})^2].
\end{aligned}$$

Since $\underline{a} \cdot \underline{a} = |\underline{a}|^2 > 0$, dividing the inequality $|\underline{w}|^2 \geq 0$ by it leaves the bracket nonnegative:

$$(\underline{a} \cdot \underline{a})(\underline{b} \cdot \underline{b}) - (\underline{a} \cdot \underline{b})^2 \geq 0 \implies (\underline{a} \cdot \underline{b})^2 \leq |\underline{a}|^2 |\underline{b}|^2.$$

Taking the (nonnegative) square root, $|\underline{a} \cdot \underline{b}| \leq |\underline{a}| |\underline{b}|$. Equality requires $\underline{w} = \underline{0}$, i.e. $\underline{a}, \underline{b}$ parallel.

Triangle inequality. Expand the squared norm and insert Cauchy–Schwarz at the marked step, $\underline{a} \cdot \underline{b} \leq |\underline{a} \cdot \underline{b}| \leq |\underline{a}| |\underline{b}|$:

$$\begin{aligned}
|\underline{a} + \underline{b}|^2 &= (\underline{a} + \underline{b}) \cdot (\underline{a} + \underline{b}) = |\underline{a}|^2 + 2 \underline{a} \cdot \underline{b} + |\underline{b}|^2 \\
&\leq |\underline{a}|^2 + 2|\underline{a}| |\underline{b}| + |\underline{b}|^2 = (|\underline{a}| + |\underline{b}|)^2.
\end{aligned}$$

Both sides are nonnegative, so $|\underline{a} + \underline{b}| \leq |\underline{a}| + |\underline{b}|$.

Pythagoras. If $\underline{a} \perp \underline{b}$, i.e. $\underline{a} \cdot \underline{b} = 0$, the middle term above drops out and $|\underline{a} + \underline{b}|^2 = |\underline{a}|^2 + |\underline{b}|^2$. $\square$

REMARK 1.102 (what the hint vector is doing). The vector $\underline{w} = (\underline{a} \cdot \underline{b})\underline{b} - (\underline{a} \cdot \underline{a})\underline{a}$ is (a positive multiple of) the part of $\underline{b}$ *perpendicular* to $\underline{a}$: subtracting the projection $(\underline{a} \cdot \underline{b}/\underline{a} \cdot \underline{a})\underline{a}$ from $\underline{b}$ and scaling by $\underline{a} \cdot \underline{a}$ gives exactly $\underline{w}$. Cauchy–Schwarz then says nothing more than “a squared length is never negative”: the inequality is tight precisely when that perpendicular part is zero, i.e. when $\underline{b}$ is parallel to $\underline{a}$. This is why Cauchy–Schwarz underlies the very definition of angle (Definition 1.8): $\cos \theta = \underline{a} \cdot \underline{b}/(|\underline{a}| |\underline{b}|)$ lands in $[-1, 1]$ exactly because $|\underline{w}|^2 \geq 0$.

PROBLEM 2 (Is the norm linear?). *Is $f(\underline{v}) = |\underline{v}|$ linear?*

Solution. No. Linearity requires both $f(\underline{u} + \underline{v}) = f(\underline{u}) + f(\underline{v})$ and $f(\alpha \underline{v}) = \alpha f(\underline{v})$; each fails.

Additivity. Take $\underline{v} = -\underline{u}$ with $\underline{u} \neq \underline{0}$:

$$f(\underline{u} + \underline{v}) = f(\underline{0}) = 0, \quad f(\underline{u}) + f(\underline{v}) = |\underline{u}| + |-\underline{u}| = 2|\underline{u}| > 0.$$

Homogeneity. For $\alpha < 0$,

$$f(\alpha \underline{v}) = |\alpha \underline{v}| = |\alpha| |\underline{v}| = -\alpha |\underline{v}| \neq \alpha |\underline{v}| = \alpha f(\underline{v})$$

whenever $\underline{v} \neq \underline{0}$.

What the norm does satisfy is $f(\alpha \underline{v}) = |\alpha| f(\underline{v})$ and $f(\underline{u} + \underline{v}) \leq f(\underline{u}) + f(\underline{v})$, the triangle inequality of Problem 1. Those two make it a norm, not a linear function. $\square$

PROBLEM 3 (A constant map is not a tensor). *For a fixed $\underline{m}$, is there a tensor $\underline{A}$ with $\underline{A}\underline{v} = \underline{m}$ for all $\underline{v}$?*

Solution. No, unless $\underline{m} = \underline{0}$. A linear map fixes the origin:

$$\underline{A}\underline{0} = \underline{A}(0 \cdot \underline{0}) = 0 \cdot \underline{A}\underline{0} = \underline{0},$$

whereas the requirement gives $\underline{A}\underline{0} = \underline{m}$. So $\underline{m} = \underline{0}$.

Second route, without using $\underline{0}$. Homogeneity with $\alpha = 2$ gives

$$\underline{m} = \underline{A}(2\underline{v}) = 2 \underline{A}\underline{v} = 2\underline{m} \implies \underline{m} = \underline{0}.$$

In the problem $\underline{m} = \frac{\sqrt{2}}{2}(\underline{e}_1 + \underline{e}_2) \neq \underline{0}$, so no such tensor exists. Contrast Problem 4, where $\underline{A}\underline{v} = (\underline{v} \cdot \underline{n})\underline{m}$ is linear: the difference is the factor $\underline{v} \cdot \underline{n}$, which vanishes with $\underline{v}$. $\square$

PROBLEM 4 (A rank-one tensor). *With $\underline{m} = \frac{\sqrt{2}}{2}(\underline{e}_1 + \underline{e}_2)$, $\underline{n} = \frac{\sqrt{2}}{2}(\underline{e}_3 - \underline{e}_1)$, find $\underline{A}$ with $\underline{A}\underline{v} = (\underline{v} \cdot \underline{n})\underline{m}$.*

Solution. The map $\underline{v} \mapsto (\underline{v} \cdot \underline{n})\underline{m}$ is linear in $\underline{v}$ (the inner product is linear in its second argument) and returns a vector, so it is a tensor. By Definition 1.32, $(\underline{m} \otimes \underline{n})\underline{v} = (\underline{n} \cdot \underline{v})\underline{m}$. Hence $\underline{A} = \underline{m} \otimes \underline{n}$.

Components. $A_{ij} = m_i n_j$ (from Example 1.34). The components of $\underline{m}$ are $m_1 = \frac{\sqrt{2}}{2}$, $m_2 = \frac{\sqrt{2}}{2}$, $m_3 = 0$; those of $\underline{n}$ are $n_1 = -\frac{\sqrt{2}}{2}$, $n_2 = 0$, $n_3 = \frac{\sqrt{2}}{2}$. Each entry $A_{ij} = m_i n_j$:

$$\begin{aligned} A_{11} &= \frac{\sqrt{2}}{2} \cdot \left(-\frac{\sqrt{2}}{2}\right) = -\frac{1}{2}, & A_{12} &= \frac{\sqrt{2}}{2} \cdot 0 = 0, & A_{13} &= \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{2}}{2} = \frac{1}{2}, \\ A_{21} &= -\frac{1}{2}, & A_{22} &= 0, & A_{23} &= \frac{1}{2}, & A_{31} &= 0, & A_{32} &= 0, & A_{33} &= 0. \end{aligned}$$

Hence

$$(A_{ij}) = \frac{1}{2} \begin{pmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$

Check. $\underline{A}$ has rank 1 (all rows are multiples of $(-1, 0, 1)$), confirming it is a rank one tensor: the image of $\underline{A}$ is the line spanned by $\underline{m}$. $\square$

PROBLEM 5 ($\varepsilon_{ijk} A_{jk} = 0 \iff \underline{A} = \underline{A}^t$).

Solution. Write $c_i := \varepsilon_{ijk} A_{jk}$ (a vector, since i is free while j, k are summed). We must show $c_i = 0$ for all i exactly when $A_{jk} = A_{kj}$.

($\Leftarrow$) *Symmetric* $\Rightarrow c_i = 0$. Suppose $A_{jk} = A_{kj}$. In $c_i = \varepsilon_{ijk} A_{jk}$ rename the two summed indices $j \leftrightarrow k$ (a dummy relabelling changes nothing):

$$c_i = \varepsilon_{ijk} A_{jk} = \varepsilon_{ikj} A_{kj}.$$

Now use symmetry $A_{kj} = A_{jk}$ and the antisymmetry of the permutation symbol under swapping its last two slots, $\varepsilon_{ikj} = -\varepsilon_{ijk}$:

$$c_i = \varepsilon_{ikj} A_{kj} = -\varepsilon_{ijk} A_{jk} = -c_i.$$

Thus $c_i = -c_i$, i.e. $2c_i = 0$, so $c_i = 0$ for every i .

($\Rightarrow$) $c_i = 0 \Rightarrow$ *symmetric*. Split $\underline{A}$ into its symmetric and skew parts, $A_{jk} = S_{jk} + W_{jk}$ with $S_{jk} = \frac{1}{2}(A_{jk} + A_{kj})$, $W_{jk} = \frac{1}{2}(A_{jk} - A_{kj})$. By the direction just proved, $\varepsilon_{ijk} S_{jk} = 0$, so

$$0 = c_i = \varepsilon_{ijk} A_{jk} = \varepsilon_{ijk} S_{jk} + \varepsilon_{ijk} W_{jk} = \varepsilon_{ijk} W_{jk}.$$

Every skew array is $W_{jk} = -\varepsilon_{jkl} w_l$ for a unique axial vector $\underline{w}$ (this is the general solution of $W_{jk} = -W_{kj}$; see Proposition 1.44). Substitute

and contract:

$$\begin{aligned}\varepsilon_{ijk}W_{jk} &= \varepsilon_{ijk}(-\varepsilon_{jkl}w_l) = -\varepsilon_{ijk}\varepsilon_{jkl}w_l \\ &= -\varepsilon_{jki}\varepsilon_{jkl}w_l = -2\delta_{il}w_l = -2w_i,\end{aligned}$$

where we wrote $\varepsilon_{ijk} = \varepsilon_{jki}$ (cyclic shift) and used the contracted identity $\varepsilon_{jki}\varepsilon_{jkl} = 2\delta_{il}$. Hence $0 = c_i = -2w_i$ gives $w_i = 0$ for all i , so $\underline{w} = \underline{0}$, so $W_{jk} = 0$, so $A_{jk} = S_{jk} = A_{kj}$: $\underline{A}$ is symmetric. $\square$

REMARK 1.103 (what the operation $A_{jk} \mapsto \varepsilon_{ijk}A_{jk}$ extracts). The computation shows $\varepsilon_{ijk}A_{jk} = -2w_i$, twice the (negated) axial vector of the skew part of $\underline{A}$. So contracting a tensor with the permutation symbol on two indices is a machine that *throws away the symmetric part and reads off the skew part as a vector*. This is exactly the device behind writing the vorticity of a velocity field, or the moment of a force, as a single axial vector; it works only in three dimensions, where a skew tensor (three independent entries) matches a vector (three components).

PROBLEM 6 (Skew/symmetric and the tensor inner product). *With converses: (a) $\underline{A}^t = -\underline{A} \Rightarrow \underline{A} \cdot (\underline{a} \otimes \underline{a}) = 0$; (b) $\underline{T} \cdot \underline{S} = 0 \ \forall \underline{S} \Rightarrow \underline{T} = \underline{0}$; (c) $\underline{T} \cdot \underline{S} = 0 \ \forall \underline{S} = \underline{S}^t \Rightarrow \underline{T}$ skew; (d) $\underline{T} \cdot \underline{S} = 0 \ \forall \underline{S} = -\underline{S}^t \Rightarrow \underline{T}$ symmetric.*

Solution. Preliminary. Recall: $\underline{A} \cdot \underline{B} = \text{tr}(\underline{A}\underline{B}^t) = A_{ij}B_{ij}$ (sum of all nine component products), and $\underline{A} \cdot (\underline{u} \otimes \underline{v}) = \underline{u} \cdot \underline{A}\underline{v}$.

Lemma (the key orthogonality). Let $\underline{S} = \underline{S}^t$ be symmetric and $\underline{W} = -\underline{W}^t$ skew. Then $\underline{S} \cdot \underline{W} = S_{ij}W_{ij}$. Relabelling $i \leftrightarrow j$: $S_{ij}W_{ij} = S_{ji}W_{ji} = S_{ij}(-W_{ij}) = -S_{ij}W_{ij}$, using $S_{ji} = S_{ij}$ and $W_{ji} = -W_{ij}$. Thus $2(\underline{S} \cdot \underline{W}) = 0$, i.e. $\underline{S} \cdot \underline{W} = 0$: a symmetric tensor is orthogonal to every skew tensor.

(a) $\underline{A} \cdot (\underline{a} \otimes \underline{a}) = \underline{a} \cdot \underline{A}\underline{a}$. Since $\underline{A}^t = -\underline{A}$, $\underline{a} \cdot \underline{A}\underline{a} = \underline{A}^t \underline{a} \cdot \underline{a} = -\underline{A}\underline{a} \cdot \underline{a} = -\underline{a} \cdot \underline{A}\underline{a}$, hence $\underline{a} \cdot \underline{A}\underline{a} = 0$. *Converse:* if $\underline{a} \cdot \underline{A}\underline{a} = 0$ for all $\underline{a}$, put $\underline{a} = \underline{e}_i + \underline{e}_j$ ($i \neq j$): $(\underline{e}_i + \underline{e}_j) \cdot \underline{A}(\underline{e}_i + \underline{e}_j) = A_{ii} + A_{ij} + A_{ji} + A_{jj} = 0$. With $\underline{a} = \underline{e}_i$: $A_{ii} = 0$; with $\underline{a} = \underline{e}_j$: $A_{jj} = 0$. Hence $A_{ij} + A_{ji} = 0$ for all $i \neq j$, together with $A_{ii} = 0$: $\underline{A}$ is skew.

(b) $\underline{T} \cdot \underline{S} = 0$ for all $\underline{S}$: choose $\underline{S} = \underline{T}$, getting $|\underline{T}|^2 = T_{ij}T_{ij} = 0$, so every $T_{ij} = 0$, i.e. $\underline{T} = \underline{O}$.

(c) $\underline{T} \cdot \underline{S} = 0$ for all symmetric $\underline{S}$. Decompose $\underline{T} = \underline{T}_s + \underline{T}_w$ (symmetric + skew). Choose $\underline{S} = \underline{T}_s$: by the Lemma, $\underline{T}_w \cdot \underline{T}_s = 0$, so $\underline{T} \cdot \underline{T}_s = \underline{T}_s \cdot \underline{T}_s = |\underline{T}_s|^2 = 0$, hence $\underline{T}_s = \underline{O}$ and $\underline{T} = \underline{T}_w$ is skew.

(d) $\underline{T} \cdot \underline{S} = 0$ for all skew $\underline{S}$. Choose $\underline{S} = \underline{T}_w$: by the Lemma, $\underline{T}_s \cdot \underline{T}_w = 0$, so $\underline{T} \cdot \underline{T}_w = |\underline{T}_w|^2 = 0$, hence $\underline{T}_w = \underline{O}$ and $\underline{T} = \underline{T}_s$ is symmetric. $\square$

PROBLEM 7 (Is $\underline{A}(\underline{u} \times \underline{v}) = \underline{A}\underline{u} \times \underline{v} + \underline{u} \times \underline{A}\underline{v}$?).

Solution. Write $D(\underline{A}) := \underline{A}(\underline{u} \times \underline{v}) - \underline{A}\underline{u} \times \underline{v} - \underline{u} \times \underline{A}\underline{v}$. The question is when $D(\underline{A}) = \underline{0}$ for all $\underline{u}, \underline{v}$. Note D is linear in $\underline{A}$.

No in general. Take $\underline{A} = \underline{I}$:

$$D(\underline{I}) = \underline{u} \times \underline{v} - \underline{u} \times \underline{v} - \underline{u} \times \underline{v} = -\underline{u} \times \underline{v} \neq \underline{0}.$$

Skew tensors do satisfy it. Let $\underline{W}\underline{x} = \underline{\omega} \times \underline{x}$, Problem 16. The Jacobi identity is

$$\underline{\omega} \times (\underline{u} \times \underline{v}) + \underline{u} \times (\underline{v} \times \underline{\omega}) + \underline{v} \times (\underline{\omega} \times \underline{u}) = \underline{0}.$$

Use $\underline{v} \times (\underline{\omega} \times \underline{u}) = -(\underline{\omega} \times \underline{u}) \times \underline{v}$ and $\underline{u} \times (\underline{v} \times \underline{\omega}) = -\underline{u} \times (\underline{\omega} \times \underline{v})$:

$$\underline{\omega} \times (\underline{u} \times \underline{v}) = (\underline{\omega} \times \underline{u}) \times \underline{v} + \underline{u} \times (\underline{\omega} \times \underline{v}),$$

that is $D(\underline{W}) = \underline{0}$.

Only skew tensors do. Suppose $D(\underline{A}) = \underline{0}$. Split $\underline{A} = \underline{S} + \underline{W}$ by Problem 14; $D(\underline{W}) = \underline{0}$, so $D(\underline{S}) = \underline{0}$ by linearity. Take the principal basis of $\underline{S}$, $\underline{S}\underline{e}_i = \lambda_i \underline{e}_i$, and put $\underline{u} = \underline{e}_i$, $\underline{v} = \underline{e}_j$ with ijk cyclic, so $\underline{e}_i \times \underline{e}_j = \underline{e}_k$:

$$\lambda_k \underline{e}_k = \lambda_i \underline{e}_i \times \underline{e}_j + \underline{e}_i \times \lambda_j \underline{e}_j = (\lambda_i + \lambda_j) \underline{e}_k \implies \lambda_k = \lambda_i + \lambda_j.$$

The three cyclic triples give

$$\lambda_3 = \lambda_1 + \lambda_2, \quad \lambda_1 = \lambda_2 + \lambda_3, \quad \lambda_2 = \lambda_3 + \lambda_1.$$

Adding, $\sum \lambda_i = 2 \sum \lambda_i$, so $\sum \lambda_i = 0$; then the first equation gives $\lambda_3 = -\lambda_3$, hence $\lambda_3 = 0$, and likewise $\lambda_1 = \lambda_2 = 0$. So $\underline{S} = \underline{Q}$ and $\underline{A} = \underline{W}$ is skew. $\square$

REMARK 1.104 (what the answer says). $D(\underline{A}) = \underline{0}$ is the Leibniz rule for $\underline{A}$ acting on the product $\underline{u} \times \underline{v}$. The result is therefore: *a tensor differentiates the cross product exactly when it is skew*, i.e. when it is an infinitesimal rotation. This is the statement that Skw is the Lie algebra of the rotation group, and it recurs in Chapter 4, where the spin $\underline{W}$ is skew for precisely this reason.

PROBLEM 8 (Reciprocal bases). *With $\underline{e}_i = a_{ij}\underline{e}'_j$, $\underline{e}'_i = a'_{ij}\underline{e}_j$, show (a) $a_{ik}a'_{kj} = \delta_{ij}$; (b) $(a') = (a)^t$.*

Solution. **(a)** Substitute the second relation into the first:

$$\underline{e}_i = a_{ij}\underline{e}'_j = a_{ij}(a'_{jk}\underline{e}_k) = a_{ij}a'_{jk}\underline{e}_k.$$

But also $\underline{e}_i = \delta_{ik}\underline{e}_k$. The $\{\underline{e}_k\}$ are independent, so the coefficients match:

$$a_{ij}a'_{jk} = \delta_{ik}.$$

(b) Dot $\underline{e}_i = a_{ij}\underline{e}'_j$ with $\underline{e}'_k$ and use $\underline{e}'_j \cdot \underline{e}'_k = \delta_{jk}$:

$$\underline{e}_i \cdot \underline{e}'_k = a_{ij}\delta_{jk} = a_{ik}.$$

Dot $\underline{e}'_i = a'_{ij}\underline{e}_j$ with $\underline{e}_k$ and use $\underline{e}_j \cdot \underline{e}_k = \delta_{jk}$:

$$\underline{e}'_i \cdot \underline{e}_k = a'_{ij}\delta_{jk} = a'_{ik}.$$

The two left sides are the same number, since the dot product is symmetric:

$$a'_{ik} = \underline{e}'_i \cdot \underline{e}_k = \underline{e}_k \cdot \underline{e}'_i = a_{ki}, \quad \text{i.e.} \quad (a') = (a)^t.$$

Part (a) then reads $a_{ij}a_{kj} = \delta_{ik}$: the matrix (a) is orthogonal, $(a)(a)^t = (a)^t(a) = (\delta)$. This is the fact used in Problem 9 and again in Problem 3 of Chapter 2. $\square$

PROBLEM 9 (Invariance of trace and dot product; a computation). (a) $A_{ii} = A'_{ii}$; (b) deduce invariance of $\underline{u} \cdot \underline{v}$; (c) compute A'_{ij} for the given data.

Solution. **(a)** With $a_{ij} = \underline{e}_i \cdot \underline{e}'_j$, so that $\underline{e}'_j = a_{kj} \underline{e}_k$, the components transform as

$$A'_{ij} = \underline{e}'_i \cdot \underline{A} \underline{e}'_j = a_{ki} \underline{e}_k \cdot \underline{A} a_{lj} \underline{e}_l = a_{ki} a_{lj} \underline{e}_k \cdot \underline{A} \underline{e}_l = a_{ki} a_{lj} A_{kl}.$$

Orthonormality of the primed basis fixes (a):

$$\delta_{ij} = \underline{e}'_i \cdot \underline{e}'_j = a_{ki} a_{lj} \underline{e}_k \cdot \underline{e}_l = a_{ki} a_{lj} \delta_{kl} = a_{ki} a_{kj},$$

so (a) is orthogonal. Set $j = i$ and sum:

$$A'_{ii} = a_{ki} a_{li} A_{kl} = \delta_{kl} A_{kl} = A_{kk}.$$

(b) Vectors carry one factor, $u'_i = \underline{e}'_i \cdot \underline{u} = a_{ki} u_k$. Hence

$$u'_i v'_i = a_{ki} a_{li} u_k v_l = \delta_{kl} u_k v_l = u_k v_k.$$

Alternatively, apply (a) to $\underline{A} = \underline{u} \otimes \underline{v}$, whose trace is $\underline{u} \cdot \underline{v}$ by Theorem 1.51.

(c) The data are

$$(A_{ij}) = \begin{pmatrix} 1 & 5 & -5 \\ 5 & 0 & 0 \\ -5 & 0 & 1 \end{pmatrix}, \quad \underline{e}'_1 = \frac{1}{\sqrt{5}}(-1, 0, 2), \quad \underline{e}'_2 = \underline{e}_2 = (0, 1, 0).$$

Fix $\underline{e}'_3$ by right-handedness, $\underline{e}'_3 = \underline{e}'_1 \times \underline{e}'_2$. The cross product $(-1, 0, 2) \times (0, 1, 0)$ has components

$$(0 \cdot 0 - 2 \cdot 1, \quad 2 \cdot 0 - (-1) \cdot 0, \quad (-1) \cdot 1 - 0 \cdot 0) = (-2, 0, -1),$$

so $\underline{e}'_3 = \frac{1}{\sqrt{5}}(-2, 0, -1)$. The change-of-basis matrix $\underline{Q} = (a_{ij})$, $a_{ij} = \underline{e}_i \cdot \underline{e}'_j$, carries the components of $\underline{e}'_j$ in its j th column:

$$\underline{Q} = \frac{1}{\sqrt{5}} \begin{pmatrix} -1 & 0 & -2 \\ 0 & \sqrt{5} & 0 \\ 2 & 0 & -1 \end{pmatrix}.$$

The law $A'_{ij} = a_{ki}a_{lj}A_{kl}$ is $\underline{A}' = \underline{Q}^t \underline{A} \underline{Q}$, which we read as $A'_{ij} = \underline{e}'_i \cdot \underline{A} \underline{e}'_j$. Compute the three columns $\underline{A} \underline{e}'_j$ (matrix times vector); for example

$$\underline{A} \underline{e}'_1 = \frac{1}{\sqrt{5}} \underline{A} \begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix} = \frac{1}{\sqrt{5}} \begin{pmatrix} 1(-1) + 5(0) - 5(2) \\ 5(-1) + 0 + 0 \\ -5(-1) + 0 + 1(2) \end{pmatrix} = \frac{1}{\sqrt{5}} \begin{pmatrix} -11 \\ -5 \\ 7 \end{pmatrix},$$

and likewise $\underline{A} \underline{e}'_2 = (5, 0, 0)^t$, $\underline{A} \underline{e}'_3 = \frac{1}{\sqrt{5}}(3, -10, 9)^t$. Now $A'_{ij} = \underline{e}'_i \cdot \underline{A} \underline{e}'_j$ entry by entry:

$$\begin{aligned} A'_{11} &= \frac{1}{\sqrt{5}}(-1, 0, 2) \cdot \frac{1}{\sqrt{5}}(-11, -5, 7) = \frac{11+0+14}{5} = 5, \\ A'_{12} &= \frac{1}{\sqrt{5}}(-1, 0, 2) \cdot (5, 0, 0) = -\frac{5}{\sqrt{5}} = -\sqrt{5}, \\ A'_{13} &= \frac{1}{\sqrt{5}}(-1, 0, 2) \cdot \frac{1}{\sqrt{5}}(3, -10, 9) = \frac{-3+0+18}{5} = 3, \\ A'_{22} &= (0, 1, 0) \cdot (5, 0, 0) = 0, \\ A'_{23} &= (0, 1, 0) \cdot \frac{1}{\sqrt{5}}(3, -10, 9) = -\frac{10}{\sqrt{5}} = -2\sqrt{5}, \\ A'_{33} &= \frac{1}{\sqrt{5}}(-2, 0, -1) \cdot \frac{1}{\sqrt{5}}(3, -10, 9) = \frac{-6+0-9}{5} = -3, \end{aligned}$$

with $A'_{21} = A'_{12}$, $A'_{31} = A'_{13}$, $A'_{32} = A'_{23}$ since $\underline{A}'$ is symmetric ($\underline{A}$ is). Therefore

$$\underline{A}' = \begin{pmatrix} 5 & -\sqrt{5} & 3 \\ -\sqrt{5} & 0 & -2\sqrt{5} \\ 3 & -2\sqrt{5} & -3 \end{pmatrix}.$$

Check: $\text{tr } \underline{A}' = 5 + 0 - 3 = 2 = 1 + 0 + 1 = \text{tr } \underline{A}$, confirming the invariance proved in part (a). $\square$

PROBLEM 10 (Projection and reflection tensors). (a) *Consider a plane P with unit normal $\underline{n}$. Let $\underline{v} \in \mathbb{E}^3$ be arbitrary and $\underline{v}_p$ its projection onto P . Verify $\underline{v} = \underline{v}_p + (\underline{v} \cdot \underline{n})\underline{n}$ and find $\mathbb{P}_{(\underline{n})}$ in $\underline{v}_p = \mathbb{P}_{(\underline{n})}\underline{v}$.* (b) *A tensor $\underline{A}$ is a projection iff $\underline{A}^t = \underline{A}$ and $\underline{A}^2 = \underline{A}$. Show $\mathbb{P}_{(\underline{n})}$ is a projection, and that $\underline{Q}$, $\underline{I}$, $\underline{n} \otimes \underline{n}$ are projections.* (c) *Let $\mathbb{R}_{(\underline{n})}$ be the reflection tensor for P . Find $\mathbb{R}_{(\underline{n})}$ and its components relative to $\{\underline{e}_i \otimes \underline{e}_j\}$ for (i) $\underline{n} = \underline{e}_1$, (ii) $\underline{n} = \frac{\sqrt{2}}{2}(\underline{e}_1 + \underline{e}_2)$.*

Solution. The normal is a unit vector and the plane through the origin with that normal consists of the vectors orthogonal to it, so throughout

$$\underline{n} \cdot \underline{n} = 1, \quad P = \{\underline{w} \in \mathbb{E}^3 : \underline{w} \cdot \underline{n} = 0\} = \{\underline{n}\}^\perp, \quad \mathcal{N} = \text{span}\{\underline{n}\}, \quad (\star_1)$$

P being two dimensional and $\mathcal{N}$ one dimensional.

(a) The shadow of $\underline{v}$ on the direction $\underline{n}$ has signed length $\underline{v} \cdot \underline{n}$, so the part of $\underline{v}$ along the normal, and what is left after removing it, are

$$\underline{v}_n := (\underline{v} \cdot \underline{n})\underline{n}, \quad \underline{v}_p := \underline{v} - (\underline{v} \cdot \underline{n})\underline{n}. \quad (\star_2)$$

That $\underline{v}_p$ lies in P is not an assumption but a computation; dot it with $\underline{n}$ and use bilinearity together with $\underline{n} \cdot \underline{n} = 1$:

$$\underline{v}_p \cdot \underline{n} = [\underline{v} - (\underline{v} \cdot \underline{n})\underline{n}] \cdot \underline{n} = \underline{v} \cdot \underline{n} - (\underline{v} \cdot \underline{n})\underline{n} \cdot \underline{n} \stackrel{(\star_1)}{=} \underline{v} \cdot \underline{n} - \underline{v} \cdot \underline{n} = 0,$$

hence $\underline{v}_p \in \{\underline{n}\}^\perp = P$, and by construction $\underline{v} = \underline{v}_p + \underline{v}_n$, which is the required identity.

The splitting is unique. Suppose $\underline{v} = \underline{w}_p + \alpha\underline{n}$ with $\underline{w}_p \in P$ and $\alpha \in \mathbb{R}$. Subtracting the two expressions for $\underline{v}$,

$$\underline{v}_p - \underline{w}_p = (\alpha - \underline{v} \cdot \underline{n})\underline{n},$$

whose left side lies in P and whose right side lies in $\mathcal{N}$. A vector of $P \cap \mathcal{N}$ satisfies both $\underline{w} = \lambda\underline{n}$ and $\underline{w} \cdot \underline{n} = 0$, so $\lambda = \lambda\underline{n} \cdot \underline{n} = \underline{w} \cdot \underline{n} = 0$ and $P \cap \mathcal{N} = \{0\}$. Therefore $\underline{w}_p = \underline{v}_p$ and $\alpha = \underline{v} \cdot \underline{n}$, and $\mathbb{E}^3 = P \oplus \mathcal{N}$.

Before writing $\underline{v}_p = \mathbb{P}_{(\underline{n})}\underline{v}$ we must know the map $\underline{v} \mapsto \underline{v}_p$ is linear, since only linear maps are tensors:

$$\begin{aligned} (\alpha\underline{u} + \beta\underline{v})_p &= \alpha\underline{u} + \beta\underline{v} - ((\alpha\underline{u} + \beta\underline{v}) \cdot \underline{n})\underline{n} \\ &= \alpha\underline{u} + \beta\underline{v} - (\alpha\underline{u} \cdot \underline{n} + \beta\underline{v} \cdot \underline{n})\underline{n} \\ &= \alpha[\underline{u} - (\underline{u} \cdot \underline{n})\underline{n}] + \beta[\underline{v} - (\underline{v} \cdot \underline{n})\underline{n}] = \alpha\underline{u}_p + \beta\underline{v}_p. \end{aligned}$$

Now recognise the subtracted term as a tensor acting on $\underline{v}$. By the definition of the tensor product, $(\underline{n} \otimes \underline{n})\underline{v} = (\underline{n} \cdot \underline{v})\underline{n} = (\underline{v} \cdot \underline{n})\underline{n}$, so $(\star_2)$ becomes $\underline{v}_p = \underline{v} - (\underline{n} \otimes \underline{n})\underline{v} = [\underline{I} - \underline{n} \otimes \underline{n}]\underline{v}$ for every $\underline{v}$, and equality of tensors is equality of their action. Hence

$$\mathbb{P}_{(\underline{n})} = \underline{I} - \underline{n} \otimes \underline{n}, \quad (\mathbb{P}_{(\underline{n})})_{ij} = \delta_{ij} - n_i n_j, \quad (\star_3)$$

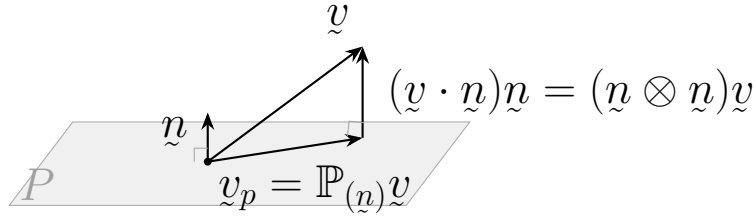


FIGURE P10.1. $v = v_p + (v \cdot \underline{n})\underline{n}$ with $v_p \cdot \underline{n} = 0$; $\mathbb{P}_{(\underline{n})}$ returns the leg in P , $\underline{n} \otimes \underline{n}$ the leg along $\underline{n}$, and $\mathbb{P}_{(\underline{n})} + \underline{n} \otimes \underline{n} = \underline{I}$.

the components following from $\underline{I} = \delta_{ij}\underline{e}_i \otimes \underline{e}_j$ and $(\underline{n} \otimes \underline{n})_{ij} = n_i n_j$.

(b) Everything rests on one property of the dyad, obtained by applying it twice to an arbitrary $\underline{v}$ and unfolding the definition at each step:

$$(\underline{n} \otimes \underline{n})^2 \underline{v} = (\underline{n} \otimes \underline{n})[(\underline{n} \cdot \underline{v})\underline{n}] = (\underline{n} \cdot \underline{v})(\underline{n} \cdot \underline{n})\underline{n} \stackrel{(\star_1)}{=} (\underline{n} \cdot \underline{v})\underline{n} = (\underline{n} \otimes \underline{n})\underline{v}, \quad (\star_4)$$

true for every $\underline{v}$, so $(\underline{n} \otimes \underline{n})^2 = \underline{n} \otimes \underline{n}$. Note where the hypothesis enters: the middle step used $\underline{n} \cdot \underline{n} = 1$, and for a normal of length other than one the dyad would not be idempotent. Its symmetry follows from the definition of the transpose, since for all $\underline{u}, \underline{v}$

$$\underline{u} \cdot (\underline{n} \otimes \underline{n})\underline{v} = \underline{u} \cdot (\underline{n} \cdot \underline{v})\underline{n} = (\underline{n} \cdot \underline{v})(\underline{u} \cdot \underline{n}) = \underline{v} \cdot (\underline{n} \cdot \underline{u})\underline{n} = \underline{v} \cdot (\underline{n} \otimes \underline{n})\underline{u},$$

which says $(\underline{n} \otimes \underline{n})^t = \underline{n} \otimes \underline{n}$. Transposition being linear, and $\underline{I}^t = \underline{I}$,

$$\mathbb{P}_{(\underline{n})}^t = \underline{I}^t - (\underline{n} \otimes \underline{n})^t = \underline{I} - \underline{n} \otimes \underline{n} = \mathbb{P}_{(\underline{n})},$$

and expanding the square, the two equal cross terms combining into one,

$$\mathbb{P}_{(\underline{n})}^2 = \underline{I} - \underline{n} \otimes \underline{n} - \underline{n} \otimes \underline{n} + (\underline{n} \otimes \underline{n})^2 \stackrel{(\star_4)}{=} \underline{I} - 2\underline{n} \otimes \underline{n} + (\underline{n} \otimes \underline{n})^2 \stackrel{(\star_4)}{=} \underline{I} - \underline{n} \otimes \underline{n} = \mathbb{P}_{(\underline{n})}.$$

The same two facts in components, which is the form used in calculations; recall $(\underline{A}\underline{B})_{ij} = A_{ik}B_{kj}$, and that a Kronecker delta collapses the sum it

is contracted with, $\delta_{ik}n_k = n_i$ and $\delta_{ik}\delta_{kj} = \delta_{ij}$:

$$\begin{aligned}
[(\underline{n} \otimes \underline{n})^2]_{ij} &= (n_i n_k)(n_k n_j) = n_i(n_k n_k)n_j = n_i n_j, \\
[\mathbb{P}_{(\underline{n})}^2]_{ij} &= (\delta_{ik} - n_i n_k)(\delta_{kj} - n_k n_j) \\
&= \delta_{ik}\delta_{kj} - \delta_{ik}n_k n_j - n_i n_k \delta_{kj} + n_i n_k n_k n_j \\
&= \delta_{ij} - n_i n_j - n_i n_j + n_i n_j \\
&= \delta_{ij} - n_i n_j = (\mathbb{P}_{(\underline{n})})_{ij}, \\
(\mathbb{P}_{(\underline{n})})_{ji} &= \delta_{ji} - n_j n_i = \delta_{ij} - n_i n_j = (\mathbb{P}_{(\underline{n})})_{ij}.
\end{aligned}$$

The remaining three are immediate from the definitions. For the zero tensor, $\underline{u} \cdot \underline{Q}\underline{v} = \underline{u} \cdot \underline{0} = 0 = \underline{v} \cdot \underline{0} = \underline{v} \cdot \underline{Q}\underline{u}$ gives $\underline{Q}^t = \underline{Q}$, and $\underline{Q}^2 \underline{v} = \underline{Q}(\underline{Q}\underline{v}) = \underline{Q}\underline{0} = \underline{0} = \underline{Q}\underline{v}$ gives $\underline{Q}^2 = \underline{Q}$; it projects onto the trivial subspace $\{0\}$. For the identity, $\underline{u} \cdot \underline{I}\underline{v} = \underline{u} \cdot \underline{v} = \underline{v} \cdot \underline{u} = \underline{v} \cdot \underline{I}\underline{u}$ gives $\underline{I}^t = \underline{I}$, and $\underline{I}^2 \underline{v} = \underline{I}(\underline{I}\underline{v}) = \underline{I}\underline{v}$ gives $\underline{I}^2 = \underline{I}$; it projects onto all of $\mathbb{E}^3$. For $\underline{n} \otimes \underline{n}$, symmetry and idempotency are exactly what was proved above, and it projects onto the normal line, since $(\underline{n} \otimes \underline{n})\underline{v} = \underline{v}_n$ by $(\star_2)$.

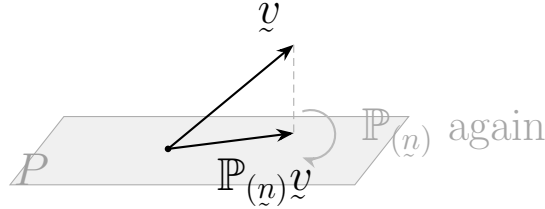


FIGURE P10.2. $\mathbb{P}_{(\underline{n})}^2 = \mathbb{P}_{(\underline{n})}$: the second application meets a vector already in P , since

$$(\mathbb{P}_{(\underline{n})}\underline{v}) \cdot \underline{n} = 0.$$

The two projections $\mathbb{P}_{(\underline{n})}$ and $\underline{n} \otimes \underline{n}$ are not independent of each other: they are *complementary*, and this is the algebraic face of $\mathbb{E}^3 = P \oplus \mathcal{N}$. Directly from $(\star_3)$ and $(\star_4)$,

$$\begin{aligned}
\mathbb{P}_{(\underline{n})} + \underline{n} \otimes \underline{n} &= \underline{I}, \\
\mathbb{P}_{(\underline{n})}(\underline{n} \otimes \underline{n}) &= (\underline{I} - \underline{n} \otimes \underline{n})(\underline{n} \otimes \underline{n}) = \underline{n} \otimes \underline{n} - (\underline{n} \otimes \underline{n})^2 \\
&\stackrel{(\star_4)}{=} \underline{Q} = (\underline{n} \otimes \underline{n})\mathbb{P}_{(\underline{n})},
\end{aligned}$$

so each annihilates the range of the other. Their traces count the dimensions of what they keep; using linearity of the trace with $\text{tr } \underline{I} = 3$ and

$$\text{tr}(\underline{u} \otimes \underline{v}) = \underline{u} \cdot \underline{v},$$

$$\text{tr } \mathbb{P}_{(\underline{n})} = \text{tr } \underline{I} - \text{tr}(\underline{n} \otimes \underline{n}) = 3 - \underline{n} \cdot \underline{n} = 3 - 1 = 2 = \dim P,$$

$$\text{tr}(\underline{n} \otimes \underline{n}) = \underline{n} \cdot \underline{n} = 1 = \dim \mathcal{N}.$$

Finally $\mathbb{P}_{(\underline{n})}$ is singular, since it kills the nonzero vector $\underline{n}$:

$$\mathbb{P}_{(\underline{n})}\underline{n} = \underline{n} - (\underline{n} \otimes \underline{n})\underline{n} = \underline{n} - (\underline{n} \cdot \underline{n})\underline{n} = \underline{n} - \underline{n} = \underline{0},$$

$$\underline{n} \neq \underline{0} \implies \det \mathbb{P}_{(\underline{n})} = 0,$$

so no projection other than $\underline{I}$ is invertible.

(c) The mirror image must leave the part lying in the plane alone and send the normal part to its negative. In the notation of $(\star_2)$ this says $\mathbb{R}_{(\underline{n})}\underline{v} = \underline{v}_p - \underline{v}_n$, and subtracting the normal part once to reach the plane and once more to pass through it produces the factor 2:

$$\mathbb{R}_{(\underline{n})}\underline{v} = \underline{v}_p - \underline{v}_n = [\underline{v} - (\underline{v} \cdot \underline{n})\underline{n}] - (\underline{v} \cdot \underline{n})\underline{n} = \underline{v} - 2(\underline{v} \cdot \underline{n})\underline{n} = [\underline{I} - 2\underline{n} \otimes \underline{n}]\underline{v}, \quad (1.1)$$

valid for every $\underline{v}$, whence

$$\mathbb{R}_{(\underline{n})} = \underline{I} - 2\underline{n} \otimes \underline{n} = \mathbb{P}_{(\underline{n})} - \underline{n} \otimes \underline{n}, \quad (\mathbb{R}_{(\underline{n})})_{ij} = \delta_{ij} - 2n_i n_j. \quad (\star_5)$$

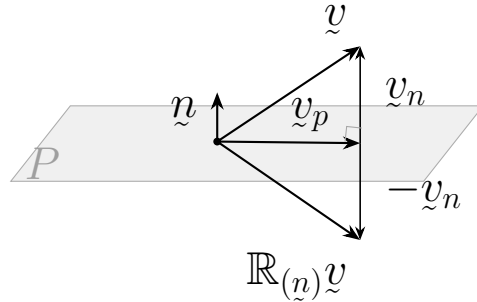


FIGURE P10.3. $\mathbb{R}_{(\underline{n})}\underline{v} = \underline{v}_p - \underline{v}_n$; the normal part is removed twice, giving the factor 2 in

$$\mathbb{R}_{(\underline{n})} = \underline{I} - 2\underline{n} \otimes \underline{n}.$$

Three properties can be established before any component is computed, and they will serve as checks on the two matrices below. Symmetry is inherited from $\underline{I}$ and $\underline{n} \otimes \underline{n}$, and the square collapses because the dyad is idempotent:

$$\mathbb{R}_{(\underline{n})}^t = \underline{I}^t - 2(\underline{n} \otimes \underline{n})^t = \underline{I} - 2\underline{n} \otimes \underline{n} = \mathbb{R}_{(\underline{n})},$$

$\mathbb{R}_{(\underline{n})}^2 = \underline{I} - 2 \underline{n} \otimes \underline{n} - 2 \underline{n} \otimes \underline{n} + 4(\underline{n} \otimes \underline{n})^2 = \underline{I} - 4 \underline{n} \otimes \underline{n} + 4(\underline{n} \otimes \underline{n})^2 \stackrel{(\star_4)}{=} \underline{I}$,
and the same in components,

$$\begin{aligned} [\mathbb{R}_{(\underline{n})}^2]_{ij} &= (\delta_{ik} - 2n_i n_k)(\delta_{kj} - 2n_k n_j) \\ &= \delta_{ij} - 2n_i n_j - 2n_i n_j + 4n_i(n_k n_k)n_j = \delta_{ij}, \end{aligned}$$

Thus $\mathbb{R}_{(\underline{n})}$ is its own inverse, $\mathbb{R}_{(\underline{n})}^{-1} = \mathbb{R}_{(\underline{n})}$: reflecting twice restores the original, as it must. Combining this with symmetry gives $\mathbb{R}_{(\underline{n})}^t \mathbb{R}_{(\underline{n})} = \mathbb{R}_{(\underline{n})}^2 = \underline{I}$, so $\mathbb{R}_{(\underline{n})}$ is orthogonal and therefore preserves every inner product, hence every length and angle:

$$\mathbb{R}_{(\underline{n})} \underline{u} \cdot \mathbb{R}_{(\underline{n})} \underline{v} = \underline{u} \cdot \mathbb{R}_{(\underline{n})}^t \mathbb{R}_{(\underline{n})} \underline{v} = \underline{u} \cdot \underline{I} \underline{v} = \underline{u} \cdot \underline{v}.$$

For the determinant, write $\mathbb{R}_{(\underline{n})} = \underline{I} + \underline{a} \otimes \underline{b}$ with $\underline{a} = -2\underline{n}$ and $\underline{b} = \underline{n}$; Problem 21(f) states $\det(\underline{I} + \underline{a} \otimes \underline{b}) = 1 + \underline{a} \cdot \underline{b}$, so

$$\det \mathbb{R}_{(\underline{n})} = 1 + (-2\underline{n}) \cdot \underline{n} = 1 - 2 \underline{n} \cdot \underline{n} = 1 - 2 = -1,$$

the negative sign saying that a reflection reverses orientation, in contrast to a rotation. The eigenvalues follow from the geometry: the normal is reversed and the plane is fixed,

$$\begin{aligned} \mathbb{R}_{(\underline{n})} \underline{n} &= \underline{n} - 2(\underline{n} \cdot \underline{n})\underline{n} = \underline{n} - 2\underline{n} = -\underline{n}, \\ \mathbb{R}_{(\underline{n})} \underline{w} &= \underline{w} - 2(\underline{w} \cdot \underline{n})\underline{n} = \underline{w} \quad (\underline{w} \in P), \end{aligned}$$

so $\lambda = -1$ once and $\lambda = +1$ twice, since $\dim P = 2$. This is confirmed by the characteristic polynomial: with

$$I_1 = \operatorname{tr} \mathbb{R}_{(\underline{n})} = \operatorname{tr} \underline{I} - 2 \operatorname{tr}(\underline{n} \otimes \underline{n}) = 3 - 2 \underline{n} \cdot \underline{n} = 1,$$

$$I_2 = \frac{1}{2} [I_1^2 - \operatorname{tr}(\mathbb{R}_{(\underline{n})}^2)] = \frac{1}{2} [1 - \operatorname{tr} \underline{I}] = \frac{1}{2} [1 - 3] = -1, \quad I_3 = \det \mathbb{R}_{(\underline{n})} = -1,$$

Problem 23 gives

$$\det(\mathbb{R}_{(\underline{n})} - \lambda \underline{I}) = -\lambda^3 + I_1 \lambda^2 - I_2 \lambda + I_3 = -\lambda^3 + \lambda^2 + \lambda - 1 = -(\lambda - 1)^2(\lambda + 1),$$

whose roots are $+1, +1, -1$ as expected.

(i) $\underline{n} = \underline{e}_1$. Here $n_i = \underline{e}_1 \cdot \underline{e}_i = \delta_{i1}$, so $n_1 = 1$ and $n_2 = n_3 = 0$, and the outer product has entries $n_i n_j = \delta_{i1} \delta_{j1}$, which vanish unless $i = j = 1$:

$$(n_i n_j) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Substituting into $(\star_5)$ and subtracting entry by entry,

$$(R_{ij}) = \delta_{ij} - 2n_i n_j = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} - 2 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Reassembling the tensor from its components, $\mathbb{R}_{(\underline{e}_1)} = R_{ij} \underline{e}_i \otimes \underline{e}_j$, the only surviving terms being the diagonal ones,

$$\mathbb{R}_{(\underline{e}_1)} = -\underline{e}_1 \otimes \underline{e}_1 + \underline{e}_2 \otimes \underline{e}_2 + \underline{e}_3 \otimes \underline{e}_3.$$

Its trace is $R_{11} + R_{22} + R_{33} = -1 + 1 + 1 = 1$ and, the matrix being diagonal, its determinant is the product of the diagonal entries, $(-1)(1)(1) = -1$, matching the general results.

(ii) $\underline{n} = \frac{\sqrt{2}}{2}(\underline{e}_1 + \underline{e}_2)$. First confirm the hypothesis $\underline{n} \cdot \underline{n} = 1$:

$$|\underline{n}|^2 = \left(\frac{\sqrt{2}}{2}\right)^2 (1^2 + 1^2 + 0^2) = \frac{1}{2} \cdot 2 = 1.$$

The components are $n_1 = n_2 = \frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}}$ and $n_3 = 0$, so the nonzero products are

$$n_1 n_1 = n_1 n_2 = n_2 n_1 = n_2 n_2 = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = \frac{1}{2},$$

every product involving the index 3 vanishing because $n_3 = 0$. Hence

$$(n_i n_j) = \frac{1}{2} \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad 2(n_i n_j) = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

and $(\star_5)$ gives, subtracting entry by entry,

$$(R_{ij}) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} - \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Reassembling $\mathbb{R}_{(\underline{n})} = R_{ij}\underline{e}_i \otimes \underline{e}_j$ with the three nonzero entries $R_{12} = R_{21} = -1$ and $R_{33} = 1$,

$$\mathbb{R}_{(\underline{n})} = -\underline{e}_1 \otimes \underline{e}_2 - \underline{e}_2 \otimes \underline{e}_1 + \underline{e}_3 \otimes \underline{e}_3.$$

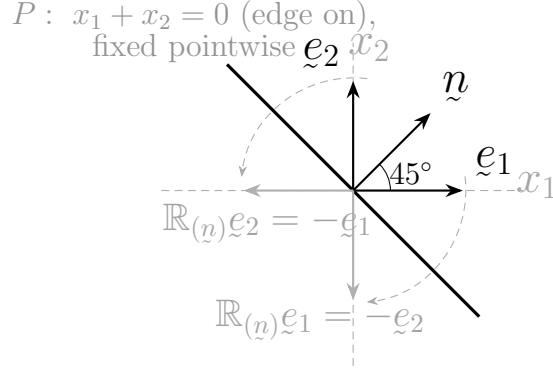


FIGURE P10.4. Case (ii): the mirror $x_1 + x_2 = 0$ seen edge on, normal at 45° . The axes are swapped and negated, which is why (R_{ij}) has zero diagonal and -1 off it.

It remains to verify that this matrix does what the general theory predicted. Compute the action on each basis vector from $\mathbb{R}_{(\underline{n})}\underline{e}_j = R_{ij}\underline{e}_i$, that is by reading the j th column and multiplying it against $\{\underline{e}_i\}$:

$$\begin{aligned}\mathbb{R}_{(\underline{n})}\underline{e}_1 &= R_{11}\underline{e}_1 + R_{21}\underline{e}_2 + R_{31}\underline{e}_3 = 0\underline{e}_1 + (-1)\underline{e}_2 + 0\underline{e}_3 = -\underline{e}_2, \\ \mathbb{R}_{(\underline{n})}\underline{e}_2 &= R_{12}\underline{e}_1 + R_{22}\underline{e}_2 + R_{32}\underline{e}_3 = (-1)\underline{e}_1 + 0\underline{e}_2 + 0\underline{e}_3 = -\underline{e}_1, \\ \mathbb{R}_{(\underline{n})}\underline{e}_3 &= R_{13}\underline{e}_1 + R_{23}\underline{e}_2 + R_{33}\underline{e}_3 = 0\underline{e}_1 + 0\underline{e}_2 + 1\underline{e}_3 = \underline{e}_3.\end{aligned}$$

The normal is reversed. Using linearity and the three images just computed,

$$\mathbb{R}_{(\underline{n})}\underline{n} = \frac{\sqrt{2}}{2}(\mathbb{R}_{(\underline{n})}\underline{e}_1 + \mathbb{R}_{(\underline{n})}\underline{e}_2) = \frac{\sqrt{2}}{2}(-\underline{e}_2 - \underline{e}_1) = -\frac{\sqrt{2}}{2}(\underline{e}_1 + \underline{e}_2) = -\underline{n}.$$

The plane is fixed pointwise. A basis of P is $\{\underline{e}_1 - \underline{e}_2, \underline{e}_3\}$; both really lie in P , since

$$(\underline{e}_1 - \underline{e}_2) \cdot \underline{n} = \frac{\sqrt{2}}{2}(1 - 1) = 0, \quad \underline{e}_3 \cdot \underline{n} = \frac{\sqrt{2}}{2}(0 + 0) = 0,$$

and both are fixed,

$$\mathbb{R}_{(\underline{n})}(\underline{e}_1 - \underline{e}_2) = \mathbb{R}_{(\underline{n})}\underline{e}_1 - \mathbb{R}_{(\underline{n})}\underline{e}_2 = -\underline{e}_2 - (-\underline{e}_1) = \underline{e}_1 - \underline{e}_2, \quad \mathbb{R}_{(\underline{n})}\underline{e}_3 = \underline{e}_3.$$

So the eigenvalues really are -1 once and $+1$ twice. *The involution, checked on the matrix.* Multiplying R by itself and writing out all nine entries,

$$\begin{aligned} R^2 &= \begin{pmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} (0)(0) + (-1)(-1) + 0 & (0)(-1) + (-1)(0) + 0 & 0 \\ (-1)(0) + (0)(-1) + 0 & (-1)(-1) + (0)(0) + 0 & 0 \\ 0 & 0 & (1)(1) \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \end{aligned}$$

so $R^2 = I$, and since $R^t = R$ this also confirms $R^t R = I$ entry by entry. *Trace and determinant.* Summing the diagonal, $\text{tr } R = R_{11} + R_{22} + R_{33} = 0 + 0 + 1 = 1$; expanding the determinant along the third row, whose only nonzero entry is $R_{33} = 1$,

$$\det R = 1 \cdot \det \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix} = 1 \cdot [(0)(0) - (-1)(-1)] = -1.$$

Both agree with $I_1 = 1$ and $I_3 = -1$ found from $(\star_5)$, and with the spectrum $\{-1, +1, +1\}$. $\square$

PROBLEM 11 (Triple products; two induced tensors). (a) *Prove $\underline{a} \times (\underline{b} \times \underline{c}) = (\underline{a} \cdot \underline{c})\underline{b} - (\underline{a} \cdot \underline{b})\underline{c}$ and $\underline{v} = (\underline{v} \cdot \underline{e})\underline{e} + \underline{e} \times (\underline{v} \times \underline{e})$ ($\underline{e}$ unit); (b) tensor for $f(\underline{v}) = \underline{e} \times (\underline{v} \times \underline{e})$; (c) tensor for $f(\underline{v}) = \underline{v} + \underline{a} \times \underline{v}$.*

Solution. **(a)** Write the inner cross product first, then the outer one:

$$(\underline{b} \times \underline{c})_k = \varepsilon_{klm} b_l c_m, \quad [\underline{a} \times (\underline{b} \times \underline{c})]_i = \varepsilon_{ijk} a_j (\underline{b} \times \underline{c})_k = \varepsilon_{ijk} \varepsilon_{klm} a_j b_l c_m.$$

Contract the two symbols. Theorem 1.29 contracts on the last index, and $\varepsilon_{ijk} = \varepsilon_{kij}$ by cyclic shift, so

$$\varepsilon_{ijk} \varepsilon_{klm} = \varepsilon_{kij} \varepsilon_{klm} = \delta_{il} \delta_{jm} - \delta_{im} \delta_{jl}.$$

Substituting and letting the deltas collapse the sums,

$$[\underline{a} \times (\underline{b} \times \underline{c})]_i = (\delta_{il} \delta_{jm} - \delta_{im} \delta_{jl}) a_j b_l c_m = a_j c_j b_i - a_j b_j c_i = (\underline{a} \cdot \underline{c}) b_i - (\underline{a} \cdot \underline{b}) c_i.$$

Now put $\underline{a} = \underline{c} = \underline{e}$ with $|\underline{e}| = 1$, so $\underline{e} \cdot \underline{e} = 1$:

$$\underline{e} \times (\underline{v} \times \underline{e}) = (\underline{e} \cdot \underline{e})\underline{v} - (\underline{e} \cdot \underline{v})\underline{e} = \underline{v} - (\underline{v} \cdot \underline{e})\underline{e},$$

which rearranges to $\underline{v} = (\underline{v} \cdot \underline{e})\underline{e} + \underline{e} \times (\underline{v} \times \underline{e})$.

(b) From (a), $f(\underline{v}) = \underline{v} - (\underline{e} \cdot \underline{v})\underline{e}$. Both terms are linear in $\underline{v}$, and $(\underline{e} \cdot \underline{v})\underline{e} = (\underline{e} \otimes \underline{e})\underline{v}$, so

$$f(\underline{v}) = (\underline{I} - \underline{e} \otimes \underline{e})\underline{v}, \quad \underline{A} = \underline{I} - \underline{e} \otimes \underline{e} = \mathbb{P}_{(\underline{e})},$$

the projection of Problem 10 onto the plane perpendicular to $\underline{e}$. Check: $f(\underline{e}) = \underline{e} - \underline{e} = \underline{0}$ and $f(\underline{w}) = \underline{w}$ for $\underline{w} \perp \underline{e}$.

(c) $f(\underline{v}) = \underline{v} + \underline{a} \times \underline{v}$. In components, $(\underline{a} \times \underline{v})_i = \varepsilon_{ijk}a_jv_k$, so the second term is $\underline{\Omega}\underline{v}$ with $\Omega_{ik} = \varepsilon_{ijk}a_j = -\varepsilon_{ikj}a_j$, skew by Problem 16. Hence

$$\underline{A} = \underline{I} + \underline{\Omega}, \quad A_{ij} = \delta_{ij} - \varepsilon_{ijk}a_k.$$

Linearity of f is then visible, and $\det \underline{A} = 1 + |\underline{a}|^2 \neq 0$ by Problem 21(g), so $\underline{A}$ is invertible for every $\underline{a}$. $\square$

PROBLEM 12 (Two-point tensors). *For orthonormal bases $\{\underline{e}_i\}$ of $\mathbb{E}^3$ and $\{\hat{\underline{E}}_A\}$ of $\hat{\mathbb{E}}^3$, show a linear $\underline{A} : \hat{\mathbb{E}}^3 \rightarrow \mathbb{E}^3$ is $\underline{A} = A_{iB}\underline{e}_i \otimes \hat{\underline{E}}_B$ with $A_{iB} = \underline{e}_i \cdot \underline{A}\hat{\underline{E}}_B$.*

Solution. Step 1. $\underline{A}\hat{\underline{E}}_B \in \mathbb{E}^3$, so resolve it on the spatial basis and name the coefficients:

$$\underline{A}\hat{\underline{E}}_B = (\underline{e}_i \cdot \underline{A}\hat{\underline{E}}_B)\underline{e}_i =: A_{iB}\underline{e}_i.$$

The dot product is legal, both arguments being in $\mathbb{E}^3$.

Step 2. For $\underline{V} = V_B\hat{\underline{E}}_B \in \hat{\mathbb{E}}^3$, by linearity,

$$\underline{A}\underline{V} = V_B \underline{A}\hat{\underline{E}}_B = A_{iB}V_B\underline{e}_i.$$

Step 3. The mixed dyad acts by $(\underline{a} \otimes \underline{b})\underline{c} = (\underline{b} \cdot \underline{c})\underline{a}$:

$$(\underline{e}_i \otimes \hat{\underline{E}}_B)\underline{V} = (\hat{\underline{E}}_B \cdot \underline{V})\underline{e}_i = V_B\underline{e}_i,$$

legal since $\hat{E}_B, V \in \hat{\mathbb{E}}^3$ while $e_i \in \mathbb{E}^3$. Hence $(A_{iB}e_i \otimes \hat{E}_B)V = A_{iB}V_B e_i = \hat{A}V$ for all V , and

$$\boxed{\hat{A} = A_{iB} e_i \otimes \hat{E}_B, \quad A_{iB} = e_i \cdot \hat{A} \hat{E}_B.}$$

Step 4, consistency. Feed $\hat{E}_C$ to the dyadic form, then dot with e_j :

$$(A_{iB}e_i \otimes \hat{E}_B)\hat{E}_C = A_{iB}\delta_{BC}e_i = A_{iC}e_i, \quad e_j \cdot \hat{A}\hat{E}_C = A_{iC}\delta_{ji} = A_{jC}.$$

□

REMARK 1.105 (which dots are legal).

$$\begin{aligned} e_i \cdot \hat{A} \hat{E}_B & \text{ defined, both in } \mathbb{E}^3, \\ \hat{E}_B \cdot V & \text{ defined, both in } \hat{\mathbb{E}}^3, \\ e_i \cdot \hat{E}_B & \text{ not defined,} \\ e_i \otimes \hat{E}_B & \text{ defined, a map } \hat{\mathbb{E}}^3 \rightarrow \mathbb{E}^3. \end{aligned}$$

Also $\hat{A}^t = A_{iB}\hat{E}_B \otimes e_i$ maps $\mathbb{E}^3 \rightarrow \hat{\mathbb{E}}^3$, so $\hat{A} + \hat{A}^t$ is meaningless: a two-point tensor is neither symmetric nor skew. The object built here is $\hat{F} = F_{iA}e_i \otimes \hat{E}_A$ of (2.39).

PROBLEM 13 (Riesz representation for tensors). *Show a linear $g : \text{Lin} \rightarrow \mathbb{R}$ is $g(V) = \hat{A} \cdot V$ for a unique $\hat{A}$.*

Solution. Existence. Define the nine numbers

$$A_{ij} := g(e_i \otimes e_j), \quad \hat{A} := A_{ij} e_i \otimes e_j.$$

Resolve an arbitrary V on the same basis, $V = V_{ij}e_i \otimes e_j$ with $V_{ij} = e_i \cdot V e_j$. Since g is linear and the V_{ij} are scalars,

$$g(V) = g(V_{ij}e_i \otimes e_j) = V_{ij}g(e_i \otimes e_j) = V_{ij}A_{ij} = \hat{A} \cdot V,$$

the last step being the definition $\hat{A} \cdot V = A_{ij}V_{ij}$ of the tensor inner product.

Uniqueness. If $\underline{A} \cdot \underline{V} = \underline{A}' \cdot \underline{V}$ for every $\underline{V}$, then $(\underline{A} - \underline{A}') \cdot \underline{V} = 0$ for every $\underline{V}$. Choose $\underline{V} = \underline{A} - \underline{A}'$:

$$0 = (\underline{A} - \underline{A}') \cdot (\underline{A} - \underline{A}') = |\underline{A} - \underline{A}'|^2 \implies \underline{A} = \underline{A}'.$$

This is the tensor version of Problem 13's vector counterpart, and the proof is the same one twice: linearity gives existence through a basis, positive definiteness of the inner product gives uniqueness. $\square$

PROBLEM 14 (Symmetric-skew decomposition). $\underline{A} = \underline{S} + \underline{W}$, $\underline{S} = \frac{1}{2}(\underline{A} + \underline{A}^t)$, $\underline{W} = \frac{1}{2}(\underline{A} - \underline{A}^t)$. (a) *uniqueness*; (b) *Sym, Skw are orthogonal subspaces*, $\mathbb{E}^9 = \text{Sym} \oplus \text{Skw}$.

Solution. Existence. With $\underline{S} = \frac{1}{2}(\underline{A} + \underline{A}^t)$ and $\underline{W} = \frac{1}{2}(\underline{A} - \underline{A}^t)$,
 $\underline{S} + \underline{W} = \frac{1}{2}(\underline{A} + \underline{A}^t) + \frac{1}{2}(\underline{A} - \underline{A}^t) = \frac{1}{2}(\underline{A} + \underline{A}^t + \underline{A} - \underline{A}^t) = \frac{1}{2}(2\underline{A}) = \underline{A}$,
and, by $(\underline{A} + \underline{B})^t = \underline{A}^t + \underline{B}^t$ and $(\underline{A}^t)^t = \underline{A}$,

$$\begin{aligned}\underline{S}^t &= \frac{1}{2}(\underline{A}^t + (\underline{A}^t)^t) = \frac{1}{2}(\underline{A}^t + \underline{A}) = \underline{S}, \\ \underline{W}^t &= \frac{1}{2}(\underline{A}^t - \underline{A}) = -\frac{1}{2}(\underline{A} - \underline{A}^t) = -\underline{W}.\end{aligned}$$

(a) Uniqueness. Let $\underline{A} = \underline{S}' + \underline{W}'$ with $(\underline{S}')^t = \underline{S}'$, $(\underline{W}')^t = -\underline{W}'$.
Transpose:

$$\underline{A}^t = (\underline{S}' + \underline{W}')^t = (\underline{S}')^t + (\underline{W}')^t = \underline{S}' - \underline{W}'.$$

Now add and subtract this from $\underline{A} = \underline{S}' + \underline{W}'$:

$$\begin{aligned}\underline{A} + \underline{A}^t &= 2\underline{S}' \implies \underline{S}' = \frac{1}{2}(\underline{A} + \underline{A}^t) = \underline{S}, \\ \underline{A} - \underline{A}^t &= 2\underline{W}' \implies \underline{W}' = \frac{1}{2}(\underline{A} - \underline{A}^t) = \underline{W}.\end{aligned}$$

(b) Orthogonal subspaces. Each is a subspace: $(\alpha \underline{S}_1 + \beta \underline{S}_2)^t = \alpha \underline{S}_1 + \beta \underline{S}_2$, and the same with a minus sign for Skw. For orthogonality, with $\underline{S} \cdot \underline{W} = S_{ij}W_{ij}$,

$$\underline{S} \cdot \underline{W} = S_{ij}W_{ij} \stackrel{(1)}{=} S_{ji}W_{ij} \stackrel{(2)}{=} S_{ij}W_{ji} \stackrel{(3)}{=} -S_{ij}W_{ij} = -\underline{S} \cdot \underline{W},$$

with (1) $S_{ij} = S_{ji}$, (2) a relabelling, (3) $W_{ji} = -W_{ij}$. Hence $\underline{S} \cdot \underline{W} = 0$.

Dimensions. $\underline{S}$ is fixed by $S_{11}, S_{22}, S_{33}, S_{12}, S_{13}, S_{23}$, so $\dim \text{Sym} = 6$. For $\underline{W}$, $W_{ii} = -W_{ii}$ gives a zero diagonal, leaving W_{12}, W_{13}, W_{23} , so $\dim \text{Skw} = 3$. Their sum is 9, and they meet only in $\underline{Q}$, since $\underline{A} = \underline{A}^t = -\underline{A}$ forces $2\underline{A} = \underline{Q}$. Therefore

$$\mathbb{E}^9 = \text{Sym} \oplus \text{Skw}.$$

□

REMARK 1.106 (three uses). (i) $\underline{a} \cdot \underline{A}\underline{a} = \underline{a} \cdot \underline{S}\underline{a}$ by Problem 6(a), so positive definiteness (3.13) sees only $\underline{S}$. (ii) Chapter 4 splits $\underline{L} = \nabla \underline{v} = \underline{D} + \underline{W}$ into stretching and spin; part (a) is what makes them unique. (iii) By Problem 16 every skew $\underline{W}$ is $\underline{W}\underline{u} = \underline{\omega} \times \underline{u}$, which is why $\dim \text{Skw} = 3$.

Expanding $\underline{A} \cdot \underline{A} = (\underline{S} + \underline{W}) \cdot (\underline{S} + \underline{W})$ and killing the cross terms by (b) gives $|\underline{A}|^2 = |\underline{S}|^2 + |\underline{W}|^2$: the splitting is an orthogonal projection, not merely a decomposition.

PROBLEM 15 (Spherical–deviatoric decomposition). $\underline{T} = \underline{A} + \alpha \underline{I}$, $\text{tr } \underline{A} = 0$, $3\alpha = \text{tr } \underline{T}$. (a) *uniqueness*; (b) *Sph, Dev orthogonal*, $\mathbb{E}^9 = \text{Sph} \oplus \text{Dev}$.

Solution. Existence. Put $\alpha = \frac{1}{3} \text{tr } \underline{T}$ and $\underline{A} = \underline{T} - \alpha \underline{I}$. Then $\underline{T} = \underline{A} + \alpha \underline{I}$ and, using $\text{tr } \underline{I} = \delta_{ii} = 3$,

$$\text{tr } \underline{A} = \text{tr } \underline{T} - \alpha \text{tr } \underline{I} = \text{tr } \underline{T} - \frac{1}{3}(\text{tr } \underline{T})(3) = 0.$$

(a) Uniqueness. Let $\underline{T} = \underline{B} + \beta \underline{I}$ with $\text{tr } \underline{B} = 0$. Take the trace:

$$\text{tr } \underline{T} = \text{tr } \underline{B} + \beta \text{tr } \underline{I} = 0 + 3\beta \implies \beta = \frac{1}{3} \text{tr } \underline{T} = \alpha,$$

and then $\underline{B} = \underline{T} - \beta \underline{I} = \underline{T} - \alpha \underline{I} = \underline{A}$.

(b) Orthogonal subspaces. Both are subspaces: *Sph* is the span of $\underline{I}$, and tr is linear so $\text{Dev} = \{\underline{D} : \text{tr } \underline{D} = 0\}$ is its kernel. For $\lambda \underline{I} \in \text{Sph}$ and $\underline{D} \in \text{Dev}$, using $\underline{I} \cdot \underline{D} = \delta_{ij} D_{ij} = D_{ii}$,

$$(\lambda \underline{I}) \cdot \underline{D} = \lambda \text{tr } \underline{D} = \lambda \cdot 0 = 0.$$

Dimensions: $\dim \text{Sph} = 1$; $\text{tr} : \mathbb{E}^9 \rightarrow \mathbb{R}$ is onto, so its kernel has $\dim \text{Dev} = 9 - 1 = 8$. Since $1 + 8 = 9$ and $\text{Sph} \cap \text{Dev} = \{\underline{\underline{O}}\}$, because $\lambda \underline{\underline{I}} \in \text{Dev}$ forces $3\lambda = 0$,

$$\mathbb{E}^9 = \text{Sph} \oplus \text{Dev}.$$

□

REMARK 1.107 (the two splittings are independent). Problems 14 and 15 split $\mathbb{E}^9$ two different ways, $\text{Sym} \oplus \text{Skw}$ and $\text{Sph} \oplus \text{Dev}$, and the two are compatible rather than competing: $\text{Sph} \subset \text{Sym}$, so a single tensor decomposes as

$$\underline{\underline{T}} = \underbrace{\alpha \underline{\underline{I}}}_1 + \underbrace{(\text{Sym } \underline{\underline{T}} - \alpha \underline{\underline{I}})}_5 + \underbrace{\text{Skw } \underline{\underline{T}}}_3, \quad \alpha = \frac{1}{3} \text{tr } \underline{\underline{T}},$$

with dimensions $1 + 5 + 3 = 9$. In Chapter 3 the middle piece is what carries distortion at constant volume and the first is what carries volume change: there $\text{tr}(\ln \underline{\underline{U}}) = \ln J$ exactly, so the spherical part of the Hencky strain is $\frac{1}{3}(\ln J)\underline{\underline{I}}$.

PROBLEM 16 (Axial vector of a skew tensor). $f(\underline{\underline{v}}) = \underline{\underline{\omega}} \times \underline{\underline{v}} = \underline{\underline{\Omega}}\underline{\underline{v}}$. (a) $\Omega_{ij} = \varepsilon_{jik}\omega_k$, so $\underline{\underline{\Omega}} \in \text{Skw}$; (b) $\omega_k = \frac{1}{2}\Omega_{ij}\varepsilon_{jik}$.

Solution. (a) Write both sides of $\underline{\underline{\Omega}}\underline{\underline{v}} = \underline{\underline{\omega}} \times \underline{\underline{v}}$ in components:

$$(\underline{\underline{\Omega}}\underline{\underline{v}})_i = \Omega_{ij}v_j, \quad (\underline{\underline{\omega}} \times \underline{\underline{v}})_i = \varepsilon_{ikj}\omega_k v_j,$$

the second by Proposition 1.24 with the summation indices named so that v carries j in both. Equating for every v ,

$$\Omega_{ij} = \varepsilon_{ikj}\omega_k.$$

Now $\varepsilon_{ikj} = -\varepsilon_{ijk} = \varepsilon_{jik}$: one swap gives a minus, a second swap gives it back. So $\Omega_{ij} = \varepsilon_{jik}\omega_k$, and interchanging i, j ,

$$\Omega_{ji} = \varepsilon_{ijk}\omega_k = -\varepsilon_{jik}\omega_k = -\Omega_{ij},$$

i.e. $\underline{\underline{\Omega}} \in \text{Skw}$.

(b) Substitute $\Omega_{ij} = \varepsilon_{jil}\omega_l$ and contract:

$$\frac{1}{2} \Omega_{ij} \varepsilon_{jik} = \frac{1}{2} \varepsilon_{jil} \varepsilon_{jik} \omega_l = \frac{1}{2} (2\delta_{lk}) \omega_l = \omega_k,$$

the middle step by Theorem 1.29: $\varepsilon_{jil} \varepsilon_{jik} = \varepsilon_{ilj} \varepsilon_{ikj} = 2\delta_{lk}$ after a cyclic shift brings the repeated pair to the front.

So $\omega \mapsto \Omega$ and $\Omega \mapsto \omega$ invert one another: Skw and $\mathbb{E}^3$ are in one-to-one correspondence, which is why $\dim \text{Skw} = 3$ in Problem 14. $\square$

PROBLEM 17 (Identities for the tensor product). *Verify (a) $(\underline{a} \otimes \underline{b})^t = \underline{b} \otimes \underline{a}$; (b) $\underline{A}(\underline{a} \otimes \underline{b}) = (\underline{A}\underline{a}) \otimes \underline{b}$; (c) $(\underline{a} \otimes \underline{b})\underline{A} = \underline{a} \otimes (\underline{A}^t \underline{b})$; (d) $\underline{A} = (\underline{A}\underline{e}_i) \otimes \underline{e}_i = \underline{e}_i \otimes (\underline{A}^t \underline{e}_i)$.*

Solution. Throughout, the definition is $(\underline{a} \otimes \underline{b})\underline{c} = (\underline{b} \cdot \underline{c})\underline{a}$: the second slot dots, the first survives.

(a) Test against arbitrary $\underline{u}, \underline{v}$ and use the symmetry of the dot product:

$$\underline{u} \cdot (\underline{a} \otimes \underline{b})\underline{v} = (\underline{b} \cdot \underline{v})(\underline{a} \cdot \underline{u}) = (\underline{a} \cdot \underline{u})(\underline{b} \cdot \underline{v}) = \underline{v} \cdot (\underline{b} \otimes \underline{a})\underline{u}.$$

By the definition of the transpose this says $(\underline{a} \otimes \underline{b})^t = \underline{b} \otimes \underline{a}$.

(b) Apply both sides to arbitrary $\underline{v}$ and use linearity of $\underline{A}$:

$$[\underline{A}(\underline{a} \otimes \underline{b})]\underline{v} = \underline{A}[(\underline{b} \cdot \underline{v})\underline{a}] = (\underline{b} \cdot \underline{v}) \underline{A}\underline{a} = [(\underline{A}\underline{a}) \otimes \underline{b}]\underline{v}.$$

(c) Same test, now moving $\underline{A}$ across the dot product:

$$[(\underline{a} \otimes \underline{b})\underline{A}]\underline{v} = (\underline{b} \cdot \underline{A}\underline{v}) \underline{a} = (\underline{A}^t \underline{b} \cdot \underline{v}) \underline{a} = [\underline{a} \otimes (\underline{A}^t \underline{b})]\underline{v}.$$

So post-multiplication touches only the second slot, pre-multiplication only the first.

(d) With $\underline{A} = A_{ij}\underline{e}_i \otimes \underline{e}_j$ one has $\underline{A}\underline{e}_i = A_{ki}\underline{e}_k$ and $\underline{A}^t \underline{e}_i = A_{ik}\underline{e}_k$. Hence, summing on i ,

$$(\underline{A}\underline{e}_i) \otimes \underline{e}_i = A_{ki} \underline{e}_k \otimes \underline{e}_i = \underline{A}, \quad \underline{e}_i \otimes (\underline{A}^t \underline{e}_i) = A_{ik} \underline{e}_i \otimes \underline{e}_k = \underline{A},$$

the second after renaming the two dummy indices. Both are (3.11) with $\underline{A}$ inserted on one side or the other. $\square$

PROBLEM 18 (Transpose of the inverse). *Show $(\underline{A}^t)^{-1} = (\underline{A}^{-1})^t =: \underline{A}^{-t}$.*

Solution. Transpose the two defining relations of $\underline{A}^{-1}$, using $(\underline{B}\underline{C})^t = \underline{C}^t \underline{B}^t$ from Problem 20 and $\underline{I}^t = \underline{I}$:

$$\underline{A}\underline{A}^{-1} = \underline{I} \implies (\underline{A}^{-1})^t \underline{A}^t = \underline{I}, \quad \underline{A}^{-1} \underline{A} = \underline{I} \implies \underline{A}^t (\underline{A}^{-1})^t = \underline{I}.$$

So $(\underline{A}^{-1})^t$ is a two-sided inverse of $\underline{A}^t$, and inverses are unique: if $\underline{P}\underline{A}^t = \underline{I} = \underline{A}^t \underline{Q}$ then $\underline{P} = \underline{P}(\underline{A}^t \underline{Q}) = (\underline{P}\underline{A}^t) \underline{Q} = \underline{Q}$. Hence $(\underline{A}^t)^{-1} = (\underline{A}^{-1})^t$, and the common value may be written $\underline{A}^{-t}$ without ambiguity. $\square$

PROBLEM 19 (Inverse of a product). $(\underline{B}\underline{A})^{-1} = \underline{A}^{-1} \underline{B}^{-1}$.

Solution. Multiply both ways, cancelling the inner pair first:

$$(\underline{A}^{-1} \underline{B}^{-1})(\underline{B}\underline{A}) = \underline{A}^{-1}(\underline{B}^{-1} \underline{B})\underline{A} = \underline{A}^{-1} \underline{I} \underline{A} = \underline{A}^{-1} \underline{A} = \underline{I},$$

$$(\underline{B}\underline{A})(\underline{A}^{-1} \underline{B}^{-1}) = \underline{B}(\underline{A}\underline{A}^{-1})\underline{B}^{-1} = \underline{B} \underline{I} \underline{B}^{-1} = \underline{B} \underline{B}^{-1} = \underline{I}.$$

By uniqueness of inverses, $(\underline{B}\underline{A})^{-1} = \underline{A}^{-1} \underline{B}^{-1}$. The order reverses, as it must: the last map applied is the first to be undone. $\square$

PROBLEM 20 (Transpose of a product). $(\underline{B}\underline{A})^t = \underline{A}^t \underline{B}^t$.

Solution. Move the two factors across the dot product one at a time, each by the definition $\underline{u} \cdot \underline{C}\underline{v} = \underline{C}^t \underline{u} \cdot \underline{v}$:

$$\underline{u} \cdot (\underline{B}\underline{A})\underline{v} = \underline{u} \cdot \underline{B}(\underline{A}\underline{v}) = \underline{B}^t \underline{u} \cdot \underline{A}\underline{v} = \underline{A}^t \underline{B}^t \underline{u} \cdot \underline{v} = \underline{u} \cdot (\underline{A}^t \underline{B}^t) \underline{v}.$$

Comparing the first and last for all $\underline{u}, \underline{v}$, and using that a tensor is determined by its bilinear form, $(\underline{B}\underline{A})^t = \underline{A}^t \underline{B}^t$. Again the order reverses. $\square$

PROBLEM 21 (Determinant identities via box products). *Verify (a) $\det(\alpha \underline{A}) = \alpha^3 \det \underline{A}$; (b) $\det(\underline{A}\underline{B}) = \det \underline{A} \det \underline{B}$; (c) $\det \underline{I} = 1$; (d) $\det(\underline{A}^{-1}) = (\det \underline{A})^{-1}$; (e) $\det(\underline{a} \otimes \underline{b}) = 0$; (f) $\det(\underline{I} + \underline{a} \otimes \underline{b}) = 1 + \underline{a} \cdot \underline{b}$; (g) $\det(\underline{A} + \underline{a} \otimes \underline{b}) = \det \underline{A} (1 + \underline{a} \cdot \underline{A}^{-t} \underline{b})$.*

Solution. The master identity is $\det \underline{A} [\underline{a}, \underline{b}, \underline{c}] = [\underline{Aa}, \underline{Ab}, \underline{Ac}]$ for all $\underline{a}, \underline{b}, \underline{c}$ (Definition 1.49₃). Every part below follows from it.

(a)

$$\begin{aligned} \det(\alpha \underline{A}) [\underline{a}, \underline{b}, \underline{c}] &= [(\alpha \underline{A})\underline{a}, (\alpha \underline{A})\underline{b}, (\alpha \underline{A})\underline{c}] \\ &= \alpha^3 [\underline{Aa}, \underline{Ab}, \underline{Ac}] = \alpha^3 \det \underline{A} [\underline{a}, \underline{b}, \underline{c}]; \end{aligned}$$

cancel $[\underline{a}, \underline{b}, \underline{c}] \neq 0$.

(b)

$$\begin{aligned} [(\underline{AB})\underline{a}, (\underline{AB})\underline{b}, (\underline{AB})\underline{c}] &= [\underline{A}(\underline{B}\underline{a}), \underline{A}(\underline{B}\underline{b}), \underline{A}(\underline{B}\underline{c})] \\ &= \det \underline{A} [\underline{B}\underline{a}, \underline{B}\underline{b}, \underline{B}\underline{c}] \\ &= \det \underline{A} \det \underline{B} [\underline{a}, \underline{b}, \underline{c}]. \end{aligned}$$

(c) $\det \underline{I} [\underline{a}, \underline{b}, \underline{c}] = [\underline{Ia}, \underline{Ib}, \underline{Ic}] = [\underline{a}, \underline{b}, \underline{c}]$, so $\det \underline{I} = 1$.

(d) From (b) and (c), $\det \underline{A} \det(\underline{A}^{-1}) = \det(\underline{A}\underline{A}^{-1}) = \det \underline{I} = 1$.

(e) $(\underline{a} \otimes \underline{b})\underline{u} = (\underline{b} \cdot \underline{u})\underline{a}$: each image is a scalar multiple of $\underline{a}$. Hence $[(\underline{a} \otimes \underline{b})\underline{e}_1, (\underline{a} \otimes \underline{b})\underline{e}_2, (\underline{a} \otimes \underline{b})\underline{e}_3] = [b_1\underline{a}, b_2\underline{a}, b_3\underline{a}]$. A box product with two parallel entries vanishes (alternation), so $\det(\underline{a} \otimes \underline{b}) = 0$.

(f) Set $\underline{T} = \underline{I} + \underline{a} \otimes \underline{b}$. Then $\underline{T}\underline{e}_i = \underline{e}_i + b_i\underline{a}$, and

$$\det \underline{T} = [(\underline{e}_1 + b_1\underline{a}), (\underline{e}_2 + b_2\underline{a}), (\underline{e}_3 + b_3\underline{a})].$$

Expand multilinearly: the box product is trilinear, so there are $2^3 = 8$ terms, classified by how many of the three factors contribute $b_i\underline{a}$:

$$\begin{aligned} &\underbrace{[\underline{e}_1, \underline{e}_2, \underline{e}_3]}_{=1} + b_1[\underline{a}, \underline{e}_2, \underline{e}_3] + b_2[\underline{e}_1, \underline{a}, \underline{e}_3] + b_3[\underline{e}_1, \underline{e}_2, \underline{a}] \\ &+ b_1b_2\underbrace{[\underline{a}, \underline{a}, \underline{e}_3]}_{=0} + b_1b_3\underbrace{[\underline{a}, \underline{e}_2, \underline{a}]}_{=0} + b_2b_3\underbrace{[\underline{e}_1, \underline{a}, \underline{a}]}_{=0} + b_1b_2b_3\underbrace{[\underline{a}, \underline{a}, \underline{a}]}_{=0}. \end{aligned}$$

Every term with $\underline{a}$ appearing twice or thrice vanishes by alternation. The surviving terms are: $1 + b_1a_1 + b_2a_2 + b_3a_3 = 1 + b_i a_i = 1 + \underline{a} \cdot \underline{b}$.

(g) For invertible $\underline{A}$, factor: $\underline{A} + \underline{a} \otimes \underline{b} = \underline{A}(\underline{I} + \underline{A}^{-1}(\underline{a} \otimes \underline{b})) = \underline{A}(\underline{I} + (\underline{A}^{-1}\underline{a}) \otimes \underline{b})$, where Lemma 1.38 gives $\underline{A}^{-1}(\underline{a} \otimes \underline{b}) = (\underline{A}^{-1}\underline{a}) \otimes \underline{b}$. Apply (b), then (f):

$$\det(\underline{A} + \underline{a} \otimes \underline{b}) = \det \underline{A} \det(\underline{I} + (\underline{A}^{-1}\underline{a}) \otimes \underline{b}) = \det \underline{A} (1 + \underline{A}^{-1}\underline{a} \cdot \underline{b}).$$

Finally, $\underline{A}^{-1}\underline{a} \cdot \underline{b} = \underline{a} \cdot (\underline{A}^{-1})^t \underline{b} = \underline{a} \cdot \underline{A}^{-t} \underline{b}$, using the definition of the transpose. $\square$

PROBLEM 22 (Cofactor identities). *Let $\underline{A}^*$ satisfy $\underline{A}^*(\underline{a} \times \underline{b}) = \underline{A}\underline{a} \times \underline{A}\underline{b}$. Show: (a) $I_2 = \text{tr } \underline{A}^*$; (b) $I_2 = \frac{1}{2}[(\text{tr } \underline{A})^2 - \text{tr}(\underline{A}^2)]$; (c) $\underline{A}^t \underline{A}^* = (\det \underline{A})\underline{I}$, hence $\underline{A}^* = (\det \underline{A})\underline{A}^{-t}$; (d) recover $\det \underline{A} = \frac{1}{6}\varepsilon_{ijk}\varepsilon_{lmn}A_{il}A_{jm}A_{kn}$.*

Solution. Step 0: is $\underline{A}^$ even well defined?* Definition 1.54 prescribes the value of $\underline{A}^*$ only on vectors that are presented to us in the form $\underline{a} \times \underline{b}$. Before the symbol may be used at all, two questions must be answered. Does the prescription reach every vector, so that at most one tensor can satisfy it? And does a tensor satisfying it exist, given that the same vector can be written as a cross product in infinitely many ways, so that the prescription might contradict itself?

Uniqueness. Every vector is a cross product. If $\underline{v} = \underline{0}$, then $\underline{v} = \underline{a} \times \underline{a}$ for any $\underline{a}$, since $\underline{a} \times \underline{a} = \underline{0}$. If $\underline{v} \neq \underline{0}$, set $\underline{u}_3 = \underline{v}/|\underline{v}|$, a unit vector, and complete it to a right-handed orthonormal triad $\{\underline{u}_1, \underline{u}_2, \underline{u}_3\}$, so that $\underline{u}_1 \times \underline{u}_2 = \underline{u}_3$. Then

$$\underline{u}_1 \times (|\underline{v}| \underline{u}_2) = |\underline{v}| (\underline{u}_1 \times \underline{u}_2) = |\underline{v}| \underline{u}_3 = \underline{v}.$$

So the defining relation does prescribe $\underline{A}^*\underline{v}$ for every $\underline{v}$, and two tensors obeying it take the same value on every vector, hence are the same tensor.

Existence. We produce $\underline{A}^*$ explicitly and verify the definition. Put

$$A_{mi}^* := \frac{1}{2} \varepsilon_{mjk} \varepsilon_{ipq} A_{jp} A_{kq}, \quad \underline{A}^* := A_{mi}^* \underline{e}_m \otimes \underline{e}_i. \quad (*1)$$

This is a tensor by construction, being a linear combination of dyads. Now compute its action on $\underline{a} \times \underline{b}$, whose components are $(\underline{a} \times \underline{b})_i = \varepsilon_{irs} a_r b_s$ by Proposition 1.24:

$$\begin{aligned} [\underline{A}^*(\underline{a} \times \underline{b})]_m &= A_{mi}^* (\underline{a} \times \underline{b})_i \\ &= \frac{1}{2} \varepsilon_{mjk} \varepsilon_{ipq} A_{jp} A_{kq} \varepsilon_{irs} a_r b_s \\ &= \frac{1}{2} \varepsilon_{mjk} A_{jp} A_{kq} (\varepsilon_{ipq} \varepsilon_{irs}) a_r b_s. \end{aligned}$$

The bracket is contracted on its *first* index, whereas Theorem 1.29 contracts on the last; the two are the same statement because ε is unchanged

by a cyclic shift of its indices, $\varepsilon_{ipq} = \varepsilon_{pqi}$ and $\varepsilon_{irs} = \varepsilon_{rsi}$, so

$$\varepsilon_{ipq}\varepsilon_{irs} = \varepsilon_{pqi}\varepsilon_{rsi} = \delta_{pr}\delta_{qs} - \delta_{ps}\delta_{qr}.$$

Substituting,

$$\begin{aligned} [\tilde{A}^*(\underline{a} \times \underline{b})]_m &= \frac{1}{2} \varepsilon_{mjk} A_{jp} A_{kq} (\delta_{pr}\delta_{qs} - \delta_{ps}\delta_{qr}) a_r b_s \\ &= \frac{1}{2} \varepsilon_{mjk} A_{jp} A_{kq} a_p b_q - \frac{1}{2} \varepsilon_{mjk} A_{jp} A_{kq} a_q b_p, \end{aligned}$$

the deltas having collapsed the sums by forcing $r = p$, $s = q$ in the first term and $s = p$, $r = q$ in the second. The two terms are in fact equal. In the second, first interchange the *names* of the summation indices p and q , which changes nothing since both are summed over 1, 2, 3:

$$-\frac{1}{2} \varepsilon_{mjk} A_{jp} A_{kq} a_q b_p = -\frac{1}{2} \varepsilon_{mjk} A_{jq} A_{kp} a_p b_q;$$

then interchange the names of the summation indices j and k :

$$-\frac{1}{2} \varepsilon_{mjk} A_{jq} A_{kp} a_p b_q = -\frac{1}{2} \varepsilon_{mkj} A_{kq} A_{jp} a_p b_q = +\frac{1}{2} \varepsilon_{mjk} A_{jp} A_{kq} a_p b_q,$$

the last equality using $\varepsilon_{mkj} = -\varepsilon_{mjk}$. Hence the two halves add rather than cancel, and

$$[\tilde{A}^*(\underline{a} \times \underline{b})]_m = \varepsilon_{mjk} (A_{jp} a_p) (A_{kq} b_q) = \varepsilon_{mjk} (\underline{Aa})_j (\underline{Ab})_k = (\underline{Aa} \times \underline{Ab})_m.$$

So $(*)_1$ does satisfy Definition 1.54, and by the uniqueness just proved it is *the* cofactor. Both the object and its components are now in hand.

(a) $I_2(\underline{A}) = I_1(\underline{A}^*)$. By Theorem 1.51, $I_1(\underline{A}^*) = \text{tr } \underline{A}^* = A_{11}^* + A_{22}^* + A_{33}^*$ with $A_{ii}^* = \underline{e}_i \cdot \underline{A}^* \underline{e}_i$ (no sum). For a right-handed orthonormal basis, $\underline{e}_1 = \underline{e}_2 \times \underline{e}_3$, $\underline{e}_2 = \underline{e}_3 \times \underline{e}_1$, $\underline{e}_3 = \underline{e}_1 \times \underline{e}_2$, so the definition of $\underline{A}^*$ evaluates on each basis vector separately. The three computations, written out in full rather than left to a cyclic instruction:

$$\begin{aligned} \underline{A}^* \underline{e}_1 &= \underline{A}^*(\underline{e}_2 \times \underline{e}_3) = \underline{A} \underline{e}_2 \times \underline{A} \underline{e}_3, & A_{11}^* &= \underline{e}_1 \cdot \underline{A} \underline{e}_2 \times \underline{A} \underline{e}_3 = [\underline{e}_1, \underline{A} \underline{e}_2, \underline{A} \underline{e}_3], \\ \underline{A}^* \underline{e}_2 &= \underline{A}^*(\underline{e}_3 \times \underline{e}_1) = \underline{A} \underline{e}_3 \times \underline{A} \underline{e}_1, & A_{22}^* &= \underline{e}_2 \cdot \underline{A} \underline{e}_3 \times \underline{A} \underline{e}_1 = [\underline{e}_2, \underline{A} \underline{e}_3, \underline{A} \underline{e}_1], \\ \underline{A}^* \underline{e}_3 &= \underline{A}^*(\underline{e}_1 \times \underline{e}_2) = \underline{A} \underline{e}_1 \times \underline{A} \underline{e}_2, & A_{33}^* &= \underline{e}_3 \cdot \underline{A} \underline{e}_1 \times \underline{A} \underline{e}_2 = [\underline{e}_3, \underline{A} \underline{e}_1, \underline{A} \underline{e}_2]. \end{aligned}$$

The box product is unchanged by a cyclic permutation of its three slots, $[\underline{u}, \underline{v}, \underline{w}] = [\underline{w}, \underline{u}, \underline{v}]$, because a cyclic permutation is two transpositions

and each transposition reverses the sign. Applying it to the second and third lines so that all three entries are ordered as in Definition 1.49₂:

$$[\underline{e}_2, \underline{A}\underline{e}_3, \underline{A}\underline{e}_1] = [\underline{A}\underline{e}_1, \underline{e}_2, \underline{A}\underline{e}_3], \quad [\underline{e}_3, \underline{A}\underline{e}_1, \underline{A}\underline{e}_2] = [\underline{A}\underline{e}_1, \underline{A}\underline{e}_2, \underline{e}_3].$$

Adding the three diagonal entries,

$$\text{tr } \underline{A}^* = [\underline{A}\underline{e}_1, \underline{A}\underline{e}_2, \underline{e}_3] + [\underline{A}\underline{e}_1, \underline{e}_2, \underline{A}\underline{e}_3] + [\underline{e}_1, \underline{A}\underline{e}_2, \underline{A}\underline{e}_3].$$

The right side is exactly Definition 1.49₂ evaluated at $(a, b, c) = (\underline{e}_1, \underline{e}_2, \underline{e}_3)$, for which $[\underline{e}_1, \underline{e}_2, \underline{e}_3] = 1$. Hence $\text{tr } \underline{A}^* = I_2(\underline{A})$, which is the assertion.

(b) $I_2(\underline{A}) = \frac{1}{2}[(\text{tr } \underline{A})^2 - \text{tr}(\underline{A}^2)]$. By (a) it suffices to contract $(*_1)$ on $m = i$:

$$I_2(\underline{A}) = \text{tr } \underline{A}^* = A_{ii}^* = \frac{1}{2} \varepsilon_{ijk} \varepsilon_{ipq} A_{jp} A_{kq}.$$

Again shifting cyclically to expose the contraction pattern of Theorem 1.29, $\varepsilon_{ijk} \varepsilon_{ipq} = \varepsilon_{jki} \varepsilon_{pqi} = \delta_{jp} \delta_{kq} - \delta_{jq} \delta_{kp}$, so

$$I_2(\underline{A}) = \frac{1}{2}(\delta_{jp} \delta_{kq} A_{jp} A_{kq} - \delta_{jq} \delta_{kp} A_{jp} A_{kq}).$$

Take the two pieces one at a time. In the first, δ_{jp} forces $p = j$ and δ_{kq} forces $q = k$, leaving

$$\delta_{jp} \delta_{kq} A_{jp} A_{kq} = A_{jj} A_{kk} = (\text{tr } \underline{A})(\text{tr } \underline{A}) = (\text{tr } \underline{A})^2.$$

In the second, δ_{jq} forces $q = j$ and δ_{kp} forces $p = k$, leaving

$$\delta_{jq} \delta_{kp} A_{jp} A_{kq} = A_{jk} A_{kj}.$$

This last expression is a trace: the components of $\underline{A}^2 = \underline{A}\underline{A}$ are $(\underline{A}^2)_{jl} = A_{jk} A_{kl}$, so setting $l = j$ and summing gives $(\underline{A}^2)_{jj} = A_{jk} A_{kj} = \text{tr}(\underline{A}^2)$. Therefore

$$I_2(\underline{A}) = \frac{1}{2}[(\text{tr } \underline{A})^2 - \text{tr}(\underline{A}^2)].$$

(c) $\underline{A}^t \underline{A}^* = (\det \underline{A}) \underline{I}$. Two independent proofs are given, the first free of components.

Without coordinates. Let $\underline{a}, \underline{b}, \underline{c}$ be arbitrary and set $\underline{v} = \underline{a} \times \underline{b}$. Using the definition of the transpose, $\underline{c} \cdot \underline{A}^t \underline{w} = \underline{A} \underline{c} \cdot \underline{w}$, and then the definition of the cofactor,

$$\begin{aligned} \underline{c} \cdot \underline{A}^t \underline{A}^* \underline{v} &= \underline{A} \underline{c} \cdot \underline{A}^* (\underline{a} \times \underline{b}) \\ &= \underline{A} \underline{c} \cdot \underline{A} \underline{a} \times \underline{A} \underline{b} \\ &= [\underline{A} \underline{c}, \underline{A} \underline{a}, \underline{A} \underline{b}] \\ &= [\underline{A} \underline{a}, \underline{A} \underline{b}, \underline{A} \underline{c}], \end{aligned}$$

the last step being the cyclic shift again. By Definition 1.49₃ together with Theorem 1.52, which identify I_3 with $\det$,

$$[\underline{A} \underline{a}, \underline{A} \underline{b}, \underline{A} \underline{c}] = (\det \underline{A}) [\underline{a}, \underline{b}, \underline{c}] = (\det \underline{A}) \underline{c} \cdot \underline{a} \times \underline{b} = (\det \underline{A}) \underline{c} \cdot \underline{v},$$

where the middle equality is one more cyclic shift, $[\underline{a}, \underline{b}, \underline{c}] = [\underline{c}, \underline{a}, \underline{b}] = \underline{c} \cdot \underline{a} \times \underline{b}$. Comparing the two displays,

$$\underline{c} \cdot [\underline{A}^t \underline{A}^* - (\det \underline{A}) \underline{I}] \underline{v} = 0 \quad \text{for every } \underline{c}.$$

A vector orthogonal to every vector is the zero vector, so $[\underline{A}^t \underline{A}^* - (\det \underline{A}) \underline{I}] \underline{v} = \underline{0}$ for every $\underline{v}$ of the form $\underline{a} \times \underline{b}$, that is, by Step 0, for every $\underline{v}$. A tensor sending every vector to $\underline{0}$ is $\underline{O}$, so $\underline{A}^t \underline{A}^* = (\det \underline{A}) \underline{I}$.

In components. Start from $(\underline{A}^t)_{ik} = A_{ki}$ and $(*)_1$:

$$(\underline{A}^t \underline{A}^*)_{ij} = (\underline{A}^t)_{ik} A_{kj}^* = A_{ki} A_{kj}^* = \frac{1}{2} \varepsilon_{kmn} \varepsilon_{j pq} A_{ki} A_{mp} A_{nq}.$$

The triple sum $\varepsilon_{kmn} A_{ki} A_{mp} A_{nq}$ is precisely Proposition 1.28 read with the matrix (A_{ij}) and columns i, p, q :

$$\varepsilon_{kmn} A_{ki} A_{mp} A_{nq} = \varepsilon_{ipq} \det \underline{A}.$$

(For completeness, here is why. The left side changes sign whenever two of i, p, q are interchanged: swapping i and p gives $\varepsilon_{kmn} A_{kp} A_{mi} A_{nq}$, and renaming the dummies $k \leftrightarrow m$ turns this into $\varepsilon_{mkn} A_{mp} A_{ki} A_{nq} = -\varepsilon_{kmn} A_{ki} A_{mp} A_{nq}$; the other two swaps are identical in form. Being trilinear and alternating in the columns, it equals a fixed multiple of ε_{ipq} by Lemma 1.25, and evaluating at $(i, p, q) = (1, 2, 3)$ identifies that

multiple as $\varepsilon_{kmn}A_{k1}A_{m2}A_{n3} = [\underline{A}\underline{e}_1, \underline{A}\underline{e}_2, \underline{A}\underline{e}_3] = \det \underline{A}$.) Substituting and using $\varepsilon_{j pq}\varepsilon_{i pq} = 2\delta_{ij}$ from Theorem 1.29,

$$(\underline{A}^t \underline{A}^*)_{ij} = \frac{1}{2} (\det \underline{A}) \varepsilon_{j pq} \varepsilon_{i pq} = \frac{1}{2} (\det \underline{A}) 2\delta_{ij} = (\det \underline{A}) \delta_{ij},$$

which is the component statement of $\underline{A}^t \underline{A}^* = (\det \underline{A}) \underline{I}$.

The consequence. Suppose now $\underline{A}$ is invertible, equivalently $\det \underline{A} \neq 0$ by Theorem 1.52. Then $\underline{A}^t$ is invertible too, and multiplying $\underline{A}^t \underline{A}^* = (\det \underline{A}) \underline{I}$ on the left by $(\underline{A}^t)^{-1}$ gives

$$\underline{A}^* = (\det \underline{A}) (\underline{A}^t)^{-1} = (\det \underline{A}) \underline{A}^{-t},$$

the last equality being $(\underline{A}^t)^{-1} = (\underline{A}^{-1})^t$, proved in Problem 18, which is what licenses the single symbol $\underline{A}^{-t}$.

(d) *The contracted determinant formula.* Take the trace of (c). On the right, $\text{tr}[(\det \underline{A}) \underline{I}] = (\det \underline{A}) \delta_{ii} = 3 \det \underline{A}$, since $\delta_{ii} = 1 + 1 + 1 = 3$. On the left, setting $j = i$ in the component computation just performed,

$$\text{tr}(\underline{A}^t \underline{A}^*) = (\underline{A}^t \underline{A}^*)_{ii} = A_{ki} A_{ki}^* = \frac{1}{2} \varepsilon_{kmn} \varepsilon_{ipq} A_{ki} A_{mp} A_{nq}.$$

Equating,

$$3 \det \underline{A} = \frac{1}{2} \varepsilon_{kmn} \varepsilon_{ipq} A_{ki} A_{mp} A_{nq}, \quad \text{so} \quad \det \underline{A} = \frac{1}{6} \varepsilon_{kmn} \varepsilon_{ipq} A_{ki} A_{mp} A_{nq}.$$

Renaming the summation indices, $k \rightarrow i$, $m \rightarrow j$, $n \rightarrow k$ in the first ε and $i \rightarrow l$, $p \rightarrow m$, $q \rightarrow n$ in the second, so that the three factors read A_{il} , A_{jm} , A_{kn} ,

$$\det \underline{A} = \frac{1}{6} \varepsilon_{ijk} \varepsilon_{lmn} A_{il} A_{jm} A_{kn},$$

in agreement with Corollary 1.30, which reached the same formula by contracting Proposition 1.28 directly. The two routes are genuinely different: there the $\frac{1}{6}$ came from $\varepsilon_{mnp} \varepsilon_{mnp} = 6$, here from $\frac{1}{2} \cdot 3$, the $\frac{1}{2}$ being built into the cofactor and the 3 being $\text{tr } \underline{I}$.

Checks. Take $\underline{A} = \underline{I}$ first. Then $\underline{I}\underline{a} \times \underline{I}\underline{b} = \underline{a} \times \underline{b}$, so $\underline{I}^* = \underline{I}$ and $\text{tr } \underline{I}^* = 3$; part (b) gives $\frac{1}{2}[3^2 - 3] = \frac{1}{2} \cdot 6 = 3$, and part (c) gives $\underline{I}^t \underline{I}^* = \underline{I} = (\det \underline{I}) \underline{I}$ since $\det \underline{I} = 1$. Part (d) gives $\frac{1}{6} \varepsilon_{ijk} \varepsilon_{lmn} \delta_{il} \delta_{jm} \delta_{kn} = \frac{1}{6} \varepsilon_{ijk} \varepsilon_{ijk} = \frac{1}{6} \cdot 6 = 1$.

Take next $\underline{A} = \alpha \underline{e}_1 \otimes \underline{e}_1 + \beta \underline{e}_2 \otimes \underline{e}_2 + \gamma \underline{e}_3 \otimes \underline{e}_3$, so $\underline{A}\underline{e}_1 = \alpha \underline{e}_1$, $\underline{A}\underline{e}_2 = \beta \underline{e}_2$, $\underline{A}\underline{e}_3 = \gamma \underline{e}_3$. From the definition,

$$\begin{aligned}\underline{A}^* \underline{e}_1 &= \underline{A}\underline{e}_2 \times \underline{A}\underline{e}_3 = \beta\gamma (\underline{e}_2 \times \underline{e}_3) = \beta\gamma \underline{e}_1, \\ \underline{A}^* \underline{e}_2 &= \underline{A}\underline{e}_3 \times \underline{A}\underline{e}_1 = \gamma\alpha (\underline{e}_3 \times \underline{e}_1) = \gamma\alpha \underline{e}_2, \\ \underline{A}^* \underline{e}_3 &= \underline{A}\underline{e}_1 \times \underline{A}\underline{e}_2 = \alpha\beta (\underline{e}_1 \times \underline{e}_2) = \alpha\beta \underline{e}_3,\end{aligned}$$

so $\text{tr } \underline{A}^* = \beta\gamma + \gamma\alpha + \alpha\beta$. Part (b) predicts $\frac{1}{2}[(\alpha + \beta + \gamma)^2 - (\alpha^2 + \beta^2 + \gamma^2)]$, and expanding the square, $(\alpha + \beta + \gamma)^2 = \alpha^2 + \beta^2 + \gamma^2 + 2(\alpha\beta + \beta\gamma + \gamma\alpha)$, the squares cancel and the $\frac{1}{2}$ removes the 2, leaving $\alpha\beta + \beta\gamma + \gamma\alpha$, which agrees. Part (c) predicts $\underline{A}^t \underline{A}^* = (\alpha\beta\gamma)\underline{I}$, and indeed $\underline{A}^t = \underline{A}$ here while $\underline{A}\underline{A}^* \underline{e}_1 = \beta\gamma \underline{A}\underline{e}_1 = \alpha\beta\gamma \underline{e}_1$, and likewise $\underline{A}\underline{A}^* \underline{e}_2 = \alpha\beta\gamma \underline{e}_2$ and $\underline{A}\underline{A}^* \underline{e}_3 = \alpha\beta\gamma \underline{e}_3$, while $\det \underline{A} = \alpha\beta\gamma$. $\square$

REMARK 1.108 (what the cofactor is for). Part (c) is not a curiosity; it is the identity that makes surface deformation computable. If $\underline{a} \times \underline{b}$ is the vector area of the parallelogram on $\underline{a}$ and $\underline{b}$, then Definition 1.54 says that $\underline{A}^*$ carries vector areas exactly as $\underline{A}$ carries vectors, and (c) converts that geometric statement into the working formula $\underline{A}^* = (\det \underline{A})\underline{A}^{-t}$. Applied in Chapter 2 to the deformation gradient $\underline{F}$ with $J = \det \underline{F}$, the same two lines become $\underline{F}^* = J\underline{F}^{-t}$ and the Piola–Nanson formula $\underline{n} \, da = \underline{F}^* \underline{N} \, dA$. Part (b) is the identity that makes I_2 computable from $\underline{A}$ alone, without ever forming $\underline{A}^*$, and it is what turns the characteristic polynomial of Problem 23 into something one can evaluate.

PROBLEM 23 (Characteristic polynomial). *Show $\det(\underline{A} - \lambda \underline{I}) = -\lambda^3 + I_1 \lambda^2 - I_2 \lambda + I_3$.*

Solution. By Definition 1.49₃ applied to the tensor $\underline{A} - \lambda \underline{I}$,

$$\begin{aligned}\det(\underline{A} - \lambda \underline{I}) [\underline{a}, \underline{b}, \underline{c}] &= [(\underline{A} - \lambda \underline{I})\underline{a}, (\underline{A} - \lambda \underline{I})\underline{b}, (\underline{A} - \lambda \underline{I})\underline{c}] \\ &= [\underline{A}\underline{a} - \lambda \underline{a}, \underline{A}\underline{b} - \lambda \underline{b}, \underline{A}\underline{c} - \lambda \underline{c}].\end{aligned}$$

Expand by linearity in each slot separately. Each slot contributes either $\underline{A}(\cdot)$ or $-\lambda(\cdot)$, so there are $2^3 = 8$ terms. Group them by how many

slots took the $-\lambda$:

$$0 \text{ slots : } [\underline{A}\underline{a}, \underline{A}\underline{b}, \underline{A}\underline{c}] = I_3[\underline{a}, \underline{b}, \underline{c}],$$

$$1 \text{ slot : } -\lambda \left[[\underline{a}, \underline{A}\underline{b}, \underline{A}\underline{c}] + [\underline{A}\underline{a}, \underline{b}, \underline{A}\underline{c}] + [\underline{A}\underline{a}, \underline{A}\underline{b}, \underline{c}] \right] = -\lambda I_2[\underline{a}, \underline{b}, \underline{c}],$$

$$2 \text{ slots : } \lambda^2 \left[[\underline{A}\underline{a}, \underline{b}, \underline{c}] + [\underline{a}, \underline{A}\underline{b}, \underline{c}] + [\underline{a}, \underline{b}, \underline{A}\underline{c}] \right] = \lambda^2 I_1[\underline{a}, \underline{b}, \underline{c}],$$

$$3 \text{ slots : } (-\lambda)^3 [\underline{a}, \underline{b}, \underline{c}] = -\lambda^3 [\underline{a}, \underline{b}, \underline{c}],$$

the three groupings being exactly Definition 1.49_{3, 2, 1} in turn. Adding and cancelling the non-zero $[\underline{a}, \underline{b}, \underline{c}]$,

$$\det(\underline{A} - \lambda \underline{I}) = -\lambda^3 + I_1(\underline{A})\lambda^2 - I_2(\underline{A})\lambda + I_3(\underline{A}).$$

The alternating signs are simply the $(-\lambda)^n$ of the grouping, and this is why the invariants enter (3.2) in that order: I_k is the coefficient produced by putting $\underline{A}$ into exactly k of the three slots. $\square$

PROBLEM 24 (Field product rules). *For scalar fields φ, ψ and vector fields $\underline{u}, \underline{v}$, show (a) $\nabla(\varphi + \psi) = \nabla\varphi + \nabla\psi$; (b) $\text{div}(\underline{u} + \underline{v}) = \text{div}\underline{u} + \text{div}\underline{v}$; (c) $\text{div}(\varphi\underline{v}) = \nabla\varphi \cdot \underline{v} + \varphi \text{div}\underline{v}$; (d) $\text{curl}(\nabla\varphi) = \underline{0}$; (e) $\text{div}(\text{curl}\underline{v}) = 0$; (f) $\nabla(\varphi\underline{u}) = \underline{u} \otimes \nabla\varphi + \varphi\nabla\underline{u}$; (g) $\text{div}(\underline{u} \otimes \underline{v}) = (\nabla\underline{u})\underline{v} + (\text{div}\underline{v})\underline{u}$.*

Solution. All follow from the Cartesian representations and the ordinary product and equality-of-mixed-partial rules; write everything in components.

(a) $[\nabla(\varphi + \psi)]_i = (\varphi + \psi)_{,i} = \varphi_{,i} + \psi_{,i}$. (b) $\text{div}(\underline{u} + \underline{v}) = (u_i + v_i)_{,i} = u_{i,i} + v_{i,i}$. (c) $\text{div}(\varphi\underline{v}) = (\varphi v_i)_{,i} = \varphi_{,i} v_i + \varphi v_{i,i} = \nabla\varphi \cdot \underline{v} + \varphi \text{div}\underline{v}$. (d) $[\text{curl}(\nabla\varphi)]_j = \varepsilon_{kij}(\varphi_{,i})_{,k} = \varepsilon_{kij}\varphi_{,ik}$; since $\varphi_{,ik} = \varphi_{,ki}$ is symmetric in (i, k) while ε_{kij} is antisymmetric under $i \leftrightarrow k$, the contraction vanishes. (e) $\text{div}(\text{curl}\underline{v}) = (\varepsilon_{kij}v_{i,k})_{,j} = \varepsilon_{kij}v_{i,kj}$; $v_{i,kj}$ is symmetric in (k, j) and ε_{kij} antisymmetric under $k \leftrightarrow j$, so the sum is 0. (f) $[\nabla(\varphi\underline{u})]_{ij} = (\varphi u_i)_{,j} = u_i \varphi_{,j} + \varphi u_{i,j} = (\underline{u} \otimes \nabla\varphi)_{ij} + \varphi(\nabla\underline{u})_{ij}$. (g) $[\text{div}(\underline{u} \otimes \underline{v})]_i = (u_i v_j)_{,j} = u_{i,j} v_j + u_i v_{j,j} = [(\nabla\underline{u})\underline{v}]_i + (\text{div}\underline{v})u_i$. (In (d),(e) twice continuous differentiability is assumed, so mixed partials commute.)

Parts (d) and (e) are the same statement twice: a symmetric array contracted with an antisymmetric one vanishes, since $A_{pq}S_{pq} = -A_{qp}S_{pq} = -A_{pq}S_{qp} = -A_{pq}S_{pq}$ after relabelling, so the quantity equals minus itself. The same argument proves the Piola identity in Problem 2 of Chapter 2. $\square$

PROBLEM 25 (Gradient in spherical coordinates). *With $\underline{x} = r\underline{e}_\rho(\theta, \varphi)$, $\underline{e}_\rho = \cos \varphi \underline{e}_r(\theta) + \sin \varphi \underline{e}_3$, $\underline{e}_\varphi = \underline{e}_\rho \times \underline{e}_\theta$, find the components of ∇f in the variable basis $\{\underline{e}_\rho, \underline{e}_\theta, \underline{e}_\varphi\}$.*

Solution. Step 1: build the frame from the Cartesian basis. Nothing is assumed about spherical coordinates; everything is constructed. The angle θ is the azimuth, measured in the equatorial plane from $\underline{e}_1$, and φ is the elevation above that plane, so φ is latitude and not the colatitude used in much of the physics literature. Start with the unit vector pointing along the projection of $\underline{x}$ onto the equatorial plane,

$$\underline{e}_r(\theta) = \cos \theta \underline{e}_1 + \sin \theta \underline{e}_2,$$

and differentiate it with respect to θ , the $\underline{e}_i$ being fixed:

$$\begin{aligned} \underline{e}_\theta(\theta) &:= \underline{e}_r'(\theta) = -\sin \theta \underline{e}_1 + \cos \theta \underline{e}_2, \\ \underline{e}_\theta'(\theta) &= -\cos \theta \underline{e}_1 - \sin \theta \underline{e}_2 = -\underline{e}_r(\theta). \end{aligned}$$

Both are unit vectors, since $\cos^2 \theta + \sin^2 \theta = 1$ in each case, and they are orthogonal:

$$\underline{e}_r \cdot \underline{e}_\theta = \cos \theta (-\sin \theta) + \sin \theta (\cos \theta) = 0.$$

Also $\underline{e}_r \cdot \underline{e}_3 = 0$ and $\underline{e}_\theta \cdot \underline{e}_3 = 0$, both vectors having no $\underline{e}_3$ component. Tilting $\underline{e}_r$ up out of the equatorial plane by the angle φ produces the radial direction given in the statement,

$$\underline{e}_\rho(\theta, \varphi) = \cos \varphi \underline{e}_r(\theta) + \sin \varphi \underline{e}_3 = \cos \varphi \cos \theta \underline{e}_1 + \cos \varphi \sin \theta \underline{e}_2 + \sin \varphi \underline{e}_3,$$

so that $\underline{x} = r\underline{e}_\rho$ has Cartesian coordinates $x_1 = r \cos \varphi \cos \theta$, $x_2 = r \cos \varphi \sin \theta$, $x_3 = r \sin \varphi$, and $|\underline{x}| = r$ as the notation intends.

Step 2: the third leg of the frame, computed rather than quoted. The statement defines $\underline{e}_\varphi = \underline{e}_\rho \times \underline{e}_\theta$. Using $(\underline{u} \times \underline{v})_1 = u_2 v_3 - u_3 v_2$ and its two companions from Proposition 1.24, with

$$\underline{e}_\rho = (\cos \varphi \cos \theta, \cos \varphi \sin \theta, \sin \varphi), \quad \underline{e}_\theta = (-\sin \theta, \cos \theta, 0),$$

one obtains

$$\begin{aligned} (\underline{e}_\rho \times \underline{e}_\theta)_1 &= (\cos \varphi \sin \theta)(0) - (\sin \varphi)(\cos \theta) \\ &= -\sin \varphi \cos \theta, \\ (\underline{e}_\rho \times \underline{e}_\theta)_2 &= (\sin \varphi)(-\sin \theta) - (\cos \varphi \cos \theta)(0) \\ &= -\sin \varphi \sin \theta, \\ (\underline{e}_\rho \times \underline{e}_\theta)_3 &= (\cos \varphi \cos \theta)(\cos \theta) - (\cos \varphi \sin \theta)(-\sin \theta) \\ &= \cos \varphi (\cos^2 \theta + \sin^2 \theta) = \cos \varphi. \end{aligned}$$

Collecting, and recognising $\cos \theta \underline{e}_1 + \sin \theta \underline{e}_2 = \underline{e}_r$,

$$\underline{e}_\varphi = -\sin \varphi \cos \theta \underline{e}_1 - \sin \varphi \sin \theta \underline{e}_2 + \cos \varphi \underline{e}_3 = -\sin \varphi \underline{e}_r(\theta) + \cos \varphi \underline{e}_3.$$

Step 3: the frame is orthonormal and right-handed. Each of the six conditions is verified, using $\underline{e}_r \cdot \underline{e}_r = 1$, $\underline{e}_\theta \cdot \underline{e}_\theta = 1$, $\underline{e}_r \cdot \underline{e}_\theta = 0$, $\underline{e}_r \cdot \underline{e}_3 = \underline{e}_\theta \cdot \underline{e}_3 = 0$, $\underline{e}_3 \cdot \underline{e}_3 = 1$:

$$\begin{aligned} \underline{e}_\rho \cdot \underline{e}_\rho &= \cos^2 \varphi \underline{e}_r \cdot \underline{e}_r + 2 \cos \varphi \sin \varphi \underline{e}_r \cdot \underline{e}_3 + \sin^2 \varphi \underline{e}_3 \cdot \underline{e}_3 \\ &= \cos^2 \varphi + \sin^2 \varphi = 1, \\ \underline{e}_\varphi \cdot \underline{e}_\varphi &= \sin^2 \varphi \underline{e}_r \cdot \underline{e}_r - 2 \sin \varphi \cos \varphi \underline{e}_r \cdot \underline{e}_3 + \cos^2 \varphi \underline{e}_3 \cdot \underline{e}_3 \\ &= \sin^2 \varphi + \cos^2 \varphi = 1, \\ \underline{e}_\theta \cdot \underline{e}_\theta &= 1, \\ \underline{e}_\rho \cdot \underline{e}_\theta &= \cos \varphi \underline{e}_r \cdot \underline{e}_\theta + \sin \varphi \underline{e}_3 \cdot \underline{e}_\theta = 0 + 0 = 0, \\ \underline{e}_\varphi \cdot \underline{e}_\theta &= -\sin \varphi \underline{e}_r \cdot \underline{e}_\theta + \cos \varphi \underline{e}_3 \cdot \underline{e}_\theta = 0 + 0 = 0, \\ \underline{e}_\rho \cdot \underline{e}_\varphi &= (\cos \varphi)(-\sin \varphi) \underline{e}_r \cdot \underline{e}_r + (\sin \varphi)(\cos \varphi) \underline{e}_3 \cdot \underline{e}_3 \\ &= -\cos \varphi \sin \varphi + \sin \varphi \cos \varphi = 0. \end{aligned}$$

For the handedness, use $\underline{u} \times (\underline{v} \times \underline{w}) = \underline{u} \cdot \underline{w} \underline{v} - \underline{u} \cdot \underline{v} \underline{w}$ from Problem 11(a) with $\underline{u} = \underline{w} = \underline{e}_\theta$ and $\underline{v} = \underline{e}_\rho$:

$$\underline{e}_\theta \times \underline{e}_\varphi = \underline{e}_\theta \times (\underline{e}_\rho \times \underline{e}_\theta) = \underline{e}_\theta \cdot \underline{e}_\theta \underline{e}_\rho - \underline{e}_\theta \cdot \underline{e}_\rho \underline{e}_\theta = 1 \cdot \underline{e}_\rho - 0 \cdot \underline{e}_\theta = \underline{e}_\rho,$$

so $[\underline{e}_\rho, \underline{e}_\theta, \underline{e}_\varphi] = \underline{e}_\rho \cdot \underline{e}_\theta \times \underline{e}_\varphi = \underline{e}_\rho \cdot \underline{e}_\rho = 1 > 0$. The triad is a right-handed orthonormal basis at every point where it is defined.

Step 4: how the frame turns. Differentiate $\underline{e}_\rho$ with respect to each angle, remembering that $\underline{e}_3$ is constant and $\underline{e}'_r = \underline{e}_\theta$:

$$\frac{\partial \underline{e}_\rho}{\partial \theta} = \cos \varphi \underline{e}'_r(\theta) + 0 = \cos \varphi \underline{e}_\theta, \quad \frac{\partial \underline{e}_\rho}{\partial \varphi} = -\sin \varphi \underline{e}_r + \cos \varphi \underline{e}_3 = \underline{e}_\varphi.$$

The second is exactly the vector computed in Step 2, which is why the problem defines $\underline{e}_\varphi$ by a cross product: the cross product and the φ derivative deliver the same direction, so the frame that is natural geometrically is also the frame that is natural for differentiation. Hence, treating r, θ, φ as independent,

$$d\underline{e}_\rho = \frac{\partial \underline{e}_\rho}{\partial \theta} d\theta + \frac{\partial \underline{e}_\rho}{\partial \varphi} d\varphi = \cos \varphi d\theta \underline{e}_\theta + d\varphi \underline{e}_\varphi.$$

Step 5: the line element. Differentiate $\underline{x} = r\underline{e}_\rho(\theta, \varphi)$ by the product rule:

$$d\underline{x} = dr \underline{e}_\rho + r d\underline{e}_\rho = \underbrace{1}_{h_r} dr \underline{e}_\rho + \underbrace{r \cos \varphi}_{h_\theta} d\theta \underline{e}_\theta + \underbrace{r}_{h_\varphi} d\varphi \underline{e}_\varphi. \quad (*_2)$$

The three coefficients $h_r = 1$, $h_\theta = r \cos \varphi$, $h_\varphi = r$ are the only content of the problem. They are lengths per unit coordinate: increasing θ by $d\theta$ slides the point along the circle of latitude, whose radius is not r but $r \cos \varphi$ (Figure 1.1), so the distance travelled is $r \cos \varphi d\theta$; increasing φ by $d\varphi$ slides it along a meridian of radius r , a distance $r d\varphi$; increasing r moves it radially a distance dr , whence $h_r = 1$.

REMARK 1.109 (the three scale factors, read off the picture). Each coefficient in $(*_2)$ is the distance moved per unit coordinate:

$$\begin{aligned} dr \text{ at fixed } \theta, \varphi : & \quad \text{outward along } \underline{e}_\rho, \text{ distance } dr, & h_r = 1, \\ d\theta \text{ at fixed } r, \varphi : & \quad \text{along the parallel, radius } \varrho = r \cos \varphi, & h_\theta = r \cos \varphi, \\ d\varphi \text{ at fixed } r, \theta : & \quad \text{along the meridian, radius } r, & h_\varphi = r. \end{aligned}$$

The middle one is the trap: the parallel has radius $\varrho = r \cos \varphi$, the distance to the *axis*, not r . Two checks. At the equator $\cos \varphi = 1$ and

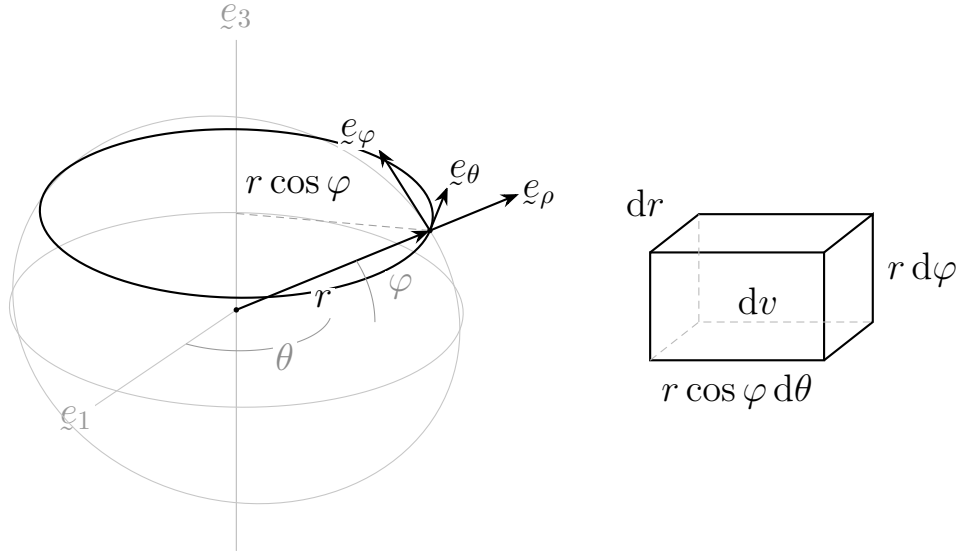


Figure 1.1: Left: the spherical frame at $\underline{x} = r\underline{e}_\rho$. The heavy circle is the parallel of latitude φ , and its radius is the dashed distance $r \cos \varphi$ from the polar axis, not r ; this single fact is the whole reason $h_\theta = r \cos \varphi$. The triad $\{\underline{e}_\rho, \underline{e}_\theta, \underline{e}_\varphi\}$ points outward, east and north, and $\underline{e}_\theta$ appears short only because it is foreshortened, pointing towards the reader. Right: the box built on the three coordinate increments, with edge lengths dr , $r \cos \varphi d\theta$ and $r d\varphi$; each edge is a coordinate increment multiplied by its scale factor.

$h_\theta = r$. At a pole $\cos \varphi \rightarrow 0$ and $h_\theta \rightarrow 0$, the degeneracy that excludes the poles below.

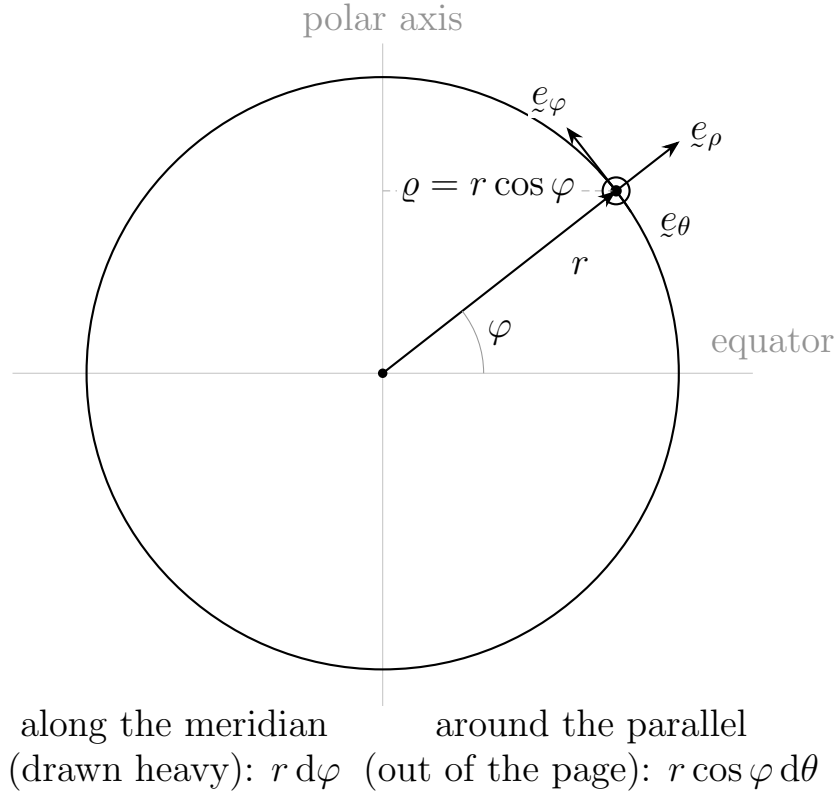


Figure 1.2: A meridional section, showing why $h_\theta = r \cos \varphi$ while $h_\varphi = r$. Since φ is measured from the equator, the point is at distance r from the origin but only $\rho = r \cos \varphi$ from the polar axis. A change of longitude sweeps it around that smaller circle, a change of latitude along a meridian of radius r . The azimuthal direction $\underline{e}_\theta$ comes out of the page, marked by the circled dot.

Step 6: invert the line element. Because $\{\underline{e}_\rho, \underline{e}_\theta, \underline{e}_\varphi\}$ is orthonormal, the three coefficients in $(*_2)$ are recovered by dotting with the corresponding leg, every cross term vanishing:

$$\begin{aligned}\underline{e}_\rho \cdot d\tilde{x} &= dr \underline{e}_\rho \cdot \underline{e}_\rho + r \cos \varphi d\theta \underline{e}_\rho \cdot \underline{e}_\theta + r d\varphi \underline{e}_\rho \cdot \underline{e}_\varphi = dr, \\ \underline{e}_\theta \cdot d\tilde{x} &= dr \underline{e}_\theta \cdot \underline{e}_\rho + r \cos \varphi d\theta \underline{e}_\theta \cdot \underline{e}_\theta + r d\varphi \underline{e}_\theta \cdot \underline{e}_\varphi = r \cos \varphi d\theta, \\ \underline{e}_\varphi \cdot d\tilde{x} &= dr \underline{e}_\varphi \cdot \underline{e}_\rho + r \cos \varphi d\theta \underline{e}_\varphi \cdot \underline{e}_\theta + r d\varphi \underline{e}_\varphi \cdot \underline{e}_\varphi = r d\varphi.\end{aligned}$$

Solving for the coordinate increments, which requires $r \neq 0$ and $\cos \varphi \neq 0$,

$$dr = \underline{e}_\rho \cdot d\tilde{x}, \quad d\theta = \frac{1}{r \cos \varphi} \underline{e}_\theta \cdot d\tilde{x}, \quad d\varphi = \frac{1}{r} \underline{e}_\varphi \cdot d\tilde{x}.$$

Step 7: the chain rule and the identification. Write $f = \hat{f}(r, \theta, \varphi)$ for the composite that assigns to the coordinates the value of the field at the corresponding point. The ordinary chain rule in three variables gives

$$df = f_{,r} dr + f_{,\theta} d\theta + f_{,\varphi} d\varphi, \quad f_{,r} := \frac{\partial \hat{f}}{\partial r}, \text{ etc.}$$

Substituting the three expressions just obtained and collecting the common factor $d\mathbf{x}$ on the right of every dot product,

$$\begin{aligned} df &= f_{,r} \underline{e}_\rho \cdot d\mathbf{x} + f_{,\theta} \frac{1}{r \cos \varphi} \underline{e}_\theta \cdot d\mathbf{x} + f_{,\varphi} \frac{1}{r} \underline{e}_\varphi \cdot d\mathbf{x} \\ &= \left(f_{,r} \underline{e}_\rho + \frac{1}{r \cos \varphi} f_{,\theta} \underline{e}_\theta + \frac{1}{r} f_{,\varphi} \underline{e}_\varphi \right) \cdot d\mathbf{x}. \end{aligned}$$

Compare with (1.114), $df = \nabla f \cdot d\mathbf{x}$. Subtracting, $\underline{w} \cdot d\mathbf{x} = 0$ where $\underline{w}$ is the difference of ∇f and the bracket. As $(dr, d\theta, d\varphi)$ ranges over $\mathbb{R}^3$ the vector $d\mathbf{x}$ in $(*_2)$ ranges over all of $\mathbb{E}^3$, precisely because the three scale factors are non-zero and the triad is a basis; so $\underline{w}$ is orthogonal to every vector and therefore $\underline{w} = \underline{0}$. Hence

$$\boxed{\nabla f = f_{,r} \underline{e}_\rho + \frac{1}{r \cos \varphi} f_{,\theta} \underline{e}_\theta + \frac{1}{r} f_{,\varphi} \underline{e}_\varphi} \quad (*_3)$$

and the components sought are $(\nabla f)_\rho = f_{,r}$, $(\nabla f)_\theta = f_{,\theta}/(r \cos \varphi)$, $(\nabla f)_\varphi = f_{,\varphi}/r$.

Where it fails. The derivation used $r \neq 0$ and $\cos \varphi \neq 0$. The first excludes the origin, where no radial direction exists; the second excludes the two poles $\varphi = \pm\pi/2$, where the circle of latitude degenerates to a point, $\underline{e}_\theta$ is undefined, and longitude carries no information. These are failures of the coordinate system, not of the field: ∇f itself is perfectly well defined there, as (1.114) makes plain, since it never mentions coordinates.

Checks. Three independent ones, each computed both ways.

Take $f = r = |\mathbf{x}|$. Then $f_{,r} = 1$ and $f_{,\theta} = f_{,\varphi} = 0$, so $(*_3)$ gives $\nabla r = \underline{e}_\rho$. Independently, from $r^2 = \mathbf{x} \cdot \mathbf{x}$ one has $2r dr = 2\mathbf{x} \cdot d\mathbf{x}$, hence $dr = (\mathbf{x}/r) \cdot d\mathbf{x}$ and $\nabla r = \mathbf{x}/r = \underline{e}_\rho$, in agreement.

Take $f = x_3 = r \sin \varphi$. Then $f_{,r} = \sin \varphi$, $f_{,\theta} = 0$, $f_{,\varphi} = r \cos \varphi$, so

$$\nabla f = \sin \varphi \underline{e}_\rho + \frac{1}{r} r \cos \varphi \underline{e}_\varphi = \sin \varphi \underline{e}_\rho + \cos \varphi \underline{e}_\varphi.$$

Substituting the Cartesian forms of $\underline{e}_\rho$ and $\underline{e}_\varphi$,

$$\begin{aligned} \nabla f &= \sin \varphi (\cos \varphi \underline{e}_r + \sin \varphi \underline{e}_3) + \cos \varphi (-\sin \varphi \underline{e}_r + \cos \varphi \underline{e}_3) \\ &= (\sin \varphi \cos \varphi - \cos \varphi \sin \varphi) \underline{e}_r + (\sin^2 \varphi + \cos^2 \varphi) \underline{e}_3 = \underline{e}_3, \end{aligned}$$

which is right, since $\nabla x_3 = \underline{e}_3$ directly from $dx_3 = \underline{e}_3 \cdot d\mathbf{x}$.

Take $f = \theta$. Then $(*_3)$ gives $\nabla \theta = \underline{e}_\theta / (r \cos \varphi)$. This is the familiar polar result $\nabla \theta = \underline{e}_\theta / \varrho$ with $\varrho = r \cos \varphi$ the distance from the polar axis, which is exactly what the dashed segment in Figure 1.1 measures, and it correctly blows up on the axis, where θ is discontinuous. $\square$

REMARK 1.110 (the scale-factor recipe, and the colatitude convention). Nothing in Steps 5 to 7 used the sphere. Whenever coordinates (q_1, q_2, q_3) produce a line element of the orthogonal form $d\mathbf{x} = \sum_k h_k dq_k \underline{f}_k$ with $\{\underline{f}_k\}$ orthonormal, the same three steps give

$$\nabla f = \sum_k \frac{1}{h_k} \frac{\partial f}{\partial q_k} \underline{f}_k,$$

the h_k being called scale factors. Cartesian coordinates have $h_1 = h_2 = h_3 = 1$ and the formula returns $\underline{f}_i \underline{e}_i$; the polar coordinates of §1.2.5 have $h_r = 1$, $h_\theta = r$, $h_z = 1$ and it returns $\underline{f}_r \underline{e}_r + r^{-1} \underline{f}_\theta \underline{e}_\theta + \underline{f}_z \underline{e}_3$. The division by h_k is not a convention but a units statement: $\partial f / \partial q_k$ is a change per unit *coordinate*, while a gradient component must be a change per unit *length*, and h_k is the number of lengths per coordinate.

One warning on conventions. Most physics texts measure the polar angle ϑ down from the north pole, the colatitude, and obtain $h_\vartheta = r$, $h_\theta = r \sin \vartheta$. Here φ is measured up from the equator, so $\varphi = \pi/2 - \vartheta$ and $\sin \vartheta = \cos \varphi$, which is why $\cos \varphi$ appears in $(*_3)$ where the familiar formula shows a sine. The two agree; only the angle being named has moved.

PROBLEM 26 (Laplace's equation between two spheres). *For $f = f(r)$, $r = |\underline{x}|$, solve $\operatorname{div}(\nabla f) = 0$ on $a \leq r \leq b$ with $f(a) = f_a$, $f(b) = f_b$.*

Solution. By Problem 25 with $f_{,\theta} = f_{,\varphi} = 0$, $\nabla f = f'(r)\underline{e}_\rho$. Put $\underline{v} = \nabla f = f'(r)\underline{e}_\rho$; then, using $d\underline{e}_\rho$ as above,

$$d\underline{v} = f''(r) dr \underline{e}_\rho + f'(r) (\cos \varphi \underline{e}_\theta d\theta + \underline{e}_\varphi d\varphi),$$

and inserting $dr = \underline{e}_\rho \cdot d\underline{x}$, $d\theta = \frac{1}{r \cos \varphi} \underline{e}_\theta \cdot d\underline{x}$, $d\varphi = \frac{1}{r} \underline{e}_\varphi \cdot d\underline{x}$ and $(\underline{e}_a \cdot d\underline{x})\underline{e}_b = (\underline{e}_b \otimes \underline{e}_a)d\underline{x}$,

$$\nabla \underline{v} = f'' \underline{e}_\rho \otimes \underline{e}_\rho + \frac{f'}{r} \underline{e}_\theta \otimes \underline{e}_\theta + \frac{f'}{r} \underline{e}_\varphi \otimes \underline{e}_\varphi, \quad \operatorname{div} \underline{v} = \operatorname{tr}(\nabla \underline{v}) = f'' + \frac{2f'}{r}.$$

So Laplace's equation is $f'' + 2f'/r = 0$, i.e. $(r^2 f')' = 0$, giving $r^2 f' = \text{const}$, $f' = \alpha/r^2$, $f(r) = \beta - \alpha/r$. Rewriting the constants through the boundary conditions,

$$\boxed{f(r) = f_a + (f_b - f_a) \frac{\frac{1}{a} - \frac{1}{r}}{\frac{1}{a} - \frac{1}{b}}}$$

which indeed gives $f(a) = f_a$, $f(b) = f_b$. □

PROBLEM 27 (A curl identity and the curl of a polar field). *Establish $\underline{u} \cdot \operatorname{curl} \underline{v} = \operatorname{div}(\underline{v} \times \underline{u}) + \underline{v} \cdot \operatorname{curl} \underline{u}$ and derive the curl of the field (1.145)'s underlying $\underline{u} = u_r \underline{e}_r + u_\theta \underline{e}_\theta + u_z \underline{k}$.*

Solution. Identity. Write $(\operatorname{curl} \underline{v})_i = \varepsilon_{ijk} v_{k,j}$. Then

$$\operatorname{div}(\underline{v} \times \underline{u}) = \varepsilon_{ijk} (v_j u_k)_{,i} = \varepsilon_{ijk} v_{j,i} u_k + \varepsilon_{ijk} v_j u_{k,i}.$$

In the first term $\varepsilon_{ijk} v_{j,i} = \varepsilon_{kij} v_{j,i} = (\operatorname{curl} \underline{v})_k$, so it is $\underline{u} \cdot \operatorname{curl} \underline{v}$; in the second $\varepsilon_{ijk} u_{k,i} = -\varepsilon_{jik} u_{k,i} = -(\operatorname{curl} \underline{u})_j$, so it is $-\underline{v} \cdot \operatorname{curl} \underline{u}$. Hence $\operatorname{div}(\underline{v} \times \underline{u}) = \underline{u} \cdot \operatorname{curl} \underline{v} - \underline{v} \cdot \operatorname{curl} \underline{u}$, which rearranges to the stated identity.

Curl of the polar field. For any field, $\operatorname{curl} \underline{u}$ is the axial vector of $\nabla \underline{u} - (\nabla \underline{u})^t$ (that is, $(\nabla \underline{u} - (\nabla \underline{u})^t)\underline{a} = (\operatorname{curl} \underline{u}) \times \underline{a}$). From (1.145),

$$\begin{aligned} \nabla \underline{u} - (\nabla \underline{u})^t &= c(\underline{e}_r \otimes \underline{e}_\theta - \underline{e}_\theta \otimes \underline{e}_r) + u_{z,r}(\underline{k} \otimes \underline{e}_r - \underline{e}_r \otimes \underline{k}) \\ &\quad + \frac{1}{r} u_{z,\theta}(\underline{k} \otimes \underline{e}_\theta - \underline{e}_\theta \otimes \underline{k}), \end{aligned}$$

with $c = \frac{1}{r}(u_{r,\theta} - u_\theta) - u_{\theta,r}$. The axial vector of $\underline{e}_a \otimes \underline{e}_b - \underline{e}_b \otimes \underline{e}_a$ is $\underline{e}_b \times \underline{e}_a$, so

$$\begin{aligned}\operatorname{curl} \underline{u} &= c(\underline{e}_\theta \times \underline{e}_r) + u_{z,r}(\underline{e}_r \times \underline{k}) + \frac{1}{r}u_{z,\theta}(\underline{e}_\theta \times \underline{k}) \\ &= \frac{1}{r}u_{z,\theta} \underline{e}_r - u_{z,r} \underline{e}_\theta + \left[u_{\theta,r} + \frac{1}{r}(u_\theta - u_{r,\theta}) \right] \underline{k},\end{aligned}$$

using $\underline{e}_\theta \times \underline{e}_r = -\underline{k}$, $\underline{e}_r \times \underline{k} = -\underline{e}_\theta$, $\underline{e}_\theta \times \underline{k} = \underline{e}_r$. □

PROBLEM 28 (Cartesian formula for the curl of a tensor). *Establish* $\operatorname{curl} \underline{A} = \varepsilon_{ilm} A_{jl,i} \underline{e}_m \otimes \underline{e}_j$.

Solution. By definition $(\operatorname{curl} \underline{A})\underline{c} = \operatorname{curl}(\underline{A}^t \underline{c})$ for fixed $\underline{c}$. Take $\underline{c} = \underline{e}_j$; then $\underline{A}^t \underline{e}_j = A_{jl} \underline{e}_l$ is a vector field with l th component A_{jl} , and with $(\operatorname{curl} \underline{w})_m = \varepsilon_{miq} w_{q,i}$,

$$\operatorname{curl}(\underline{A}^t \underline{e}_j) = \varepsilon_{miq} A_{jq,i} \underline{e}_m = \varepsilon_{ilm} A_{jl,i} \underline{e}_m$$

(renaming $q \rightarrow l$ and using $\varepsilon_{miq} = \varepsilon_{qmi}$, i.e. writing the antisymmetric symbol in the cyclic order ilm). Writing $\operatorname{curl} \underline{A} = C_{mj} \underline{e}_m \otimes \underline{e}_j$ and matching $(\operatorname{curl} \underline{A})\underline{e}_j = C_{mj} \underline{e}_m$ gives $C_{mj} = \varepsilon_{ilm} A_{jl,i}$, i.e. $\operatorname{curl} \underline{A} = \varepsilon_{ilm} A_{jl,i} \underline{e}_m \otimes \underline{e}_j$. □

Appendix A

Dictionary to the book

Book	Here	Book	Here
(1.1)–(1.3)	Def. 1.2 , Thm. 1.4	(1.79)–(1.84)	Thm. 1.51
ip axioms	Def. 1.5	(1.85), (1.86)	Def. 1.54 , Prop. 1.55
(1.4)–(1.6)	Thm. 1.6 –Cor. 1.7	(1.87)–(1.93)	Def. 1.57 , Thm. 1.58
(1.7)–(1.16)	Def. 1.9 –Prop. 1.11	(1.95)	Prop. 1.63
(1.17)–(1.19)	Thm. 1.15	(1.96), (1.97)	Def. 1.68 , Def. 1.66
(1.20)–(1.22)	Thm. 1.19	(1.98), (1.99)	Thm. 1.69
(1.23)–(1.30)	Def. 1.21 –Prop. 1.24	(1.100)–(1.108)	Thm. 1.73 –Thm. 1.75
(1.31)–(1.39)	Def. 1.27 –Cor. 1.30	(1.109)–(1.115)	Thm. 1.77 –Ex. 1.80
(1.40)–(1.50)	Def. 1.31 –Ex. 1.34	(1.116)–(1.125)	Ex. 1.81 , Ex. 1.82
(1.55)–(1.58)	Thm. 1.39	(1.126)–(1.134)	Def. 1.84 –Cor. 1.86
(1.59)–(1.62)	Prop. 1.42 , Prop. 1.44	(1.135)–(1.145)	Ex. 1.87 –Ex. 1.89
(1.63)–(1.72)	Prop. 1.46	(1.146)–(1.152)	Def. 1.90 –Prop. 1.94
(1.73)–(1.77)	Prop. 1.48	(1.153)–(1.162)	Ex. 1.98
(1.78)	Def. 1.49	(1.163)–(1.166)	Def. 1.95 –Rem. 1.97
sym/skw, sph/dev	Def. 1.59	(1.167), (1.168)	Thm. 1.99 , Thm. 1.101

Lemma [1.25](#) (one dimensionality of alternating 3 forms) replaces the book's Appendix A computations in Propositions [1.28](#), [1.50](#) and Theorem [1.52](#). All twenty eight chapter problems are solved in §[1.3](#).